Forcing global Π^2_n uniformization, with and without projective determinacy

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Abstract

We construct a set-forcing extension of L which satisfies the Π^2_n-uniformization property for every finite $n \geq 1$. Assuming ω Woodin cardinals and a measurable cardinal above them, we also obtain an extension of $L(\mathbb{R})^{M_\omega}$ which adds no reals and satisfies projective determinacy together with Π^2_n uniformization for every finite $n \geq 1$.

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1 Introduction

A *uniformization* of $A \subseteq \omega^\omega \times \omega^\omega$ is a function $U \subseteq A$ with $\operatorname{dom} U = \operatorname{dom} A$. A pointclass has the uniformization property if each of its relations has a uniformization in the pointclass. We work with second-order pointclasses over HC: first-order quantifiers range over HC, class quantifiers over all subsets of HC.

Definable uniformizations classically come from scales and determinacy; see Moschovakis [17]. At the second-order level the two sides behave differently. Under $\mathsf{AD} + V = L(\mathbb{R})$, the scale analysis of Steel [19] and Martin and Steel [15] gives the scale property, and hence uniformization, for Σ^2_1. Woodin extended the positive scale result to $\mathsf{ZF} + \mathsf{AD}^+ + V = L(\mathcal{P}(\mathbb{R}))$; see Steel and Trang [24, Theorem 2]. An argument of Kechris and Solovay, similar to the classical argument of Lévy, gives, under $\mathsf{AD} + V = L(\mathbb{R})$, a Π^2_1 relation which has no uniformization in $L(\mathbb{R})$ at all [21, Theorem 3.11 and the following discussion].

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Positive results on the Π side come from games of length ω_1. Building on [22], Steel [23] showed that ω_1-open-projective determinacy gives the scale property for the open game quantifier on analytical trees: the games have length ω_1 and integer moves, and the trees of countable positions are definable over $(\mathrm{HC}, \in)$ from parameters. Under CH this pointclass is Π^2_1, so the hypothesis yields Π^2_1 uniformization [23, Corollary 0.6 and Theorem 2.1].

The relevant determinacy hypothesis follows from Neeman's work [18]: in the form quoted by Steel [23, Theorem 0.1], it suffices that every real belongs to a countable, (ω_1+1)-iterable mouse satisfying ZFC^-+ "there is a measurable Woodin cardinal." Steel distinguishes this mouse-existence hypothesis from an ordinary large-cardinal assumption and notes that deriving it from the latter requires iterability results which are not yet known [23, footnote 1]. Neeman also records Steel's result that definable open length-ω_1 determinacy fails in the minimal iterable model with a class of indiscernible Woodin cardinals [18, p. 83].

On the forcing side, a precursor of our work is the construction of Abraham and Shelah [1]: without adding reals, they obtain a Δ^2_2 wellorder of the reals and, by the same method, code prescribed relations on the reals. This is an earlier example of a forcing which changes second-order definability while leaving the reals unchanged.

We obtain the Π side by forcing over L, at every finite level simultaneously.

Theorem 1.1. *There is a cardinal-preserving set-forcing extension $L[G]$ of L which adds no reals and satisfies Π^2_n-UP for every $n \geq 1$.*

Since the forcing adds no reals and preserves ω_1, HC is unchanged, but the new subsets of HC change the meaning of the class quantifiers.

The construction is also compatible with projective determinacy.

Theorem 1.2. *Assume there are ω Woodin cardinals and a measurable cardinal above them, and put $\mathcal{W} = L(\mathbb{R})^{M_\omega}$. There is a forcing $\mathbb{P} \in \mathcal{W}$ which preserves all cardinals of $\mathcal{W}$, adds no reals, and forces $\mathsf{PD} + \mathsf{CH}$ and Π^2_n-UP for every finite $n \geq 1$.*

The ground model $\mathcal{W}$ is its own $L(\mathbb{R})$ and satisfies $\mathsf{ZFC} + \mathsf{CH} + \mathsf{PD}$. A forcing which adds no reals changes neither $L(\mathbb{R})$ nor the projective sets, and adds no strategies for games on ω; hence it preserves PD. As before, the second-order pointclasses change only because new subsets of HC are added.

Both ground models show the opposite pattern. Their canonical wellorders of the subsets of ω_1 are Σ^2_1-good (in $\mathcal{W}$ with respect to a definable predicate A), and choosing least witnesses gives Σ^2_n uniformization for every finite $n \geq 1$. Since the two dual properties cannot hold at the same level, we obtain the following.

Corollary 1.3. *Under the hypothesis of Theorem 1.2, the models* $\mathcal{W} \subseteq \mathcal{W}[G]$ *both satisfy* $\mathsf{ZFC} + \mathsf{PD} + \mathsf{CH}$*, have the same* HC*, and satisfy, for every finite* $n \geq 1$*,*

$$\mathcal{W} \models \Sigma^2_n\text{-UP} \ \wedge \ \neg\mathbf{\Pi}^2_n\text{-UP},$$
$$\mathcal{W}[G] \models \Pi^2_n\text{-UP} \ \wedge \ \neg\mathbf{\Sigma}^2_n\text{-UP}.$$

The analogous statement holds for L *and the extension* $L[G]$ *of Theorem 1.1.*

Thus, relative to the stated large-cardinal hypothesis, both of these patterns of second-order uniformization are consistent with $\mathsf{PD} + \mathsf{CH}$. The forcing turns one into the other while preserving the reals, HC and all projective truth; only the collection of classes over HC changes.

The good wellorders which give the Σ pattern in the ground models are related to those studied by Lücke, Schindler and Schlicht [14]. Their Theorem 5.2 gives a good $\Sigma_1(\omega_1)$-definable wellorder of H_{ω_2} in M_1. They also prove that infinitely many Woodin cardinals and a measurable above them preclude a $\Sigma_1(\omega_1)$-definable uniformization of the club filter on ω_1 [14, Theorem 4.19]; there the inputs and outputs are subsets of ω_1, rather than reals.

Lücke and Müller [13, Theorem 1.6] show that the same large-cardinal hypothesis, or Woodin's axiom $(*)$, precludes a Σ_1-definable wellorder of any subset of $\mathcal{P}(\omega_1)$ of size greater than $\aleph_1$, with parameters in $\mathrm{HC} \cup \{\omega_1\}$. At the second-order level, a theorem of Woodin rules out a Σ^2_1 wellorder of the reals under CH and a measurable Woodin cardinal [1, p. 2].

These results leave open what happens above the first level in the presence of large cardinals. Steel's theorem gives Π^2_1 uniformization from CH and the mouse-existence hypothesis stated above, and it does not concern Π^2_n for $n \geq 2$. The models constructed in this paper do not decide the matter either. In both of them the reals have a Δ^2_1 wellorder: in the extension of L this is the constructible wellorder of the reals, and in the extension of $\mathcal{W}$ it is the wellorder $\prec$ of Section 10, which is Δ^2_1 there by Lemmas 10.5 and 10.6. By Woodin's theorem, such a wellorder is incompatible with CH and a measurable Woodin cardinal. Since our construction starts from a ground model with a definable wellorder of this kind and adds no reals, it cannot be carried out in the presence of a measurable Woodin cardinal. We therefore ask the following.

Question 1.4. *Is* Π^2_2-UP *consistent with* CH *and the existence of a measurable Woodin cardinal? More generally, is the conjunction of* Π^2_n-UP *for all finite* $n \geq 1$ *consistent with these assumptions?*

The method. Fix a recursive injection ℓ which codes a triple (e, x, y), consisting of a natural number and two reals, by a real $\ell(e, x, y)$. Let $A_{n,e}(x, y)$ be the Π^2_n formula with Gödel number e. In the final extension we define

the uniformizing function by

$$U_{n,e}(x,y) \iff A_{n,e}(x,y) \wedge \neg\,\mathrm{Code}_n(\ell(e,x,y)).$$

Each real t in the range of ℓ indexes a member S^n_t of a canonical family of stationary subsets of ω_n in L. Given a sequence $\vec{T} = (T_n)_{n\geq 1}$ of sets of such reals, we construct a forcing $F_{\vec{T}}$ in whose extension

$$\mathrm{Code}_n(t) \iff S^n_t \text{ is nonstationary in } \omega_n \iff t \in T_n.$$

For $n = 1$, $\mathrm{Code}_1(t)$ simply asserts that there is a club in ω_1 disjoint from S^1_t. For $n \geq 2$, a club in ω_n is too large to be a class over HC. Here $\mathrm{Code}_n(t)$ asserts that there is a set $X \subseteq \omega_1$ from which iterated almost disjoint decoding produces a set $C_X \subseteq \omega_n$[1] with the following property: every suitable model N which is likewise coded by a subset of ω_1, and which contains the relevant set decoded from X, believes that $C_X \cap \omega^N_n$ is a club in ω^N_n disjoint from $S^n_t \cap \omega^N_n$. The ordinals ω^N_n are cofinal in ω_n, and all these models see initial segments of the same set C_X; hence such a set X yields a club in ω_n disjoint from S^n_t in the final model. Section 7 expresses this property by a parameter-free Σ^2_n formula. Thus $U_{n,e}$ is Π^2_n, and it uniformizes $A_{n,e}$ provided that every nonempty section $\{y : A_{n,e}(x,y)\}$ contains exactly one y with $\ell(e,x,y) \notin T_n$.

The difficulty is that the sections themselves depend on $\vec{T}$. The forcing adds no reals, so every candidate y is already in L, but new subsets of HC can change the truth value of $A_{n,e}(x,y)$. To determine in L which statements $A_{n,e}(x,y)$ a given forcing $F_{\vec{T}}$ forces, we reduce Π^2_n truth in the full extension to $\Pi^2_1(H_{\omega_n})$ truth in the intermediate extension by the club shooting forcing. Cone homogeneity of the club shooting forcing then shows that each such statement with real parameters from the ground model is decided by the trivial condition.

We start with $T_n = I$ for every n, where I is the range of ℓ, and treat $n = 1, 2, \ldots$ in increasing order. At step n, take a pair (e,x) to which no value has been assigned yet and for which $F_{\vec{T}} \Vdash A_{n,e}(x,y)$ for some y. Choose one such y, assign it to (e,x), and remove only $\ell(e,x,y)$ from T_n. Shrinking T_n while all other T_m are fixed preserves every Π^2_n statement which is already forced, so y remains in its section. A section which was empty can become nonempty, so we repeat this step, taking intersections of the sets T_n at limit stages, until every nonempty section has been assigned a value. Each step assigns a value to a new pair (e,x), so by CH the recursion stops before ω_2. Once $T_1, \ldots, T_n$ are fixed, changing the sets T_m for $m > n$, even all of them simultaneously, does not change which Π^2_n statements are forced. Thus each

[1] For coding by stationary sets see Friedman and Magidor [5], and for localization see Fischer and Friedman [3] and Fischer, Friedman and Zdomskyy [4]. A related application is given in [9].

nonempty section in the final model contains its assigned value y, and this y is the only element of the section with $\ell(e, x, y) \notin T_n$.

For Theorem 1.2 we write $\mathcal{W} = L[A]$, where $A \subseteq \omega_1^{\mathcal{W}}$ codes the natural enumeration of the reals, and relativize the construction to $(\mathrm{HC}, \in, A)$. Membership and nonmembership in A have parameter-free Σ^2_1 definitions which remain correct in all relevant extensions; they allow us to eliminate A from the definitions of the uniformizing functions without increasing their complexity.

Organization. Section 2 collects preliminaries. Section 3 constructs the stationary sets and the club shooting forcing. Subsection 3.7 sketches the case $n = 2$, which already shows one of the main ideas: we pass from larger models to smaller ones by removing codes, and this process is carried out entirely in the ground model. Section 4 defines the coding forcing and proves its basic properties. Section 5 shows that its bookkeeping handles all relevant names and gives a first-order characterization of the class of forcings used in the reduction of complexity. Sections 6 and 7 give the reduction of complexity and the formulas Code_n. Sections 8 and 9 compare the forcings $F_{\vec{T}}$ for different sequences $\vec{T}$ and complete the proof of Theorem 1.1. Section 10 relativizes the construction to $L[A]$, proves Theorem 1.2 and derives Corollary 1.3.

2 Preliminaries

2.1 Second-order pointclasses

Let H be a transitive set. We refer to subsets of H as *classes over* H. A formula is $\Sigma^2_n(H)$ if it has the form

$$\exists X_1 \forall X_2 \cdots Q X_n \, \varphi,$$

where the class quantifiers range over $\mathcal{P}(H)$ and φ is first order over $(H, \in)$, with the class variables as additional predicates. Define $\Pi^2_n(H)$ dually. If no domain is mentioned, it is HC.

The pointclasses Π^2_n and Σ^2_n are lightface: their definitions use no parameters. Their boldface counterparts $\mathbf{\Pi}^2_n$ and $\mathbf{\Sigma}^2_n$ allow real parameters.

Definition 2.1. Π^2_n-UP *means that every* Π^2_n *relation* $A \subseteq \omega^\omega \times \omega^\omega$ *has a* Π^2_n *uniformization. Define* Σ^2_n-UP *analogously.*

Until Section 10, the extensions constructed for the main theorem add no reals and preserve ω_1. Thus they have the same HC as L and satisfy CH. Their collections of classes over HC may differ.

2.2 Cardinals

Until Section 10, the ground model is L. We write ω_n and $\aleph_\alpha$ for its cardinals, with $\omega_0 = \omega$. As proved below, the forcing extensions used in the construction preserve all cardinals, so this notation has the same interpretation in each of them.

2.3 A criterion for simultaneous uniformization

Fix a recursive injection ℓ which maps every triple (e, x, y), where $e < \omega$ and x, y are reals, to a real $\ell(e, x, y) \in 2^\omega$, and let

$$I = \{\ell(e, x, y) : e < \omega,\ x, y \in \mathbb{R}^L\}$$

be its range in L. We call the elements of I *codes* and regard them as subsets of ω. Thus $|I| = \omega_1$ in L.

For $\vec{R}, \vec{T} \in (\mathcal{P}(I))^\omega$, write $\vec{R} \leq \vec{T}$ if $R_n \subseteq T_n$ for every $n \geq 1$. The following criterion is proved in Section 9.

Theorem 2.2. *Suppose that $\vec{T} \mapsto F_{\vec{T}}$ is a function, definable in L, which assigns a set forcing $F_{\vec{T}} \in L$ to every $\vec{T} \in (\mathcal{P}(I))^\omega \cap L$, and that the following hold for all such sequences $\vec{R}, \vec{T}$ and every $n \geq 1$.*

(A1) $F_{\vec{T}}$ preserves cardinals and adds no reals, and for every second-order formula φ and all reals a in L, either $1_{F_{\vec{T}}} \Vdash \varphi(a)$ or $1_{F_{\vec{T}}} \Vdash \neg\varphi(a)$.

(A2) There is a parameter-free Σ^2_n formula Code_n, not depending on $\vec{T}$, such that $F_{\vec{T}} \Vdash (\mathrm{Code}_n(t) \leftrightarrow t \in T_n)$ for every $t \in I$.

(A3) If $\vec{R} \leq \vec{T}$ and $R_m = T_m$ for all $m \neq n$, then every Π^2_n statement with real parameters which is forced by $F_{\vec{T}}$ is forced by $F_{\vec{R}}$.

(A4) If $R_m = T_m$ for all $m \leq n$, then $F_{\vec{R}}$ and $F_{\vec{T}}$ force the same Π^2_n statements with real parameters; here R_m and T_m may differ for infinitely many $m > n$.

Assume moreover that the relation "$1_{F_{\vec{T}}} \Vdash \varphi(a)$" is definable in L, as a relation between Gödel numbers of second-order formulas φ, sequences $\vec{T}$ and finite sequences a of reals. Then there is $\vec{T}_* \in L$ such that $F_{\vec{T}_*}$ forces Π^2_n-UP for every $n \geq 1$.

2.4 Forcing terminology

We use the forcing and iteration terminology of [10, Chapters 14–16]. Let ρ be a regular cardinal and let R be a forcing.

- R is $< \rho$-*closed* if every descending sequence of length $< \rho$ has a lower bound, and $< \rho$-*distributive* if it adds no new sequences of ordinals of length $< \rho$.

- A set of conditions is *centered* if each of its finite subsets has a common lower bound. R is $< \rho$-*compact* if every centered set of fewer than ρ conditions has a lower bound.

- R is ρ-*linked* if it is the union of ρ many sets of pairwise compatible conditions. A function $c : R \to \rho$ such that any two conditions with the same value are compatible is called a *linking function*.

- R is *cone homogeneous* if for any two conditions p, q there are $p' \leq p$ and $q' \leq q$ such that the cones $\{r \in R : r \leq p'\}$ and $\{r \in R : r \leq q'\}$ are isomorphic.

- Following [2], R is *countably parallel-closed* if, whenever $\langle p_n : n < \omega \rangle$ and $\langle q_n : n < \omega \rangle$ are descending sequences in R such that p_n and q_n are compatible for every $n < \omega$, there is $r \in R$ with $r \leq p_n, q_n$ for all $n < \omega$.

- For a regular uncountable cardinal ρ, R satisfies the *stationary* ρ^+*-chain condition* if for every sequence $\langle p_\alpha : \alpha < \rho^+ \rangle$ in R there are a club $C \subseteq \rho^+$ and a regressive function $f : C \cap \{\alpha < \rho^+ : \mathrm{cf}(\alpha) = \rho\} \to \rho^+$ such that p_α and p_β are compatible whenever $\alpha, \beta \in C$ have cofinality ρ and $f(\alpha) = f(\beta)$.

A *projection* $\pi : P \to Q$ preserves the order and the largest condition and, whenever $q \leq \pi(p)$, admits $p' \leq p$ with $\pi(p') \leq q$. For the restriction maps considered below we shall prove more: whenever $q \leq \pi(p)$, there is $p' \leq p$ with $\pi(p') = q$ which agrees with p outside the coordinates to which we restrict.

An isomorphism between two cones of a forcing R induces a bijection between the corresponding names which maps $\check{a}$ to $\check{a}$ for every set a of the ground model. This gives the following standard fact.

Fact 2.3. *Let R be cone homogeneous, let φ be a formula, and let a be a set in the ground model. Then either $1_R \Vdash \varphi(\check{a})$ or $1_R \Vdash \neg\varphi(\check{a})$.*

Proof. Otherwise there are conditions p, q with $p \Vdash \varphi(\check{a})$ and $q \Vdash \neg\varphi(\check{a})$. Choose $p' \leq p$ and $q' \leq q$ with isomorphic cones. Then $p' \Vdash \varphi(\check{a})$ implies $q' \Vdash \varphi(\check{a})$, a contradiction. □

The following fact combines two results. The chain condition is the preservation theorem [2, Theorem 1.2]; the closure is the standard fact that a $< \rho$-support iteration of $< \rho$-closed forcings is $< \rho$-closed.

Fact 2.4. *Let ρ be regular uncountable with $\rho^{<\rho} = \rho$. A $< \rho$-support iteration whose iterands are forced to be $< \rho$-compact and to satisfy the stationary ρ^+-chain condition is $< \rho$-closed and satisfies the stationary ρ^+-chain condition, hence the ρ^+-chain condition.*

Here compactness gives the closure of the iterands and the countable parallel closure which are required by [2, Theorem 1.2]. Indeed, a $< \rho$-compact forcing is $< \rho$-closed. Moreover, if $\langle p_n : n < \omega \rangle$ and $\langle q_n : n < \omega \rangle$ are descending sequences such that p_n and q_n are compatible for every n, then $\{p_n : n < \omega\} \cup \{q_n : n < \omega\}$ is centered, so countable compactness gives a common lower bound. Similarly, ρ-linkedness implies the stationary ρ^+-chain condition: if c is a linking function, then $\alpha \mapsto c(p_\alpha)$ is regressive on the club of all $\alpha \geq \rho$. Both implications are recorded in [2, Lemma 1.5]. We therefore obtain the following consequence, which is the form used below.

Corollary 2.5. *Let ρ be regular uncountable with $\rho^{<\rho} = \rho$. Every $< \rho$-support iteration whose iterands are forced to be $< \rho$-compact and ρ-linked is $< \rho$-closed and satisfies the stationary ρ^+-chain condition; in particular it is ρ^+-cc.*

Note that only the iterands are assumed to be ρ-linked; an iteration of ρ-linked forcings need not be ρ-linked.

We shall verify the preservation of cardinals by means of the following lemma.

Lemma 2.6. *Let $\mathbb{P}$ be a forcing. Suppose that for every $n \geq 1$ there are a $< \omega_n$-distributive forcing Q_n and a Q_n-name $\dot{R}_n$ for an ω_n-cc forcing such that $\mathbb{P}$ is forcing equivalent to $Q_n * \dot{R}_n$. Then $\mathbb{P}$ preserves all cardinals $\leq \aleph_{\omega+1}$. If $\mathbb{P}$ is also $\aleph_{\omega+2}$-cc, then it preserves all cardinals.*

Proof. Let G be $\mathbb{P}$-generic over V, fix $n \geq 1$, and let $M = V[G_{Q_n}]$ be the corresponding intermediate model. Since Q_n adds no sequences of ordinals of length $< \omega_n$, all cardinals $\leq \omega_n$ of V are cardinals in M, ω_n is regular in M, and

$$\mathrm{cf}^M(\aleph_{\omega+1}) \geq \omega_n.$$

The forcing $R_n = \dot{R}_n^{G_{Q_n}}$ is ω_n-cc in M and thus preserves all cardinals $\geq \omega_n$ of M. Hence ω_n is a cardinal in $V[G]$. As n was arbitrary, all cardinals $\leq \aleph_\omega$ are preserved.

Now let $f : \eta \to \aleph_{\omega+1}$ be a function in $V[G]$, where $\eta < \omega_n$, and let $\dot{f} \in M$ be an R_n-name for f. For $\zeta < \eta$ let $X_\zeta \in M$ be the set of all α such that some condition of R_n forces $\dot{f}(\zeta) = \check{\alpha}$. By the ω_n-chain condition, in M,

$$|X_\zeta| < \omega_n, \qquad X = \bigcup_{\zeta<\eta} X_\zeta \in [\aleph_{\omega+1}]^{<\omega_n}, \qquad \sup X < \aleph_{\omega+1},$$

and the range of f is contained in X. Consequently, in $V[G]$,

$$\mathrm{cf}(\aleph_{\omega+1}^V) \geq \omega_n \qquad (n \geq 1).$$

Since $\aleph_\omega$ is singular in $V[G]$, the cofinality of $\aleph_{\omega+1}^V$ in $V[G]$ is a regular cardinal greater than $\aleph_\omega$. If $\aleph_{\omega+1}^V$ were not a cardinal in $V[G]$, it would have cardinality $\aleph_\omega$ there, and hence cofinality at most $\aleph_\omega$. Thus $\aleph_{\omega+1}^V$ remains a regular cardinal. Finally, the $\aleph_{\omega+2}$-chain condition preserves all cardinals $\geq \aleph_{\omega+2}$. □

2.5 Club shooting and almost disjoint coding

For a regular ρ and $S \subseteq \rho$ with $\rho \setminus S$ stationary, $\mathbb{C}(\rho \setminus S)$ is the forcing of closed bounded subsets of $\rho \setminus S$, ordered by end extension.

Fix a regular ρ with $\rho^{<\rho} = \rho$, let $\sigma = \rho^+$, and fix a constructible almost disjoint family $\langle B_\eta : \eta < \sigma \rangle$ of unbounded subsets of ρ, that is, a family with pairwise bounded intersections. For $D \subseteq \sigma$, the forcing $\mathbb{A}_\rho(D)$ consists of pairs (s, F) with $s \in 2^{<\rho}$ and $F \in [D]^{<\rho}$, where $(s', F') \leq (s, F)$ iff $s' \supseteq s$, $F' \supseteq F$, and $s'(\alpha) = 0$ whenever $\alpha \in \operatorname{dom} s' \setminus \operatorname{dom} s$ and $\alpha \in B_\eta$ for some $\eta \in F$. We call s the *stem* and F the *restraint set* of (s, F). If G is generic and $Z = \{\alpha < \rho : s(\alpha) = 1 \text{ for some } (s, F) \in G\}$, then

$$\eta \in D \iff Z \cap B_\eta \text{ is bounded in } \rho \qquad (\eta < \sigma).$$

In this situation we say that *Z codes D by almost disjoint coding*. The forcing $\mathbb{A}_\rho(D)$ is $< \rho$-compact and ρ-linked, since conditions with the same stem are compatible.

2.6 Absoluteness

Lemma 2.7. *Let ρ be regular, let R be $< \rho$-distributive, and let P have a dense subset of size $< \rho$. Then R is still $< \rho$-distributive after forcing with P. That is, if $G_P \times G_R$ is generic for the product $P \times R$, formed in V, then $V[G_P][G_R]$ and $V[G_P]$ have the same sequences of ordinals of length $< \rho$.*

Proof. For $\rho = \omega$ the assertion is immediate. Otherwise, a fixed dense suborder of P of size $< \rho$ remains ρ-cc in V^R. Apply [25, Lemma 4.7]. □

Fact 2.8 (Silver)**.** *Let ρ be regular with $\rho^{<\rho} = \rho$ and let R be $< \rho$-closed. Then $\Sigma^2_1(H_\rho)$ statements with parameters in $\mathcal{P}(H_\rho) \cap V$ are absolute between V and V^R.*

Proof. Since $|H_\rho| = \rho$ and $H_\rho^{V^R} = H_\rho^V$, code $(H_\rho, \in, \vec{A})$ by a structure on ρ in V. Apply Silver's absoluteness lemma for existential second-order formulas [6, Lemma 3.5]; see also [12, p. 298, (I6)]. □

2.7 Term forcing

Let ρ be regular uncountable, $\sigma = \rho^+$, $\rho^{<\rho} = \rho$ and $\sigma^{<\rho} = \sigma$. Let P be ρ-cc with a fixed enumeration of a dense subset P_0 of cardinality at most ρ, and let $\dot{Q}$ be a P-name with $P \Vdash \dot{Q} = \mathbb{A}_\rho(\dot{B})$, where $\dot{B}$ is a P-name for a subset of σ.

Definition 2.9. *The* term forcing $\mathrm{Term}(P, \dot{Q})$ *consists of all P-names $\tau = (\dot{s}, \dot{F})$ with $1_P \Vdash \tau \in \dot{Q}$, where $\dot{s}$ and $\dot{F}$ are nice P_0-names for the stem and the restraint set. Put $\tau \leq \upsilon$ iff $1_P \Vdash \tau \leq \upsilon$.*

Note that the stems of the elements of $\mathrm{Term}(P, \dot{Q})$ are P-names, not elements of the ground model.

Theorem 2.10. $\mathrm{Term}(P, \dot{Q})$ *is $< \rho$-compact, ρ-linked and has a dense subset of size at most σ. The map $(p, \tau) \mapsto (p, \tau)$ is a projection*

$$P \times \mathrm{Term}(P, \dot{Q}) \twoheadrightarrow P * \dot{Q}.$$

Proof. The ρ-cc of P and the regularity of ρ give

$$1_P \Vdash \dot{\delta} < \rho \quad \Longrightarrow \quad \exists \beta < \rho \ (1_P \Vdash \dot{\delta} < \beta).$$

Thus the lengths of the stems and the sizes of the restraint sets of elements of $\mathrm{Term}(P, \dot{Q})$ are bounded below ρ. Using nice names for stems and for enumerations of restraint sets, choose sets $\mathcal{S}, \mathcal{N}$ of representatives for stems and conditions with

$$|\mathcal{S}| \leq (\rho \cdot |P_0|)^{<\rho} = \rho,$$
$$|\mathcal{N}| \leq (\sigma \cdot |P_0|)^{<\rho} = \sigma.$$

Every element of $\mathrm{Term}(P, \dot{Q})$ is equivalent to a member of $\mathcal{N}$, so $\mathcal{N}$ is dense. For two elements with the same name for the stem,

$$1_P \Vdash (\dot{s}, \dot{F} \cup \dot{F}') \leq (\dot{s}, \dot{F}), (\dot{s}, \dot{F}').$$

Hence the map which sends $(\dot{s}, \dot{F})$ to the index of the representative of $\dot{s}$ in $\mathcal{S}$ is a linking function.

Let $\{\tau_\alpha : \alpha < \eta\}$ be centered in $\mathrm{Term}(P, \dot{Q})$, where $\eta < \rho$. Then 1_P forces that $\{\tau_\alpha : \alpha < \eta\}$ is centered in $\dot{Q}$. Since P preserves the regularity of ρ, the compactness of $\dot{Q}$ gives

$$1_P \Vdash \exists q \in \dot{Q} \ \forall \alpha < \eta \ (q \leq \tau_\alpha).$$

By the maximum principle there is $\tau \in \mathrm{Term}(P, \dot{Q})$ with $\tau \leq \tau_\alpha$ for all $\alpha < \eta$.

Put $\pi(p, \tau) = (p, \tau)$. Given $(p', \dot{q}) \leq_{P * \dot{Q}} (p, \tau)$, choose a maximal antichain $A \subseteq P$ containing p' and a name τ' such that

$$p' \Vdash \tau' = \dot{q}, \qquad a \Vdash \tau' = \tau \quad (a \in A \setminus \{p'\}).$$

Then $\tau' \in \mathrm{Term}(P, \dot{Q})$, $1_P \Vdash \tau' \leq \tau$, and

$$(p', \tau') \leq_{P \times \mathrm{Term}(P,\dot{Q})} (p, \tau), \qquad \pi(p', \tau') \leq_{P * \dot{Q}} (p', \dot{q}).$$

The map π preserves the order and the largest condition, so it is a projection. □

Remark 2.11. The same proof applies to P-names for arbitrary forcings which are $< \rho$-compact and have a given linking function. To apply term forcing to an iteration of such forcings, which need not be ρ-linked, one lifts the names for its iterands successively along the projections from products to two-step iterations.

Remark 2.12. In our applications, the term forcing is defined in a model in which $\rho^{<\rho} = \rho$ holds, whereas the P-generic filter which evaluates its elements is generic over a larger model. All applications below use the conditions and the order of the term forcing as defined in the smaller model.

3 Stationary sets and club shooting

Fix $n \geq 1$. In this section we define stationary subsets of ω_n which are indexed by the reals in I, and the forcing which adds clubs disjoint from some of them.

The construction is related to the stationary sets used in [8, Section 2] and [4, Section 2, Steps 0–1]. We give the definitions and proofs needed here for every finite n. For $n = 1$, only the diamond sequence and its stationary sets are needed.

3.1 Constructible definitions

Fix the standard good Σ_1 wellorder $<_L$ of L, which orders sets first by the level of the constructible hierarchy at which they appear and then by their least definition; see [10, Chapter 13]. Write $\mathrm{Init}(p, J)$ for $p \in L$ and $J = \{q : q <_L p\}$, and fix a bounded formula q_{init} with

$$\mathrm{Init}(p, J) \iff \exists w \, q_{\mathrm{init}}(p, J, w).$$

For $p \in L_{\omega_n}$, both J and a witness w belong to L_{ω_n}.

For $\delta < \omega_n$, we say that a strictly increasing cofinal map $\omega_{n-1} \to \delta$ *witnesses* $\mathrm{cf}(\delta) = \omega_{n-1}$. We say that $\mathrm{cf}(\delta) \neq \omega_{n-1}$ is witnessed if $\delta = 0$, or by a cofinal map $\gamma \to \delta$ with $\gamma < \omega_{n-1}$. Here a map is cofinal in δ if every $\eta < \delta$ is at most some value of the map; thus a map with domain 1 is a witness for a successor ordinal. These witnesses lie in $H^L_{\omega_n} = L_{\omega_n}$. Collecting them for $\delta \leq a$ gives a Σ_1 definition of

$$A_a = (a + 1) \cap \mathrm{cof}(\omega_{n-1}) \qquad (a < \omega_n).$$

We use these witnesses in the bounded formulas below.

3.2 A canonical Σ_1-definable diamond sequence

Put $\rho = \omega_{n-1}$ and $A = \omega_n \cap \mathrm{cof}(\rho)$.

Lemma 3.1. *In L, there is a canonical $\diamondsuit(A)$-sequence $\langle d_\delta : \delta \in A\rangle$ whose graph has a functional Σ_1 definition $\Phi_D(\omega_{n-1}, \delta, d)$, uniformly in n. More precisely, there is a fixed bounded formula q_D such that*

$$\Phi_D(\omega_{n-1}, \delta, d) \iff \exists w\, q_D(\omega_{n-1}, \delta, d, w),$$

and, for every $\delta \in A$, a witness for $\Phi_D(\omega_{n-1}, \delta, d_\delta)$ belongs to L_{ω_n}.

Proof. For $\gamma \in A$ let (d_γ, c_γ) be the $<_L$-least pair (u, c) with $u \subseteq \gamma$, c club in γ, and

$$\forall \xi \in c \cap A \quad u \cap \xi \neq d_\xi.$$

Such a pair exists: the range of a continuous cofinal sequence in γ of length ρ, with successor ordinals at its initial and successor steps, is a club in γ which is disjoint from $A \cap \gamma$. Jensen's condensation argument gives $\diamondsuit(A)$ [11, Lemma 6.5]. The clubs of order type ρ just mentioned do not affect this argument. Indeed, let (X, C) be the $<_L$-least counterexample, let M be an elementary submodel which contains it, with $\gamma = M \cap \omega_n \in A$, and let $L_{\bar\beta}$ be the transitive collapse of M. Then γ is regular in $L_{\bar\beta}$, so no club in γ of order type ρ belongs to $L_{\bar\beta}$. All $<_L$-predecessors of $(X \cap \gamma, C \cap \gamma)$ belong to $L_{\bar\beta}$, and the displayed condition is absolute for $L_{\bar\beta}$. Hence $(d_\gamma, c_\gamma) = (X \cap \gamma, C \cap \gamma)$ as in the usual argument, and $\gamma \in C$ gives a contradiction.

A witness for $\Phi_D(\rho, \delta, d)$ consists of the recursion through δ, the cofinality witnesses for A_δ, and the $<_L$-initial segments and the witnesses for leastness at each stage. The displayed condition and its failure for each predecessor are bounded. The recursion determines $d = d_\delta$ uniquely, also in cardinal-preserving extensions: the cofinality partition below ω_n and the constructible initial segments are absolute. Each component has hereditary size at most ρ in L, and there are at most ρ stages, so the witness belongs to L_{ω_n}. □

For $t \subseteq \omega$, put

$$\mathrm{Tr}(d, t) \iff \omega \in d \ \wedge\ d \cap \omega = t,$$

$$S_t = \{\delta \in A : \mathrm{Tr}(d_\delta, t)\}, \qquad S_* = \{\delta \in A : d_\delta = \varnothing\}.$$

The set S_t is stationary, because the diamond sequence guesses the set $t \cup \{\omega\}$ on a stationary set; similarly S_* is stationary, because the sequence guesses the empty set on a stationary set. Distinct sets S_t are disjoint, and all of them are disjoint from S_*. Being a subset of ω, the real t is fixed by every transitive collapse. When n varies we write S_t^n and S_*^n for these sets.

3.3 A Σ_1 formula for clubs disjoint from S_t

For ordinal parameters ρ, ν, let $\psi_{\mathrm{club}}(\rho, \nu, C, t)$ assert the existence of a set K and functions U, F satisfying the following conditions:

1. $C \subseteq \nu$ is club in ν, and $K \subseteq \nu$.
2. $\operatorname{dom} U = \nu$. For $\delta \in K$, $U(\delta)$ is a strictly increasing cofinal map from ρ into δ. For $\delta \in \nu \setminus K$, either $\delta = 0$ or $U(\delta)$ is a cofinal map from some $\gamma < \rho$ into δ.
3. $\operatorname{dom} F = K$. Writing $F(\delta) = (a_\delta, w_\delta)$, we require
$$\forall \delta \in K \quad q_D(\rho, \delta, a_\delta, w_\delta).$$
4. $\forall \delta \in C \cap K \ \neg \mathrm{Tr}(a_\delta, t)$.

The formula following $\exists K\, \exists U\, \exists F$ is bounded, so ψ_{club} is Σ_1. For $\rho = \omega_{n-1}$ and $\nu = \omega_n$, the witnesses for the cofinalities give $K = A$, and the functionality of Φ_D gives $a_\delta = d_\delta$ for every $\delta \in K$. Consequently,

$$\psi_{\mathrm{club}}(\omega_{n-1}, \omega_n, C, t) \iff C \text{ is club in } \omega_n \text{ and } C \cap S_t = \varnothing.$$

3.4 The constructible almost disjoint family

Enumerate $\mathcal{P}(\omega_{n-1})^L$ in the order $<_L$ as $\langle U_\eta : \eta < \omega_n \rangle$, identifying subsets with their characteristic functions. Let $h(s)$ be the $<_L$-index of s among the constructible binary strings of length $< \omega_{n-1}$; these strings are ordered by $<_L$ in order type ω_{n-1}. Put

$$B_\eta = \{ h(U_\eta \restriction \zeta) : \zeta < \omega_{n-1} \}.$$

Since $<_L$ is a good Σ_1 wellorder and L_{ω_n} satisfies Collection, the function $\eta \mapsto B_\eta$ has a functional Σ_1 definition $\Phi_B(\omega_{n-1}, \eta, B)$ over L_{ω_n}. Each B_η is unbounded in ω_{n-1}, and for $\eta \neq \eta'$ the intersection $B_\eta \cap B_{\eta'}$ is bounded. We always use this fixed family for almost disjoint coding at ω_{n-1}. When n varies we write B^{n-1}_η for B_η; thus $\langle B^r_\eta : \eta < \omega_{r+1} \rangle$ is an almost disjoint family of subsets of ω_r.

3.5 Coding clubs

Let $n \geq 2$. A club in ω_n disjoint from S_t will be recognized by its initial segments in transitive models of the following kind.

Definition 3.2. *Let* ZFC^- *denote ZFC without the Power Set axiom, with Collection and Choice. A transitive model* $N \models \mathsf{ZFC}^-$ *is* suitable for n *if* $\omega^N_{n-1} = \omega_{n-1}$,

$$N \models \text{“}\omega_{n+1} \text{ exists and } \omega_{n+1} = \omega^L_{n+1}\text{”},$$

and the following hold in N, *with* $\beta = \omega^N_n$.

1. *over the set L_β, Φ_D is total and functional at every $\delta \in \beta \cap \mathrm{cof}(\omega_{n-1})$;*

2. *over L_β, Φ_B is total and functional at every $\eta < \beta$.*

Since N is transitive, its L_β is the true L_β. As Φ_D and Φ_B are Σ_1 and define functions in L, upward absoluteness shows that a model N which is suitable for n computes d_δ and B_η correctly for all $\delta \in \beta \cap \mathrm{cof}(\omega_{n-1})$ and all $\eta < \beta$. In every cardinal-preserving extension of L, the transitive collapse of an elementary submodel of a sufficiently large H_Θ which has size ω_{n-1} and contains ω_{n-1} as a subset is suitable for n, by elementarity.

Put

$$\mathrm{Even}(P) = \{\alpha : 2 \cdot \alpha \in P\}, \qquad \mathrm{Odd}(P) = \{\alpha : 2 \cdot \alpha + 1 \in P\}.$$

Let $C \subseteq \omega_n \setminus S_t$ be club. In $L[C]$ choose canonically a club $E \subseteq \mathrm{Lim}(C)$ with

$$\min E > \omega_{n-1}, \qquad d(c) := \min(E \setminus (c+1)) \geq c + \omega_{n-1} \quad (c \in E \cup \{0\}).$$

For $\omega_{n-1} \leq d < \omega_n$ let $w_d \subseteq \omega_{n-1}$ be the $<_L$-least code of a wellorder of ω_{n-1} of type d, using the ordinal pairing obtained by ordering pairs first by their maximum and then lexicographically. Define the set $P = P_t^n(C) \subseteq \omega_n$ by

$$\begin{aligned} \mathrm{Even}(P) &= C, \\ \mathrm{Odd}(P) &= \bigcup_{c \in E \cup \{0\}} \{c + \xi : \xi \in w_{d(c)}\}. \end{aligned}$$

This defines the map $C \mapsto P_t^n(C)$ uniformly in $L[C]$. Thus $P_t^n(C)$ codes C on its even part, and its odd part codes the successor function of the club E.

Lemma 3.3. *Let $C \subseteq \omega_n \setminus S_t$ be club and put $P = P_t^n(C)$. If N is suitable for n, $\beta = \omega_n^N < \omega_n$, and $t, P \cap \beta \in N$, then*

$$N \models \psi_{\mathrm{club}}(\omega_{n-1}, \beta, C \cap \beta, t).$$

Proof. We first show that $\beta \in E$. Otherwise there are successive elements $c < d$ of $E \cup \{0\}$ with $c < \beta < d$. Since β is a cardinal of N above ω_{n-1}, we have

$$c + \omega_{n-1} < \beta, \qquad w_d = \{\xi < \omega_{n-1} : c + \xi \in \mathrm{Odd}(P \cap \beta)\} \in N.$$

The ordinal pairing is absolute to N, which computes the order type coded by w_d correctly. In N this order type has cardinality ω_{n-1}, giving

$$d < \omega_n^N = \beta < d.$$

Thus $\beta \in E \subseteq \mathrm{Lim}(C)$, so $C \cap \beta = \mathrm{Even}(P \cap \beta)$ is club in β also in N. Put $\rho = \omega_{n-1}$. In N, take $K = \{\delta < \beta : \mathrm{cf}(\delta) = \rho\}$ and choose the

cofinality witnesses U. For $\delta \in K$, a strictly increasing cofinal map from the true ρ into δ witnesses $\mathrm{cf}(\delta) = \rho$ in the universe and hence in L, since $\delta < \omega_n$. Suitability gives the correct d_δ and a witness to $\Phi_D(\rho, \delta, d_\delta)$ over L_β. By Collection we obtain F in N. Since $C \cap S_t = \varnothing$, the resulting K, U, F witness the assertion in N. □

We shall apply Lemma 3.3 to models N which contain a set $Z \subseteq \omega_{n-1}$ coding P by almost disjoint coding. For such a model N, the correctness of Φ_B in N and the fact that N has the true ω_{n-1} give

$$\{\eta < \beta : Z \cap B_\eta \text{ is bounded in } \omega_{n-1}\} = P \cap \beta \in N,$$

so the lemma applies. For $t \in I$ we have $t \in L_\beta \subseteq N$. For the converse direction we use the following observation.

Lemma 3.4. *Let $C \subseteq \omega_n$ and suppose that the set*

$$D = \{\beta < \omega_n : C \cap \beta \text{ is a club in } \beta \text{ disjoint from } S_t\}$$

is unbounded in ω_n. Then C is a club in ω_n disjoint from S_t.

Proof. Since D is unbounded and $C \cap \beta$ is unbounded in β for every $\beta \in D$, the set C is unbounded in ω_n. Every element of C belongs to $C \cap \beta$ for some $\beta \in D$, so $C \cap S_t = \varnothing$. Finally, if $0 < \delta < \omega_n$ and $\sup(C \cap \delta) = \delta$, choose $\beta \in D$ with $\beta > \delta$. Then δ is a limit point of $C \cap \beta$ below β, so $\delta \in C \cap \beta \subseteq C$. □

In the application, $C = \mathrm{Even}(P)$ for the set P coded by some $Z \subseteq \omega_{n-1}$, and the elements of D will be the ordinals ω_n^N of transitive collapses N of elementary submodels of a sufficiently large H_Θ which have size ω_{n-1} and contain Z as an element and ω_{n-1} as a subset. These ordinals are cofinal in ω_n, the model N contains the true L_β for $\beta = \omega_n^N$, and the collapsing map fixes Z and every B_η with $\eta < \beta$.

3.6 The product of the club shooting forcings

Recall that

$$I = \{\ell(e, x, y) : e < \omega,\ x, y \in \mathbb{R}^L\} \subseteq 2^\omega$$

is the set of codes of triples consisting of a natural number and two constructible reals.

For a sequence $\vec{T} = \langle T_n : n \geq 1\rangle \in L$ with $T_n \subseteq I$, let $C_n(T_n)$ be the product of the forcings $\mathbb{C}(\omega_n \setminus S_t^n)$, $t \in T_n$, with countable support for $n = 1$ and full support for $n \geq 2$. Let

$$C_{\vec{T}} = \prod_{n \geq 1} C_n(T_n) \quad \text{(full support)}, \qquad B_{\vec{T}} = L[G_{C_{\vec{T}}}].$$

All these products are formed in L; in particular the factors $C_n(T_n)$ are not redefined in intermediate extensions. For $\vec{R} \leq \vec{T}$, $C_{\vec{R}}$ is a complete subforcing of $C_{\vec{T}}$ via the natural projection of the product.

Lemma 3.5. *For every $n \geq 1$ and $T \subseteq I$, the forcing $C_n(T)$ has cardinality at most ω_n, is $< \omega_n$-distributive, and preserves the stationarity of S^n_* and of every S^n_t with $t \notin T$. For $n \geq 2$ it is also $< \omega_{n-1}$-closed.*

Proof. Put $\rho = \omega_{n-1}$. The size bound follows from GCH and the support bound ρ. For $n \geq 2$, a descending sequence of length $< \rho$ has a lower bound by taking unions and adding their suprema at each coordinate, since $S^n_t \subseteq \mathrm{cof}(\rho)$.

Choose $M \prec H_\Theta$ containing the parameters and the initial condition, with $\rho \subseteq M$, $|M| = \rho$, $M^{<\rho} \subseteq M$, and $\delta = M \cap \omega_n$ in the stationary set to be preserved. For distributivity use S^n_*. Such models are obtained at limits of cofinality ρ of a continuous elementary chain whose successor models are closed under $< \rho$-sequences. Fix a cofinal sequence $\langle \alpha_\eta : \eta < \rho \rangle$ in δ. Construct a descending sequence in M. At step η meet the next open dense set in M, or force a point of the named club in (α_η, δ) for the stationarity argument. Then end-extend the condition, at every coordinate of its support, to a common maximum $\gamma_\eta \in M \cap S^n_*$ above α_η and above all current maxima. At limits below ρ take lower bounds in M.

At the end, take the union at each coordinate in the union of the supports and add δ. This is a lower bound because δ is outside every S^n_t for $t \in T$, and the maxima at each such coordinate are cofinal in δ. The union of the supports is the support of a condition: it is countable if $n = 1$, and the product has full support if $n \geq 2$. The lower bound meets all the given dense sets, or forces δ into the named club, respectively. □

For $n \geq 1$ write

$$Q_{\geq n} = \prod_{m \geq n} C_m(T_m), \qquad Q_{>n} = \prod_{m > n} C_m(T_m), \qquad Q_{<n} = \prod_{m < n} C_m(T_m).$$

Lemma 3.6. *For every $n \geq 1$, the forcing $Q_{\geq n}$ is $< \omega_n$-distributive. The forcing $C_{\vec{T}}$ preserves all cardinals and the GCH and adds no countable sequences of ordinals. The set S^n_* and every S^n_t with $t \notin T_n$ remain stationary. $C_{\vec{T}}$ is cone homogeneous.*

Proof. Fix n. By Lemma 3.5, each factor $C_m(T_m)$ with $m > n$ is $< \omega_{m-1}$-closed, so $Q_{>n}$ is $< \omega_n$-closed. Hence $Q_{>n}$ adds no elements of H_{ω_n} and preserves the stationarity of S^n_* and of all sets S^n_t. In particular $C_n(T_n)$ has the same conditions in the extension by $Q_{>n}$, and the proof of Lemma 3.5 applies in this extension. Thus

$$Q_{\geq n} \text{ is } < \omega_n\text{-distributive}, \qquad |Q_{<n}| < \omega_n,$$

and S^n_* and every S^n_t with $t \notin T_n$ are stationary in the extension by $Q_{\geq n}$. Since $Q_{<n}$ has size $< \omega_n$, it preserves the stationarity of these sets.

The case $n = 1$ shows that $C_{\vec{T}} = Q_{\geq 1}$ adds no countable sequences of ordinals. For the preservation of cardinals we apply Lemma 2.6 with $Q_n = Q_{\geq n}$ and $R_n = Q_{<n}$; note that $Q_{<n}$ is ω_n-cc because of its size, and that, in L,

$$|C_{\vec{T}}| \leq \aleph_\omega^\omega \leq 2^{\aleph_\omega} = \aleph_{\omega+1},$$

so $C_{\vec{T}}$ is $\aleph_{\omega+2}$-cc.

Next we verify the GCH in the extension. For $i \geq 1$, the forcing $Q_{>i} = Q_{\geq i+1}$ adds no subsets of ω_i, and $|Q_{\leq i}| \leq \omega_i$, where $Q_{\leq i} = Q_{<i+1}$. Counting nice $Q_{\leq i}$-names for subsets of ω_i in L therefore gives

$$2^{\omega_i} \leq \left((2^{\omega_i})^{\omega_i}\right)^L = \omega_{i+1}.$$

CH holds because no reals are added. In the extension fix bijections $e_i : \omega_{i+1} \to \mathcal{P}(\omega_i)$ $(i \geq 1)$. The map

$$X \longmapsto \langle e_i^{-1}(X \cap \omega_i) : i \geq 1\rangle \qquad (X \subseteq \aleph_\omega)$$

is injective and takes values in $(\prod_{i\geq 1} \omega_{i+1})^L$, since no countable sequences of ordinals are added. Thus $2^{\aleph_\omega} \leq (\aleph_\omega^\omega)^L \leq \aleph_{\omega+1}$, hence $2^{\aleph_\omega} = \aleph_{\omega+1}$. Finally, for every cardinal $\tau \geq \aleph_{\omega+1}$, nice names coded by subsets of $\tau \times C_{\vec{T}}$ give $2^\tau \leq (2^{\tau\cdot\aleph_{\omega+1}})^L = (\tau^+)^L$.

For cone homogeneity, let $p, q \in C_{\vec{T}}$. Extend them to $p' \leq p$ and $q' \leq q$ which have the same support and such that, for every n, there is an ordinal $\gamma_n \in S^n_*$ which is the maximum of $p'(n,t)$ and of $q'(n,t)$ for every t in the support at n. This is possible because S^n_* is unbounded in ω_n and disjoint from every S^n_t. The map which, for a condition below p', replaces the initial segment $p'(n,t)$ by $q'(n,t)$ at every coordinate (n,t) of the support of p' is an isomorphism between the cones below p' and q'. □

For later reference we summarize the properties of the stationary sets.

Proposition 3.7. *In L, for every $n \geq 1$ there are, uniformly definably in n:*

(S1) pairwise disjoint stationary sets $S^n_t \subseteq \omega_n \cap \operatorname{cof}(\omega_{n-1})$, $t \in I$, and a stationary set $S^n_* \subseteq \omega_n \cap \operatorname{cof}(\omega_{n-1})$ disjoint from all of them, defined from the canonical constructible diamond sequence on $\omega_n \cap \operatorname{cof}(\omega_{n-1})$;

(S2) for $n = 1$, the function $(t, \alpha) \mapsto S^1_t \cap \alpha$ $(\alpha < \omega_1)$ is first-order definable over HC *in every outer model with the same reals, by the definition in Lemma 3.1, whose witnesses belong to* HC*;*

(S3) *for $n \geq 2$, a definable map $C \mapsto P^n_t(C) \subseteq \omega_n$, computed in $L[C]$ from a club C disjoint from S^n_t, such that* $\mathrm{Even}(P^n_t(C)) = C$ *and such that the following holds: if N is suitable for n, $\beta = \omega_n^N < \omega_n$, and $P^n_t(C) \cap \beta \in N$, then*

$$N \models \psi_{\mathrm{club}}(\omega_{n-1}, \beta, \mathrm{Even}(P^n_t(C) \cap \beta), t);$$

(S4) *in the extension by the forcing $C_{\vec{T}}$ of Subsection 3.6, every S^n_t with $t \notin T_n$ remains stationary.*

These properties follow from the definitions above, Lemma 3.3, and Lemmas 3.5 and 3.6.

3.7 The case $n = 2$

We illustrate the main idea of our construction by sketching how to obtain Π^2_2 uniformization. The main difficulty is the following. A coding forcing enlarges the universe, which fits well with Σ statements, whereas our goal is Π^2_2 uniformization, and Π statements behave well when the universe shrinks. This creates a tension with iterated forcing; it is the abstract reason why combining forcing arguments with Π uniformization is technically challenging and interesting.

Recall that $\vec{T} = \langle T_n : n \geq 1 \rangle$ determines the forcing $C_{\vec{T}}$: for $t \in T_n$, the factor $\mathbb{C}(\omega_n \setminus S^n_t)$ of $C_n(T_n)$ adds a club disjoint from S^n_t. In the recursion for $n = 2$ we vary only T_2 and keep every T_m with $m \neq 2$ fixed.

We begin with $T_2 = I$, so that the first model is the largest one. We then define the uniformizing function by successively removing factors $\mathbb{C}(\omega_2 \setminus S^2_t)$ from $C_2(T_2)$, that is, by shrinking T_2; in this way we pass to smaller and smaller models. Values which were chosen at earlier stages remain in their sections throughout this process.

Put $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$, where $D_{\vec{T}}$ is the coding forcing constructed below. This forcing depends on the entire sequence $\vec{T}$. Proposition 7.8 gives a single Σ^2_2 formula Code_2, independent of $\vec{T}$, such that for every such sequence

$$F_{\vec{T}} \Vdash (\mathrm{Code}_2(t) \longleftrightarrow t \in T_2) \qquad (t \in I).$$

The aim is to arrange that every nonempty section contains exactly one y with $\ell(e, x, y) \notin T_2$; this y will be the value of the uniformizing function.

As long as there is a pair (e, x) to which no value has been assigned and whose section is forced to be nonempty by the current forcing $F_{\vec{T}}$, choose the $<_L$-least such pair and the $<_L$-least y with $F_{\vec{T}} \Vdash A_{2,e}(x, y)$, assign y to (e, x), and replace

$$\vec{T} \longmapsto \langle T_1,\ T_2 \setminus \{\ell(e, x, y)\},\ T_3, \ldots \rangle.$$

The forcing adds no reals, so every candidate y belongs to L. The reduction of complexity below and cone homogeneity allow all these choices to be made in L.

The chosen value remains in its section because the relevant statements are downward absolute. Indeed, suppose that $R_2 \subseteq T_2$ and $R_m = T_m$ for $m \neq 2$, and that the generic filter for $C_{\vec{R}}$ is the restriction of the generic filter for $C_{\vec{T}}$. Then

$$B_{\vec{R}} \subseteq B_{\vec{T}}, \qquad H_{\omega_2}^{B_{\vec{R}}} = H_{\omega_2}^{B_{\vec{T}}}.$$

Proposition 6.7 expresses Π^2_2 truth in the full extension as Π^2_1 truth over H_{ω_2} in $B_{\vec{T}}$. Passing from $B_{\vec{T}}$ to $B_{\vec{R}}$ leaves H_{ω_2} unchanged and removes some of its subsets. A $\Pi^2_1(H_{\omega_2})$ statement therefore remains true. Consequently,

$$F_{\vec{T}} \Vdash A_{2,e}(x,y) \quad \Longrightarrow \quad F_{\vec{R}} \Vdash A_{2,e}(x,y),$$

as proved in Proposition 8.2. Every value which was chosen earlier is therefore preserved when further codes are removed.

A section which was empty can become nonempty when we pass to a smaller model. We therefore repeat the choice until every nonempty section has been assigned a value, taking intersections of the sets T_2 at limit stages. There are only ω_1 pairs (e,x), and each step assigns a value to a new one, so the recursion stops before ω_2. For the resulting sequence $\vec{T}$,

$$U_{2,e}(x,y) \quad \Longleftrightarrow \quad A_{2,e}(x,y) \wedge \neg\, \mathrm{Code}_2(\ell(e,x,y))$$

is a Π^2_2 uniformization of $A_{2,e}$: each nonempty section contains its chosen value y, and this is the only element of the section with $\ell(e,x,y) \notin T_2$. Subsequent changes of the sets T_m for $m > 2$ preserve this uniformization, by Proposition 8.3 and the equivalence between $\mathrm{Code}_2(t)$ and $t \in T_2$. Section 9 carries out this recursion for every finite n.

4 The coding forcing and its quotients

Fix $\vec{T}$ and recall that

$$B_{\vec{T}} = L[G_{C_{\vec{T}}}],$$

where $G_{C_{\vec{T}}}$ is $C_{\vec{T}}$-generic over L and $C_{\vec{T}} = \prod_{n\geq 1} C_n(T_n)$ is the product of club shooting forcings defined in Subsection 3.6. Write $B = B_{\vec{T}}$. In this section the ground model is B. By Lemma 3.6, B has the same cardinals as L and satisfies the GCH.

The forcing $D = D_{\vec{T}}$ is an iteration of length $\aleph_{\omega+2}$ whose coordinates are assigned levels $i \geq 1$. The iterand at a coordinate of level i is a $< \omega_i$-compact, ω_i-linked forcing on ω_{i+1}, and it is defined in the extension by only a part of the preceding coordinates. There are two kinds of iterands. First, at level i we force with the iterations of such forcings which are named

over suitable subforcings (the Γ_i-iterations defined below), and with their restrictions below conditions. Secondly, we force with term forcings for almost disjoint coding; these are used to code sets which belong to extensions by coordinates of lower levels. A bookkeeping function enumerates the names for all these forcings. We first describe the general form of such an iteration, in which each coordinate depends only on some of the earlier coordinates, together with its conditions and order. We then define D by recursion along the bookkeeping function, and finally prove its basic properties.

4.1 Γ_i-iterations

Definition 4.1. *Fix $i \geq 1$. A* Γ_i-iteration *is a $< \omega_i$-support iteration*

$$\langle P_\alpha, \dot{Q}_\alpha : \alpha < \delta \rangle, \qquad |\delta| \leq \omega_{i+1},$$

such that for every $\alpha < \delta$ there is a P_α-name $\dot{\leq}_\alpha$ for a binary relation on ω_{i+1} with

$$P_\alpha \Vdash \text{“}\dot{Q}_\alpha = (\check{\omega}_{i+1}, \dot{\leq}_\alpha) \text{ is a } < \omega_i\text{-compact and } \omega_i\text{-linked preorder.”}$$

We take 0 as the largest condition of each iterand.

Let Γ_i consist of the last forcings P_δ of these iterations, given with their iteration codes. The codes are described in Subsection 5.4.

By Corollary 2.5, in every model satisfying $\omega_i^{<\omega_i} = \omega_i$, every Γ_i-iteration is $< \omega_i$-closed and satisfies the ω_{i+1}-chain condition. If $q \in P_\delta$, the cone $P_\delta \restriction q$ is forcing equivalent to a Γ_i-iteration. At stage α, restrict the iterand below $q(\alpha)$ and re-index it with largest condition 0. By induction this gives the cone below $q \restriction \alpha$ at each stage. At limits, the union of the support of a condition with that of q still has size $< \omega_i$.

4.2 Iterations with partial memory

The forcing D is an iteration in which the iterand at a coordinate ξ is defined from the generic filter on a set $m(\xi) \subseteq \xi$ of earlier coordinates only, the *memory* of ξ. Such iterations are often called iterations with partial memory. Our use of memory is inspired by the admissible support pairs in [7, Subsection 2.6]. In addition we assign levels to the coordinates, which determine the sizes of the supports and the degree of closure. We first describe the general form of these iterations and their conditions. The bookkeeping which chooses the iterands follows in Subsection 4.3.

Definition 4.2. *Let γ be an ordinal. An* iteration with partial memory of length γ *assigns to every $\xi < \gamma$*

- *a* level $i(\xi) \geq 1$,

- *a* memory $m(\xi) \subseteq \xi$*, and*
- *an* iterand*: a* $D_{m(\xi)}$*-name* $\dot{Q}_\xi$ *for a* $< \omega_{i(\xi)}$*-compact,* $\omega_{i(\xi)}$*-linked preorder* $(\omega_{i(\xi)+1}, \dot{\leq}_\xi)$ *with largest element* 0*, together with a name for a linking function.*

Writing $I_k = \{\xi < \gamma : i(\xi) = k\}$*, we require for every* $\xi < \gamma$*:*

(a) $\eta \in m(\xi)$ *implies* $m(\eta) \subseteq m(\xi)$*;*

(b) $m(\xi) \subseteq \bigcup_{k \geq i(\xi)} I_k$*;*

(c) $|m(\xi) \cap I_k| \leq \omega_{k+1}$ *for every* k*.*

A set $M \subseteq \gamma$ *is* closed *if* $m(\xi) \subseteq M$ *for every* $\xi \in M$*.*

Thus the memory of a coordinate contains only coordinates of its own level or of higher levels, and only few of them at each level. Every initial segment of γ is closed, since $m(\xi) \subseteq \xi$. By (a) every memory is closed, and by (b) so is $\bigcup_{k \geq j} I_k$ for every j. The forcings D_M, for closed M, are given by the next definition; we also write $D \restriction M$ for D_M. Since $m(\xi) \subseteq \xi$, the name $\dot{Q}_\xi$ refers only to a forcing which has already been defined. As for ordinary iterations, Definitions 4.2 and 4.3 form a single simultaneous recursion on γ.

In the next definition, the value $p(\xi)$ of a condition is taken from a set $\mathcal{N}_\xi$ of names which is small enough for the size and chain condition estimates below, and which is closed under canonical lower bounds, so that the forcing is sufficiently closed.

Definition 4.3. *Fix an iteration with partial memory of length* γ*, and let* $\xi < \gamma$ *have level* i*. For* $\eta < \omega_i$ *and a sequence* $\vec{\tau} = \langle \tau_\zeta : \zeta < \eta \rangle$ *of* $D_{m(\xi)}$*-names forced by* $1_{D_{m(\xi)}}$ *to be ordinals below* ω_{i+1}*, let* $b_{\xi,\eta}(\vec{\tau})$ *be a fixed* $D_{m(\xi)}$*-name for*

$$\min\{b < \omega_{i+1} : \forall \zeta < \eta \ (b \dot{\leq}_\xi \tau_\zeta)\}$$

if $\vec{\tau}$ *is* $\dot{\leq}_\xi$*-descending, and for* 0 *otherwise. (Whether* $\vec{\tau}$ *is descending is decided in the* $D_{m(\xi)}$*-generic extension, after the names* τ_ζ *and* $\dot{\leq}_\xi$ *have been evaluated.) Let* $\mathcal{N}_\xi$ *be the least set of* $D_{m(\xi)}$*-names which contains* $\check{a}$ *for every* $a < \omega_{i+1}$ *and is closed under the operations* $b_{\xi,\eta}$*,* $\eta < \omega_i$*.*

For closed $M \subseteq \gamma$*, the conditions of* D_M *are the functions* p *with*

$$\operatorname{supp}(p) \subseteq M, \qquad p(\xi) \in \mathcal{N}_\xi \quad (\xi \in \operatorname{supp}(p)),$$
$$|\operatorname{supp}(p) \cap I_k| < \omega_k \quad (k \geq 1).$$

Setting $p(\xi) = \check{0}$ *for* $\xi \notin \operatorname{supp}(p)$*, put*

$$q \leq p \iff \forall \xi \in M \quad q \restriction m(\xi) \Vdash_{D_{m(\xi)}} q(\xi) \dot{\leq}_\xi p(\xi).$$

Remark 4.4. The minimum in the definition of $b_{\xi,\eta}$ exists because a descending sequence is centered and $\dot{Q}_\xi$ is $< \omega_i$-compact. Since ω_i is regular, the closure defining $\mathcal{N}_\xi$ is reached after ω_i steps. Equivalently, $\mathcal{N}_\xi$ consists of the names given by well-founded terms of size $< \omega_i$ built from the constants $\check{a}$ and the operations $b_{\xi,\eta}$. Two features of the definition are used throughout.

1. Closure under the operations $b_{\xi,\eta}$, together with the bound on the supports at each level, gives coordinatewise lower bounds of short descending sequences in the ground model (Lemma 4.6).

2. Under GCH, $|\mathcal{N}_\xi| \leq \omega_{i+1}^{<\omega_i} = \omega_{i+1}$, which gives the bound on the size of the forcing and the chain condition (Lemma 4.12).

All values are ordinals below ω_{i+1}, so $1_{D_{m(\xi)}} \Vdash \tau \in \dot{Q}_\xi$ for every $\tau \in \mathcal{N}_\xi$. For closed M, the order of D_M refers only to the forcings $D_{m(\xi)}$ with $\xi \in M$. Hence D_M is the suborder of D_γ consisting of the conditions whose support is contained in M.

Formally, D_M is a preorder; we pass to its separative quotient without further mention.

4.3 The bookkeeping

The forcing D is $D_{\aleph_{\omega+2}}$ for an iteration with partial memory of length $\aleph_{\omega+2}$ which is defined by recursion below. Its coordinates come in blocks. Each block consists either of the iterands of a Γ_i-iteration, for some i, or of a single term forcing.

We code conditions and names by sets of ordinals. A forcing of size at most ω_{i+1} is identified with a preorder on ω_{i+1} whose largest condition is 0; several ordinals may represent the same condition. A nice name is then coded by the ordinals representing the conditions and the values which occur in it. The details of this coding are given in Subsection 5.4.

Let $\mathcal{B}$ be the set of all sets of ordinals below $\aleph_{\omega+2}$ of size at most $\aleph_{\omega+1}$. By GCH,

$$|\mathcal{B}| \leq \aleph_{\omega+2}^{\aleph_{\omega+1}} = \aleph_{\omega+2}.$$

Fix a function $F : \aleph_{\omega+2} \to \mathcal{B}$ such that

$$\sup F^{-1}\{b\} = \aleph_{\omega+2} \qquad (b \in \mathcal{B}).$$

The elements of $\mathcal{B}$ are regarded as candidates for codes of the quadruples $(i, M, p, \dot{X})$ considered in Definition 4.5; whether $F(\beta)$ actually codes such a quadruple is checked only at stage β of the recursion.

For $p \in D_M$ and a name $\dot{R}$ which p forces to be a forcing, let $\dot{R}^p$ be a name for the forcing which is equal to $\dot{R}$ if p belongs to the generic filter for D_M, and trivial otherwise. We use this notation for names over D_M, over D_N for closed sets $N \supseteq M$, and over two-step iterations $D_M * \dot{P}$. In the

last two cases p is identified with its canonical image in the larger forcing, and the generic filter for D_M is the one induced by the generic filter for the larger forcing.

Definition 4.5. *We define an iteration with partial memory of length $\aleph_{\omega+2}$ by recursion, appending one block of coordinates at each stage $\beta < \aleph_{\omega+2}$. Let γ_β be the length of the iteration defined before stage β; thus $\gamma_0 = 0$ and $\gamma_\lambda = \sup_{\beta<\lambda} \gamma_\beta$ for limit ordinals λ.*

At stage β, suppose that $F(\beta)$ codes a quadruple $(i, M, p, \dot{X})$ with the following properties.

(i) $i \geq 1$, and $M \subseteq \gamma_\beta$ is a closed set such that $M \subseteq \bigcup_{k\geq i} I_k$ and $|M \cap I_k| \leq \omega_{k+1}$ for all $k \geq i$.

(ii) $p \in D_M$, and $\dot{X}$ is a D_M-name.

(iii) One of the following two cases applies.

Case 1 (iterations). *p forces that $\dot{X}$ is a code for a Γ_i-iteration $\langle P_\alpha, \dot{R}_\alpha : \alpha < \delta\rangle$ of fixed length δ with $|\delta| \leq \omega_{i+1}$.*

Case 2 (term forcings). *p forces that $\dot{X} = (P, \dot{E})$, where P is ω_i-cc with a given dense subset of size at most ω_i, $P \Vdash \dot{E} \subseteq \omega_{i+1}$, and $\omega_i^{<\omega_i} = \omega_i$ and $\omega_{i+1}^{<\omega_i} = \omega_{i+1}$. In this case put $\delta = 1$ and let $\dot{R}_0$ be a name for the term forcing $\mathrm{Term}(P, \mathbb{A}_{\omega_i}(\dot{E}))$, defined in the extension by D_M and identified with a preorder on ω_{i+1}.*

If moreover $\delta > 0$, append the coordinates $\xi_\alpha = \gamma_\beta + \alpha$, for $\alpha < \delta$, with

$$i(\xi_\alpha) = i, \qquad m(\xi_\alpha) = M \cup \{\xi_\zeta : \zeta < \alpha\}, \qquad \dot{Q}_{\xi_\alpha} = \dot{R}^p_\alpha,$$

and put $\gamma_{\beta+1} = \gamma_\beta + \delta$. Here $\dot{R}_\alpha$ is a name over $D_M * \dot{P}^p_\alpha$, where P_α is the iteration of the iterands $\dot{R}_\zeta$, $\zeta < \alpha$, and P_0 is trivial. By induction on α, the map

$$r \longmapsto (r \restriction M,\ \langle r(\xi_\zeta) : \zeta < \alpha\rangle)$$

shows that $D_{m(\xi_\alpha)}$ is forcing equivalent to $D_M * \dot{P}^p_\alpha$; this is verified by the argument in the proof of Proposition 4.9, applied to this block over D_M. We use this equivalence to regard $\dot{R}^p_\alpha$ as a $D_{m(\xi_\alpha)}$-name.

In all other cases, append a single coordinate of level 1 with empty memory and trivial iterand, and put $\gamma_{\beta+1} = \gamma_\beta + 1$.

Finally, put $D = D_{\aleph_{\omega+2}}$.

In Case 2, the forcing P and the name $\dot{E}$ are given by their codes. Coordinates of levels below i may occur in these codes, but only as ordinals; they need not belong to M. In both cases p forces every $\dot{R}_\alpha$ to be a $< \omega_i$-compact, ω_i-linked preorder on ω_{i+1} with a linking function (Definition 4.1,

Theorem 2.10). Whether $F(\beta)$ codes a quadruple with the properties (i)–(iii) is a statement about the forcing relation of D_M, which has already been defined at stage β.

The requirements of Definition 4.2 hold for every block which is appended in this way.

(a) holds because M is closed and the memories within a block increase.

(b) holds because the block has level i and $M \subseteq \bigcup_{k \geq i} I_k$.

(c) holds because the block adds at most $|\delta| \leq \omega_{i+1}$ coordinates of level i.

In particular $|m(\xi)| \leq \aleph_\omega$ for every ξ. Every block has cardinality at most $\omega_{i+1} < \aleph_\omega$ for some i. Since $\aleph_{\omega+2}$ is regular, it follows that

$$\gamma_\beta < \aleph_{\omega+2} \quad (\beta < \aleph_{\omega+2}), \qquad \sup_{\beta < \aleph_{\omega+2}} \gamma_\beta = \aleph_{\omega+2}.$$

The support of a condition has cardinality at most $\aleph_\omega$ and may meet infinitely many levels. For $j \geq 1$, let $D_{\geq j}$ be the restriction of D to $\bigcup_{k \geq j} I_k$, and put $D_{>j} = D_{\geq j+1}$. We prove below that D_M is a complete subforcing of D for every closed set M. Given a D-generic filter G over B, we write G_M for its restriction to D_M, and

$$W_j = B[G_{D_{\geq j}}] \qquad (j \geq 1).$$

The construction can be carried out uniformly in $\vec{T}$, in L. Indeed, every element of $\mathcal{B}$ has a nice $C_{\vec{T}}$-name in L, which can be coded by a set of ordinals of size at most $\aleph_{\omega+1}$, because $|C_{\vec{T}}| \leq \aleph_{\omega+1}$. Since $\aleph_{\omega+2}^{\aleph_{\omega+1}} = \aleph_{\omega+2}$ in L, there are at most $\aleph_{\omega+2}$ such names. Take the $<_L$-least enumeration of these names of length $\aleph_{\omega+2}$ in which every name occurs unboundedly often, and let $F(\beta)$ be the value of the β-th name. At each stage, whether Case 1 or Case 2 applies is expressed by the forcing relation of the part of the forcing which has already been defined. The same definable recursion therefore produces a $C_{\vec{T}}$-name $\dot{D}_{\vec{T}}$ for every $\vec{T} \in L$. This uses only the usual names of set forcing and the wellorder $<_L$.

The results of the remainder of this section use only the properties of the iterands, the supports and the memories stated in Definitions 4.2 and 4.3. They therefore hold for every iteration with partial memory, in particular for the iteration constructed up to any stage of the recursion; they do not depend on which names the bookkeeping has handled.

4.4 Complete subforcings and closure

Lemma 4.6. *Consider an iteration with partial memory, and let $K \subseteq M$ be closed sets of coordinates.*

1. *The restriction map $p \mapsto p \restriction K$ is a projection from D_M onto D_K. More precisely, if $p \in D_M$ and $q \leq p \restriction K$ in D_K, then $p' = q \cup (p \restriction (M \setminus K))$ is a condition in D_M with $p' \leq p$ and $p' \restriction K = q$. In particular D_K is a complete subforcing of D_M.*

2. *Suppose that all coordinates in M have level at least j. Let $\langle p_\zeta : \zeta < \eta \rangle$ be a descending sequence in D_M, where $\eta < \omega_j$, and let $r \in D_K$ be a lower bound of $\langle p_\zeta \restriction K : \zeta < \eta \rangle$. Then $\langle p_\zeta : \zeta < \eta \rangle$ has a lower bound $q \in D_M$ with $q \restriction K = r$. In particular D_M is $< \omega_j$-closed.*

Proof. We argue by induction on the length of the iteration, simultaneously for all closed sets of coordinates.

(1) Let p' be as in the statement. For $\xi \in K$ we have $m(\xi) \subseteq K$, so $p' \restriction m(\xi) = q \restriction m(\xi)$ forces $q(\xi) \dot{\leq}_\xi p(\xi)$, because $q \leq p \restriction K$. For $\xi \in M \setminus K$ we have $p'(\xi) = p(\xi)$, and the required inequality holds because $\dot{\leq}_\xi$ is forced to be reflexive. Thus $p' \leq p$ and $p' \restriction K = q$. The restriction map preserves the order, again because $m(\xi) \subseteq K$ for $\xi \in K$. The order of D_M is transitive: if $r \leq q \leq p$, then $r \restriction m(\xi) \leq q \restriction m(\xi)$ for every $\xi \in M$, so $r \restriction m(\xi)$ forces both $r(\xi) \dot{\leq}_\xi q(\xi)$ and $q(\xi) \dot{\leq}_\xi p(\xi)$, and $\dot{\leq}_\xi$ is forced to be transitive.

(2) Define q by $q \restriction K = r$ and, for $\xi \in M \setminus K$ in the union of the supports of the conditions p_ζ,

$$q(\xi) = b_{\xi,\eta}(\langle p_\zeta(\xi) : \zeta < \eta \rangle).$$

Since $\eta < \omega_j$ and all coordinates in M have level at least j, the regularity of the cardinals ω_k shows that $|\operatorname{supp}(q) \cap I_k| < \omega_k$ for every k and that $q(\xi) \in \mathcal{N}_\xi$ for every ξ. Thus $q \in D_M$. We show by induction on $\xi \in M$ that $q \restriction m(\xi) \leq p_\zeta \restriction m(\xi)$ and $q \restriction m(\xi) \Vdash q(\xi) \dot{\leq}_\xi p_\zeta(\xi)$ for all $\zeta < \eta$. For $\xi \in K$ this holds because $m(\xi) \subseteq K$ and r is a lower bound of the restrictions to K. Let $\xi \in M \setminus K$. By the induction hypothesis, $q \restriction m(\xi)$ is below all $p_\zeta \restriction m(\xi)$. Hence it forces the sequence $\langle p_\zeta(\xi) : \zeta < \eta \rangle$ to be descending, so that $b_{\xi,\eta}$ yields a lower bound of this sequence. Therefore $q \leq p_\zeta$ for all $\zeta < \eta$, and $q \restriction K = r$. □

Lemma 4.7. *1. Let τ be regular, let R be $a < \tau$-closed forcing and let G_R be R-generic over V. Then every $X \subseteq G_R$ of size $< \tau$ in $V[G_R]$ has a lower bound in G_R.*

2. *Let $K \subseteq M$ be closed sets of coordinates of levels at least j. Then D_M/G_K is $< \omega_j$-closed in $B[G_K]$.*

Proof. (1) We have $X \in V$, since R adds no sequences of ordinals of length $< \tau$. Choose $r \in G_R$ with $r \Vdash \check{X} \subseteq \dot{G}_R$. For each $a \in X$, the set $\{q \leq r : q \leq a\}$ is dense below r. By $< \tau$-closure, the intersection of these sets is dense below r and hence meets G_R.

(2) Let $\langle p_\zeta : \zeta < \eta \rangle$ be a descending sequence in D_M/G_K, where $\eta < \omega_j$. Since D_K is $< \omega_j$-closed, the sequence belongs to B, and by (1) there is $r \in G_K$ with $r \le p_\zeta \restriction K$ for all $\zeta < \eta$. Lemma 4.6(2) gives a lower bound q of the sequence with $q \restriction K = r$. Then $q \in D_M/G_K$. □

4.5 The quotients at level i

Fix $i \ge 1$ and put $\kappa = \omega_i$, $\sigma = \omega_{i+1}$,

$$U_i = \bigcup_{k>i} I_k, \qquad R = D_{>i} = D_{U_i}.$$

By Lemma 4.6, R is $< \sigma$-closed. Let G_R be R-generic over B and work in $W = B[G_R]$; thus $W = W_{i+1}$. We call $D_{\ge i}/G_R$ the *quotient of D at level i*.

For an ordinal γ which is at most the length of the iteration, let $S_\gamma = S_{i,\gamma} \in B$ be the set of all conditions of D whose support is contained in $I_i \cap \gamma$. In W, define a preorder on S_γ by

$$q \preceq_{G_R} p \quad \iff \quad \exists r \in G_R \ (r \cup q \le_D p).$$

By Lemma 4.6(1), the preorder $(S_\gamma, \preceq_{G_R})$ is forcing equivalent to the quotient $D_{U_i \cup (I_i \cap \gamma)}/G_R$: a condition p of this quotient is equivalent to its restriction $p \restriction I_i \in S_\gamma$, since $p \restriction U_i \in G_R$.

More generally, for $C \subseteq I_i$ in W with $U_i \cup C$ closed, use the conditions supported on C with order $\preceq_{G_R}$. We denote this forcing by $D_{U_i \cup C}/G_R$ up to forcing equivalence. A restriction to C belongs to B, since it is coded by a sequence of length $< \kappa$ and R adds no such sequences. The amalgamation of Lemma 4.6(1) therefore gives a complete subforcing also for $C \notin B$.

The lower-bound construction of Lemma 4.6(2) also works for $\preceq_{G_R}$. For a short descending sequence and a lower bound of its restrictions to C, first take a common lower bound in G_R of the witnesses to the order comparisons, using Lemma 4.7(1). Then use the prescribed values on C and the coordinatewise lower bounds off C. All the data on the union of the level-i supports belong to B. Thus Lemma 4.7(2) applies over W to these restrictions.

Lemma 4.8. *Let $\xi \in I_i$ and put $A = U_i \cup (I_i \cap \xi)$. Then A is closed and contains $m(\xi)$, and the evaluation of $\dot{Q}_\xi$ by $G_{m(\xi)}$ is $< \kappa$-compact and κ-linked in $B[G_A]$, with the same linking function and the same lower bounds.*

Proof. The set A is closed by (b), and $m(\xi) \subseteq A$. By Lemma 4.7(2), $B[G_A]$ is an extension of $B[G_{m(\xi)}]$ by a $< \kappa$-closed forcing. Hence every family of fewer than κ conditions of the iterand, which is a family of ordinals below σ, already belongs to $B[G_{m(\xi)}]$. If such a family is centered in $B[G_A]$, it is centered in $B[G_{m(\xi)}]$, where it has a lower bound. Similarly, conditions with the same value of the linking function are compatible in $B[G_{m(\xi)}]$ and hence in $B[G_A]$. □

Proposition 4.9. *For every γ, the preorder $(S_\gamma, \preceq_{G_R})$ is forcing equivalent in W to the $< \kappa$-support iteration P_γ of the iterands $\dot{Q}_\xi$, $\xi \in I_i \cap \gamma$; more precisely, the coordinatewise interpretation of the elements of S_γ as conditions of P_γ preserves and reflects the order and has dense range. The forcing P_γ satisfies the stationary σ-chain condition.*

Proof. We argue by induction on γ, constructing P_γ and the interpretation of S_γ in P_γ simultaneously. Let $\xi \in I_i \cap \gamma$. The forcing $D_{m(\xi)}$ is a complete subforcing of $D_{U_i \cup (I_i \cap \xi)}$, and by the induction hypothesis the quotient of the latter by G_R is forcing equivalent to P_ξ. Hence $\dot{Q}_\xi$ can be regarded as a P_ξ-name in W. By Lemma 4.8, it is forced to be $< \kappa$-compact and κ-linked. A condition $p \in S_\gamma$ is interpreted as the condition of P_γ whose value at ξ is the name $p(\xi)$, regarded as a P_ξ-name.

The order. By induction, the interpretation preserves $\preceq_{G_R}$. Conversely, let $p, q \in S_\gamma$ with $p \leq_{P_\gamma} q$, and put $A = \operatorname{supp}(p) \cup \operatorname{supp}(q)$. Fix $\xi \in A$, write

$$K = m(\xi) \cap U_i, \qquad C_\xi = m(\xi) \cap I_i,$$

and let θ_ξ be the statement $p(\xi) \leq_{\dot{Q}_\xi} q(\xi)$ of the forcing language of $D_{m(\xi)}$. The set K is closed, and $G_K = G_R \cap D_K$ is D_K-generic over B. In B, let E be the set of all $s \in D_K$ for which one of the following holds:

(i) $s \cup (p \restriction C_\xi) \Vdash_{D_{m(\xi)}} \theta_\xi$;

(ii) there is $a \in D_{m(\xi)}$ with $a \leq s \cup (p \restriction C_\xi)$, $a \restriction K = s$ and $a \Vdash_{D_{m(\xi)}} \neg\theta_\xi$.

The set E is dense in D_K. Indeed, if $s_0 \in D_K$ does not satisfy (i), choose $a \leq s_0 \cup (p \restriction C_\xi)$ which forces $\neg\theta_\xi$; then $s = a \restriction K \leq s_0$ satisfies (ii). Choose $s \in G_K \cap E$.

We claim that s satisfies (i). Otherwise let a be as in (ii), and let $p^* \in S_\xi$ be the condition which agrees with a on C_ξ and with p on $(I_i \cap \xi) \setminus C_\xi$. Since $a \leq s \cup (p \restriction C_\xi)$ and $a \restriction K = s$, we have $s \cup p^* \leq_D p \restriction \xi$, and therefore $p^* \preceq_{G_R} p \restriction \xi$. By the induction hypothesis for ξ, the interpretation of p^* is below that of $p \restriction \xi$ in P_ξ, and every P_ξ-generic filter over W which contains it induces a $D_{m(\xi)}$-generic filter over B which contains $s \cup (p^* \restriction C_\xi) = a$. In the corresponding extension θ_ξ fails. On the other hand, $p \leq_{P_\gamma} q$ implies that $p \restriction \xi$ forces θ_ξ over P_ξ, a contradiction.

Hence for every $\xi \in A$ there is $s_\xi \in G_R$ such that

$$s_\xi \cup (p \restriction C_\xi) \Vdash_{D_{m(\xi)}} p(\xi) \leq_{\dot{Q}_\xi} q(\xi).$$

Since $|A| < \kappa$, Lemma 4.7(1) gives $r \in G_R$ below all s_ξ. Then $r \cup p \leq_D q$, so $p \preceq_{G_R} q$.

Density. We show that for every $p \in P_\gamma$ there is $q \in S_\gamma$ whose interpretation is below p. If $\gamma = \xi + 1$ and $\xi \in I_i$, first extend $p \restriction \xi$ to a condition which decides $p(\xi) = \check{c}$ for some $c < \sigma$. By the induction hypothesis there

is $q \in S_\xi$ below this condition, and then $q \cup \{(\xi, \check{c})\} \in S_\gamma$ is below p; recall that $1 \Vdash \check{c} \in \dot{Q}_\xi$. If $\gamma = \xi + 1$ and $\xi \notin I_i$, there is nothing to show. If γ is a limit ordinal with $\mathrm{cf}(\gamma) \geq \kappa$, then $\sup \mathrm{supp}(p) < \gamma$, and the induction hypothesis applies.

Now let γ be a limit ordinal with $\tau = \mathrm{cf}(\gamma) < \kappa$. Choose a continuous increasing sequence $\langle \gamma_\eta : \eta \leq \tau \rangle$ with $\gamma_0 = 0$ and $\gamma_\tau = \gamma$, and construct

$$q_\eta \in S_{\gamma_\eta}, \qquad q_\eta \leq_{P_{\gamma_\eta}} p \restriction \gamma_\eta \quad (\eta \leq \tau),$$
$$q_\zeta \preceq_{G_R} q_\eta \quad (\eta < \zeta \leq \tau).$$

Start with the trivial condition q_0. At successor steps apply the induction hypothesis to the condition $q_\eta \cup (p \restriction [\gamma_\eta, \gamma_{\eta+1}))$ of $P_{\gamma_{\eta+1}}$; the resulting condition $q_{\eta+1}$ is below q_η in $P_{\gamma_{\eta+1}}$ and hence $q_{\eta+1} \preceq_{G_R} q_\eta$ by the first part of the proof. At a limit $\nu \leq \tau$, use Lemma 4.7(1) to choose $r \in G_R$ such that $r \cup q_\zeta \leq_D q_\eta$ for all $\eta < \zeta < \nu$. Then $\langle r \cup q_\eta : \eta < \nu \rangle$ is a descending sequence in D, and it belongs to B because R adds no sequences of length $< \sigma$ of elements of B. Lemma 4.6(2) yields a condition s with

$$s \restriction U_i = r, \qquad s \leq_D r \cup q_\eta \quad (\eta < \nu).$$

Put $q_\nu = s \restriction I_i \in S_{\gamma_\nu}$. Then $q_\nu \preceq_{G_R} q_\eta$ for all $\eta < \nu$, and since $\gamma_\nu = \sup_{\eta<\nu} \gamma_\eta$ we obtain $q_\nu \leq_{P_{\gamma_\nu}} p \restriction \gamma_\nu$. In particular, $q_\tau \leq_{P_\gamma} p$.

The chain condition. Since R is $< \sigma$-closed, $\kappa^{<\kappa} = \kappa$ and $\sigma = \kappa^+$ hold in W. The iterands of P_γ are forced to be $< \kappa$-compact and κ-linked, so Corollary 2.5 applies to P_γ. □

In Definition 4.5, a Γ_i-iteration is appended as a block consisting of its iterands. The proposition therefore applies directly to these coordinates; it is not required that the whole Γ_i-iteration is compact or linked.

Definition 4.10. *Let $i \geq 1$ and work in $W = W_{i+1}$. Let $\mathcal{C}_i$ be the set of all $C \subseteq I_i$ such that $|C| \leq \omega_{i+1}$ and $U_i \cup C$ is closed. For $C \subseteq I_i$ such that $U_i \cup C$ is closed, put*

$$P_C = D_{U_i \cup C} / G_{D_{>i}},$$

let G_C be the P_C-generic filter induced by the D-generic filter, and let $N_C = W[G_C]$.

Remark 4.11. A set $C \in \mathcal{C}_i$ need not be closed and need not belong to B: the memories of its elements contain coordinates of higher levels, and C is only required to belong to W. As shown above, P_C is a complete subforcing of the quotient at level i. All statements below about the sizes of the forcings P_C and about the genericity of filters on them are made in W.

4.6 Size and preservation of cardinals

Lemma 4.12. *The forcing D of Definitions 4.5 and 4.3 satisfies*

$$|D| \le \aleph_{\omega+2}, \qquad D \text{ is } \aleph_{\omega+2}\text{-cc.}$$

Proof. Since B satisfies GCH,

$$\aleph_\omega^{\aleph_\omega} = 2^{\aleph_\omega} = \aleph_{\omega+1},$$
$$\aleph_{\omega+1}^{\aleph_\omega} = \aleph_{\omega+1}, \qquad \aleph_{\omega+2}^{\aleph_\omega} = \aleph_{\omega+2},$$
$$|\mathcal{N}_\xi| \le \omega_{i+1}^{<\omega_i} = \omega_{i+1} < \aleph_\omega \quad (\xi \in I_i).$$

Since $|\operatorname{supp}(p)| \le \aleph_\omega$ for every condition p, we obtain $|D| \le \aleph_{\omega+2}^{\aleph_\omega} \cdot \aleph_\omega^{\aleph_\omega} = \aleph_{\omega+2}$.

Given $\langle p_\alpha : \alpha < \aleph_{\omega+2}\rangle$ in D, put

$$A_\alpha = \operatorname{supp}(p_\alpha), \qquad U_\alpha = \bigcup_{\beta<\alpha} A_\beta,$$
$$S = \{\alpha < \aleph_{\omega+2} : \operatorname{cf}(\alpha) = \aleph_{\omega+1}\}.$$

For $\alpha \in S$, choose $f(\alpha) < \alpha$ with $A_\alpha \cap U_\alpha \subseteq U_{f(\alpha)}$; this is possible since $|A_\alpha \cap U_\alpha| \le \aleph_\omega < \operatorname{cf}(\alpha)$. Fodor's lemma gives a stationary $S' \subseteq S$ with $f[S'] = \{\gamma\}$. As $|U_\gamma| \le \aleph_{\omega+1}$ and $|[U_\gamma]^{\le\aleph_\omega}| \le \aleph_{\omega+1}^{\aleph_\omega} = \aleph_{\omega+1} < \aleph_{\omega+2}$, there are $H \in [S']^{\aleph_{\omega+2}}$ and a such that $A_\alpha \cap U_\alpha = a$ for all $\alpha \in H$. Hence

$$a \subseteq A_\alpha \cap A_\beta \subseteq A_\beta \cap U_\beta = a \quad (\alpha < \beta,\ \alpha, \beta \in H).$$

There are at most $\aleph_\omega^{|a|} \le \aleph_{\omega+1}$ restrictions $p_\alpha \restriction a$. Choose $K \in [H]^{\aleph_{\omega+2}}$ and u with $p_\alpha \restriction a = u$ for all $\alpha \in K$. For distinct $\alpha, \beta \in K$, the union $r = p_\alpha \cup p_\beta$ is a function and

$$|\operatorname{supp}(r) \cap I_i| \le |A_\alpha \cap I_i| + |A_\beta \cap I_i| < \omega_i \quad (i \ge 1).$$

Since $r(\xi) = p_\epsilon(\xi)$ whenever $\epsilon \in \{\alpha, \beta\}$ and $\xi \in A_\epsilon$, every inequality required for $r \le_D p_\epsilon$ is an instance of reflexivity and is forced by the trivial condition; hence $r \in D$ and $r \le_D p_\alpha, p_\beta$. □

Lemma 4.13. *The forcing D preserves all cardinals and adds no reals.*

Proof. By Lemma 4.6, $D = D_{\ge 1}$ is countably closed, so it adds no reals. For the preservation of cardinals we apply Lemma 2.6. Let $j \ge 1$. Then D is forcing equivalent to $D_{\ge j} * (D/\dot{G}_{D_{\ge j}})$, and $D_{\ge j}$ is $< \omega_j$-closed by Lemma 4.6. The quotient $D/G_{D_{\ge j}}$ is the iteration of the quotients at the levels $j-1, j-2, \dots, 1$. By Proposition 4.9, the quotient at level k is ω_{k+1}-cc and hence ω_j-cc for $k < j$. Each of them preserves ω_j, so their finite iteration is ω_j-cc. Finally D is $\aleph_{\omega+2}$-cc by Lemma 4.12. □

The next lemma shows that a Γ_i-iteration of the ground model remains a Γ_i-iteration, up to forcing equivalence, after a sufficiently closed forcing.

Lemma 4.14. *Let $\kappa > \omega$ be regular, $\sigma = \kappa^+$ and $\kappa^{<\kappa} = \kappa$. Suppose R is $< \kappa$-closed and preserves σ. Let $\langle Q_\alpha, \dot{S}_\alpha : \alpha < \delta \rangle \in V$ be a $< \kappa$-support iteration such that*

$$Q_\alpha \Vdash \text{“}\dot{S}_\alpha = (\check{\sigma}, \dot{\leq}_\alpha) \text{ is } < \kappa\text{-compact and } \kappa\text{-linked,”}$$

given together with names for linking functions and for lower bounds of centered families. Let G_R be R-generic over V, and let $\langle \widehat{Q}_\alpha, \dot{S}_\alpha : \alpha < \delta \rangle$ be the $< \kappa$-support iteration defined in $V[G_R]$ from the same names for the orders of the iterands. Then, in $V[G_R]$:

1. *for every $\alpha \leq \delta$, the coordinatewise interpretation $e_\alpha : Q_\alpha^V \to \widehat{Q}_\alpha$ preserves and reflects the order and has dense range; in particular Q_δ^V is forcing equivalent to $\widehat{Q}_\delta$;*
2. *the iterands of $\widehat{Q}_\delta$ are forced to be $< \kappa$-compact and κ-linked, with the same linking functions and lower bounds;*
3. *if $\kappa = \omega_i$, $|\delta| \leq \sigma$, and $D_\alpha \subseteq Q_\alpha$ are dense sets of size at most σ in V, then $\langle \widehat{Q}_\alpha, \dot{S}_\alpha : \alpha < \delta \rangle$ is a Γ_i-iteration with dense sets $e_\alpha[D_\alpha]$ of size at most σ. In particular Q_δ^V is forcing equivalent to a member of Γ_i.*

Proof. Write $W = V[G_R]$. Since R and every Q_α add no sequences of ordinals of length $< \kappa$,

$$R \Vdash Q_\alpha^V \text{ is } < \kappa\text{-closed}, \qquad Q_\alpha \Vdash R^V \text{ is } < \kappa\text{-closed}.$$

We construct the maps e_α by induction on α. For ground-model names, the forcing relation for statements about the orders of the iterands is absolute between V and W, so the maps e_α preserve and reflect the order. In the product $R^V \times Q_\alpha^V$, the closure of R^V over $V[G_{Q_\alpha}]$ gives

$$[\sigma]^{<\kappa} \cap V[G_{Q_\alpha}][G_R] = [\sigma]^{<\kappa} \cap V[G_{Q_\alpha}].$$

The order of the iterand and its linking function on σ are unchanged, and whether finitely many conditions have a common lower bound is absolute. Thus compactness and linkedness are preserved, which proves (2).

For the density of the range of $e_{\alpha+1}$, let $p \in \widehat{Q}_{\alpha+1}$. Choose $p' \leq p \restriction \alpha$ which decides $p(\alpha) = \check{c}$ for some $c < \sigma$, and $q \in Q_\alpha^V$ with $e_\alpha(q) \leq p'$. Then $e_{\alpha+1}(q^\frown\langle \check{c} \rangle) \leq p$. Let γ be a limit ordinal and $p \in \widehat{Q}_\gamma$. If $\mathrm{cf}^W(\gamma) \geq \kappa$, then $\sup \operatorname{supp}(p) < \gamma$, and the induction hypothesis applies. Otherwise let $\tau = \mathrm{cf}^W(\gamma) < \kappa$, choose a continuous increasing sequence $\langle \gamma_\eta : \eta \leq \tau \rangle$ with $\gamma_0 = 0$ and $\gamma_\tau = \gamma$, and construct

$$q_\eta \in Q_{\gamma_\eta}^V, \qquad e_{\gamma_\eta}(q_\eta) \leq p \restriction \gamma_\eta \quad (\eta \leq \tau),$$
$$q_\zeta \leq q_\eta \quad (\eta < \zeta \leq \tau).$$

Start with the trivial condition q_0, and at successor steps apply the induction hypothesis to the condition $e_{\gamma_\eta}(q_\eta) \cup (p \restriction [\gamma_\eta, \gamma_{\eta+1}))$. At limits $\nu \leq \tau$, we have $\langle q_\eta : \eta < \nu \rangle \in V$, since R adds no such sequences; let q_ν be a lower bound in the $< \kappa$-closed forcing $Q^V_{\gamma_\nu}$. Comparing coordinates, we obtain $e_\gamma(q_\tau) \leq p$. Since the maps e_α reflect the order and have dense range, they also preserve incompatibility. This proves (1).

For (3), note that $W \models \kappa^{<\kappa} = \kappa \wedge \sigma = \kappa^+$, so that Corollary 2.5 applies to $\widehat{Q}_\delta$ in W and gives the σ-chain condition. For each dense set $D_\alpha \subseteq Q_\alpha$, the set $e_\alpha[D_\alpha]$ is dense in $\widehat{Q}_\alpha$, and $|\delta|, |e_\alpha[D_\alpha]| \leq \sigma$ in W. Restricting e_δ to the conditions below a given condition gives the assertion for the restrictions of Q^V_δ. □

5 The bookkeeping and a first-order characterization of Γ_i

The forcing D and its bookkeeping function F are now fixed. We first show that the names which are needed later are names over D_M for bounded closed sets M of coordinates, and are therefore handled at some stage of the recursion. We then show that the Γ_i-iterations can be characterized by a first-order formula over $H_{\omega_{i+1}}$; this is used in the reduction of complexity in Section 6.

5.1 Names depend on few coordinates

Throughout, for a fixed $i \geq 1$ we write $U_i = \bigcup_{k>i} I_k$ and $W = W_{i+1} = B[G_{D_{>i}}]$, and we use the notation $\mathcal{C}_i$, P_C, G_C and $N_C = W[G_C]$ of Definition 4.10. Recall that for $C \in \mathcal{C}_i$ the set $U_i \cup C$ is closed, but C itself need not be closed.

We shall use Easton's lemma in the following form.

Lemma 5.1. *Let σ be regular uncountable, let R be $< \sigma$-closed and let P be σ-cc, and let $G_R \times G_P$ be generic for $R \times P$ over V. Then every sequence of ordinals of length $< \sigma$ in $V[G_R][G_P]$ belongs to $V[G_P]$, and hence*

$$H^{V[G_P]}_\sigma = H^{V[G_R][G_P]}_\sigma.$$

Consequently, first-order statements about H_σ with predicates from $V[G_P]$ have the same truth value in $V[G_P]$ and in $V[G_R][G_P]$.

Proof. Let $r_0 \in R$ and let $\dot{f}$ be an $R \times P$-name for a function from some $\eta < \sigma$ to the ordinals. For every $\zeta < \eta$ in turn, we construct by recursion on ξ conditions $p_{\zeta,\xi} \in P$ and $r_{\zeta,\xi} \leq r_0$ and ordinals $\alpha_{\zeta,\xi}$ such that, for fixed ζ, the conditions $p_{\zeta,\xi}$ are pairwise incompatible, the conditions $r_{\zeta,\xi}$ are descending throughout the whole construction, and

$$(r_{\zeta,\xi}, p_{\zeta,\xi}) \Vdash \dot{f}(\zeta) = \check{\alpha}_{\zeta,\xi}.$$

As long as there is a condition q which is incompatible with all $p_{\zeta,\xi'}$ chosen so far, take a lower bound r of all conditions of R chosen so far, and extend (r,q) to a condition which decides $\dot f(\zeta)$. By the chain condition, the construction for ζ stops after $\theta_\zeta < \sigma$ steps, with a maximal antichain $A_\zeta = \{p_{\zeta,\xi} : \xi < \theta_\zeta\}$. Since σ is regular, $\sum_{\zeta<\eta}\theta_\zeta < \sigma$, so the required lower bounds exist by the closure of R, and there is a lower bound r_* of all the conditions $r_{\zeta,\xi}$. The antichains A_ζ and the ordinals $\alpha_{\zeta,\xi}$ define a P-name $\dot g \in V$ with $(r_*, 1_P) \Vdash \dot f = \dot g$. Since r_0 was arbitrary, this proves the first assertion. Every element of H_σ is coded by a well-founded relation on an ordinal below σ; this gives the equality of the two sets, and the last assertion follows. □

Lemma 5.2. *Let $A \in B$ be a set of coordinates with $|A \cap I_k| \le \omega_{k+1}$ for every k. Then $\bar A = A \cup \bigcup_{\xi\in A} m(\xi)$ is the least closed set containing A, and it satisfies the same bounds. It has size at most $\aleph_\omega$ and is bounded in $\aleph_{\omega+2}$. If A is contained in an ordinal α, then so is $\bar A$.*

Proof. Each $m(\xi)$ is closed, so $\bar A$ is closed. By (b), a coordinate of level k can belong to $m(\xi)$ only if ξ has one of the finitely many levels $1,\dots,k$. There are at most ω_{k+1} such coordinates $\xi \in A$, and each of them contributes at most ω_{k+1} coordinates of level k. Hence $|\bar A \cap I_k| \le \omega_{k+1}$ and $|\bar A| \le \aleph_\omega$. Since $\aleph_{\omega+2}$ is regular, $\bar A$ is bounded in $\aleph_{\omega+2}$. The last assertion holds because $m(\xi) \subseteq \xi$. □

Proposition 5.3. *Let $i \ge 1$ and $\alpha \le \aleph_{\omega+2}$, and let $D_{\alpha,>i}$ be the restriction of D to the coordinates $\xi < \alpha$ of level greater than i. Let $u \in B[G_{D_{\alpha,>i}}]$ be a sequence of ordinals of length at most ω_{i+1}. Then there is a closed set $H \in B$ with*

$$H \subseteq \{\xi < \alpha : i(\xi) > i\}, \qquad |H \cap I_k| \le \omega_{k+1} \quad (k > i),$$

such that $u \in B[G_H]$. Moreover, the range of u is contained in a set in B of size at most ω_{i+1}.

Proof. Put $R = D_{\alpha,>i+1}$. By Lemma 4.6, R is $<\omega_{i+2}$-closed, and by Proposition 4.9 (for level $i+1$) the quotient $D_{\alpha,>i}/G_R$ is forcing equivalent to $(S_{i+1,\alpha}, \preceq_{G_R})$ and is ω_{i+2}-cc. Recall that $S_{i+1,\alpha} \in B$.

In $B[G_R]$, choose a nice name for u over this quotient. By the chain condition it consists of at most ω_{i+1} triples, each consisting of an argument, a condition in $S_{i+1,\alpha}$ and an ordinal value. Replace the conditions by their indices in a fixed enumeration of $S_{i+1,\alpha}$ which belongs to B. The resulting sequence of ordinals has length at most ω_{i+1}. Since R is $<\omega_{i+2}$-closed, this sequence belongs to B, although its evaluation u need not.

The supports of these conditions together contain at most ω_{i+1} coordinates, all of level $i+1$. Let H be the closure of their union under the sets $m(\xi)$; by Lemma 5.2, H has the required properties. Each of these conditions is a condition of D with support contained in H, and it belongs to the

generic filter for the quotient if and only if it belongs to G_H. Thus the name can be evaluated using G_H, and $u \in B[G_H]$. The set of all ordinal values which occur in the name belongs to B and has size at most ω_{i+1}; it contains the range of u. □

Lemma 5.4. *Let $C \in \mathcal{C}_i$, let $Q \in \Gamma_i$ in N_C, and let $q \in Q$. There is a bounded closed set $M \in B$ with*

$$M \subseteq \bigcup_{k \geq i} I_k, \qquad |M \cap I_k| \leq \omega_{k+1} \ (k \geq i), \qquad C \subseteq K = M \cap I_i.$$

There are $p \in G_M$ and a D_M-name $\dot{R}$ such that p forces $\dot{R}$ to code a Γ_i-iteration of positive length. The last forcing of the iteration coded by $\dot{R}^{G_M}$ in $B[G_M]$ is forcing equivalent in N_K to $Q \restriction q$.

Proof. Put $\rho = \omega_i$, $\sigma = \omega_{i+1}$ and $W = W_{i+1}$. By Proposition 5.3, C is covered by a set in B of size at most σ. Intersect this set with I_i and close it under $\xi \mapsto m(\xi) \cap I_i$ to obtain $K \in B$ with $|K| \leq \sigma$ and $U_i \cup K$ closed. In B, fix enumerations of H_σ^B and of the conditions supported on K. There are at most σ such conditions by the bounds on $\mathcal{N}_\xi$ and the supports. Write $P = P_K$ for the quotient, represented as the set of these conditions with the order $\preceq_{G_{D_{>i}}}$ of Subsection 4.5. The $< \rho$-closure and σ-cc of P give $(*)$ in N_K by counting nice names.

By Lemma 5.7(a), whose proof uses only the structural properties of the quotients, P/G_C is forcing equivalent to a $< \rho$-closed, σ-cc forcing. In $N_K = W[G_K]$, lift the iteration for Q by Lemma 4.14 and restrict it below the image of q; by the remark on cones following Definition 4.1, the result is again given by a Γ_i-iteration. Choose a code $R \subseteq \sigma$ for the resulting iteration, adding a trivial iterand if needed, and choose χ with $\mathrm{Wit}(R, \chi)$ by Proposition 5.9.

The σ-cc of P gives nice names for R and χ coded together by a sequence u of at most σ ordinals. Indeed, a nice name for an element of $H_\sigma^{N_K}$ has a code in $H_\sigma^W = H_\sigma^B$. For each such code use antichains of size at most ρ to decide membership in the two classes, using the fixed enumerations for the indices of names and conditions. By Proposition 5.3 and Lemma 5.2, choose a closed $H \in B$, $H \subseteq U_i$, with the required bounds, containing $m(\xi) \cap U_i$ for every $\xi \in K$, such that $u \in Z = B[G_H]$. Put $M = H \cup K$.

In Z, the quotient D_M/G_H is represented by the same preorder P as in W. For conditions s, t supported on K, a witness $r \in G_{D_{>i}}$ to $r \cup s \leq_D t$ can be restricted to H, since $m(\xi) \subseteq M$ for $\xi \in K$. Conversely, a witness in G_H gives one in $G_{D_{>i}}$. Put $R' = D_{>i}/G_H$. It is $< \sigma$-closed in Z, and P is σ-cc there because its order is the same in W. Applying Lemma 5.1 to $P \times R'$ over Z gives

$$H_\sigma^{B[G_M]} = H_\sigma^{Z[G_K]} = H_\sigma^{N_K}.$$

The sequence u defines the same R, χ in $B[G_M]$, so the first-order assertion $\mathrm{Wit}(R, \chi)$ holds there. Proposition 5.9 and the forcing theorem give p and a nice name $\dot{R}$ for a subset of σ.

Since P adds no sequences of length $< \rho$ over Z, R' is still $< \rho$-closed after forcing with P. It preserves σ by Lemma 5.1. By Lemma 4.14, the iteration given by R in $B[G_M]$ is forcing equivalent to its lift to N_K. The antichain names in R translate along the dense embeddings of that lemma, so the lift is the iteration chosen in N_K. □

Lemma 5.5. *Fix $j \geq 1$ and work in $W_j = B[G_{D_{\geq j}}]$. Let C be a set of coordinates of levels below j, with $|C \cap I_k| \leq \omega_{k+1}$ for $k < j$, such that $C \cup \bigcup_{k \geq j} I_k$ is closed. Present the quotient $D_{C \cup \bigcup_{k \geq j} I_k} / G_{D_{\geq j}}$ by the conditions supported on C with the quotient order. There are at most ω_j such conditions by the bounds on the supports and the $\mathcal{N}_\xi$. Enumerate them to give a preorder P on ω_j, and let $\dot{E}$ be a nice P-name for a subset of ω_{j+1}. Then there is a closed set $H \in B$ with*

$$H \subseteq \bigcup_{k \geq j} I_k, \qquad |H \cap I_k| \leq \omega_{k+1} \ (k \geq j), \qquad \sup H < \aleph_{\omega+2},$$

such that P and $\dot{E}$ belong to $B[G_H]$. Moreover, there is a stage β of the recursion with $\sup(H \cup C) < \gamma_\beta$ at which Case 2 of Definition 4.5 applies with $i = j$, $M = H$ and a name for $(P, \dot{E})$; thus the term forcing $\mathrm{Term}(P, \mathbb{A}_{\omega_j}(\dot{E}))$, as defined in $B[G_H]$, is appended at stage β.

Proof. The forcing P is ω_j-cc, since the quotient is a finite iteration of the quotients at the levels $j-1, \ldots, 1$ restricted to C, each of which is ω_j-cc by Proposition 4.9. The order of P is a subset of $\omega_j \times \omega_j$, and each antichain A_α in the nice name $\dot{E}$ has size $< \omega_j$. Thus a sequence of ordinals u coding P and $\dot{E}$ has length

$$|u| \leq \omega_j + \sum_{\alpha < \omega_{j+1}} |A_\alpha| \leq \omega_{j+1}.$$

In $B[G_{D_{>j}}]$, a nice name for u over the quotient at level j consists of at most ω_{j+1} conditions and ordinals. Apply Proposition 5.3 to a sequence of ordinals coding this name, add the supports of the conditions which occur in the name (these conditions are contained in a set in B of size at most ω_{j+1}, by the last assertion of Proposition 5.3), and take the closure, using Lemma 5.2. These supports contain at most ω_{j+1} coordinates of level j. This gives the required set H, with $u \in Z = B[G_H]$ and $|H| \leq \aleph_\omega$; since $\aleph_{\omega+2}$ is regular, $\sup H < \aleph_{\omega+2}$.

The coordinates in C, which have levels below j, occur in u only as ordinals; the generic filter on these coordinates is not needed to define P and $\dot{E}$. A coordinate ξ at which the term forcing is appended may therefore

have the memory

$$m(\xi) = H \subseteq \bigcup_{k \geq j} I_k.$$

In Z, P is the same preorder on ω_j, and it is ω_j-cc there because it is ω_j-cc in the larger model W_j. Moreover $\omega_j^{<\omega_j} = \omega_j$ and $\omega_{j+1}^{<\omega_j} = \omega_{j+1}$ hold in Z, since D_H is $< \omega_j$-closed. Theorem 2.10 therefore applies to

$$T = \mathrm{Term}(P, \mathbb{A}_{\omega_j}(\dot{E}))^Z.$$

In all later arguments we use this forcing T with its conditions and order as defined in Z, also after the generic filter for P has been added.

Choose $p \in G_H$ and a D_H-name $\dot{X}$ for $(P, \dot{E})$ such that p forces the properties required in Case 2. The quadruple $(j, H, p, \dot{X})$ is coded by an element of $\mathcal{B}$, which is equal to $F(\beta)$ for unboundedly many β. Since $H \cup C$ is bounded, there is such a β with $\sup(H \cup C) < \gamma_\beta$. At this stage Case 2 applies and the term forcing T is appended. Note that whether Case 2 applies depends only on the forcing D_H, which has been defined before stage β. □

5.2 The bookkeeping handles all relevant names

Proposition 5.6. *1. Let $C \in \mathcal{C}_i$, $Q \in \Gamma_i$ in N_C, and $q \in Q$. There are $M, p, \dot{R}$ as in Lemma 5.4 and a stage β with $M \subseteq \gamma_\beta$ at which Case 1 applies with $\dot{X} = \dot{R}$.*

2. Let $j \geq 1$ and let C, P and $\dot{E}$ be as in Lemma 5.5. Then the term forcing $\mathrm{Term}(P, \mathbb{A}_{\omega_j}(\dot{E}))$ is appended at a stage after all coordinates in C and all coordinates on which P and $\dot{E}$ depend. The generic filter induced on P is generic over W_j, which contains the generic filter for the term forcing.

Proof. For a closed set $M \in B$ with $|M \cap I_k| \leq \omega_{k+1}$ for all $k \geq 1$, we have

$$|M| \leq \aleph_\omega, \qquad |D_M| \leq \aleph_\omega^{\aleph_\omega} = \aleph_{\omega+1}.$$

A nice D_M-name given by maximal antichains therefore consists of at most $\aleph_{\omega+1}$ pairs of conditions and ordinals. These are coded by ordinals which represent conditions, coordinates below $\aleph_{\omega+2}$, and ordinal values below ω_{i+1}. Thus every pair $(p, \dot{X})$ of a condition and such a name, together with i and M, is coded by an element of the set $\mathcal{B}$.

(1) Take $M, p, \dot{R}$ from Lemma 5.4. Strengthen p in G_M to decide the length of the iteration. The code of $(i, M, p, \dot{R})$ occurs unboundedly often, so it occurs at a stage β with $\sup M < \gamma_\beta$. The recursion appends the iterands of the iteration given by this code.

(2) Lemma 5.5 provides a bounded closed set H of coordinates of levels at least j and a stage at which Case 2 applies and the term forcing is appended.

By Lemma 4.6, applied to the restriction to the closed set $C \cup \bigcup_{k \geq j} I_k$, the filter induced on P is generic over W_j, and W_j contains the generic filter for the term forcing. Hence the projection of Theorem 2.10 from the product to the two-step iteration can be applied. □

5.3 Subforcings of the quotients

Lemma 5.7. *Let $i \geq 1$, work in $W = W_{i+1}$, and let $C \in \mathcal{C}_i$.*

(a) For every $X \subseteq H_{\omega_i}$ in W_i there is $C' \in \mathcal{C}_i$ with $C \subseteq C'$ and $X \in N_{C'}$. Moreover, for all $C \subseteq C'$ in $\mathcal{C}_i$, the quotient $P_{C'}/G_C$ is forcing equivalent, in N_C, to a Γ_i-iteration.

(b) Let $Q \in \Gamma_i$ in N_C and $q \in Q$. There are $C_0, C' \in \mathcal{C}_i$ with $C \subseteq C_0 \subseteq C'$ and a Q-generic filter H over N_{C_0} with $q \in H$ and $N_{C'} = N_{C_0}[H]$.

Proof. (a) Choose a nice name for X over the quotient at level i. Since H_{ω_i} has cardinality ω_i and every antichain of the quotient has cardinality at most ω_i, the name consists of at most ω_i conditions. Let C' be the closure of the union of C and the supports of these conditions under the maps $\xi \mapsto m(\xi) \cap I_i$. As in the proof of Lemma 5.2, $|C'| \leq \omega_{i+1}$. Then $C' \in \mathcal{C}_i$ and $X \in N_{C'}$.

Now let $C \subseteq C'$ be elements of $\mathcal{C}_i$. To describe $P_{C'}/G_C$, we order the coordinates of C first and then the coordinates of $C' \setminus C$ in increasing order. Since $U_i \cup C'$ is closed, every element of $m(\xi)$, for $\xi \in C' \setminus C$, belongs to $U_i \cup C$ or precedes ξ in this order. For $\xi \in C' \setminus C$, the model obtained at this stage lies between $B[G_{m(\xi)}]$ and W_i. By Lemma 4.7, W_i adds no sequences of length $< \omega_i$ over $B[G_{m(\xi)}]$, so the iterand remains $< \omega_i$-compact and keeps its linking function. The proof of Proposition 4.9, carried out for conditions whose restriction to C belongs to G_C, shows that the quotient is forcing equivalent to the $< \omega_i$-support iteration of the iterands at the coordinates in $C' \setminus C$. This iteration has length of cardinality at most ω_{i+1} and a dense subset of that cardinality, so it is a Γ_i-iteration. If a set in B which covers C' is needed, apply Proposition 5.3 to an enumeration of C', as in the proof of Lemma 5.4.

(b) Take $M, p, \dot{R}$ and the block from Proposition 5.6(1). Let $C_0 = M \cap I_i$ and let C' be its union with the block. Then $C_0, C' \in \mathcal{C}_i$ by the bounds on M and by the definition of the memories of the block. By (a), the quotient from N_{C_0} to $N_{C'}$ is given by the iterands of the block. Since $p \in G_M$, these are the iterands specified by the code $\dot{R}^{G_M}$, lifted to N_{C_0}. Lemma 5.4 gives a forcing equivalence with $Q \restriction q$ there. Under this equivalence, the generic filter on the block corresponds to a Q-generic filter H over N_{C_0} with $q \in H$, and $N_{C'} = N_{C_0}[H]$. □

5.4 A first-order characterization of Γ_i-iterations

Fix a regular uncountable cardinal ρ and put $\sigma = \rho^+$. We consider the following hypothesis on the universe:

$$(*) \qquad \rho^{<\rho} = \rho \quad \text{and} \quad 2^\rho = \sigma.$$

It implies $|H_\sigma| = \sigma$ and $\sigma^\rho = \sigma$. We shall apply the results of this subsection with $\rho = \omega_i$ in the models W_{i+1} and N_C, for $C \in \mathcal{C}_i$, where $(*)$ holds. We do not use them in W_i: forcing with the whole quotient at level i increases 2^ρ.

In this subsection, a Γ-iteration is an iteration as in Definition 4.1, with ρ in place of ω_i. To quantify over these iterations, we code them, and names for them, by sets of ordinals. Let $\langle P_\alpha, \dot{Q}_\alpha : \alpha < \delta\rangle$ be a Γ-iteration, and assume $(*)$. Choose a wellorder $(J, \prec)$ with

$$J \subseteq \sigma, \qquad \mathrm{otp}(J, \prec) = \delta.$$

Recall that $|\delta| \leq \sigma$; the ordinal δ itself may be larger than σ. Using the σ-chain condition and $\sigma^\rho = \sigma$, one constructs by recursion, using nice names, dense subsets and enumerations

$$e_\alpha : \sigma \twoheadrightarrow D_\alpha \subseteq P_\alpha, \qquad D_\alpha \text{ dense in } P_\alpha \quad (\alpha \leq \delta).$$

By means of these enumerations, the conditions which occur in a name can be replaced by ordinals. The enumerations need not be injective. Similarly, each iterand is a preorder on σ with largest condition 0.

The *code* of the iteration consists of $(J, \prec)$, of names for the orders of the iterands, for their linking functions and for lower bounds of centered families, and of the enumerations e_α. A nice name is coded by the ordinals which represent the conditions of its antichains, together with the corresponding values. A name for an ordinal which is the value of a condition at one coordinate consists of at most ρ conditions. Since a condition has support of size $< \rho$, its code has hereditary size at most ρ and belongs to H_σ. In the iteration code, each value $e_\alpha(\xi)$ is given by the code of this condition, that is, by its support and, for each coordinate of the support, by the antichain and the values of the corresponding nice name. The stages are indexed by the elements of J, and the last stage by one additional index. The names for the orders and for the functions are coded in the same way; where an argument is a set of size $< \rho$, its elements are listed. Coding finite tuples of ordinals below σ by ordinals below σ, we take the iteration code to be a subset of σ. A condition used in a forcing statement can be strengthened to one in D_δ. The iteration codes and the codes for names of subsets of H_σ are classes over H_σ.

Let $\mathcal{T}$ be the following theory in the language of set theory with additional constants for ρ, σ and a code R. It consists of ZFC^- (that is, ZFC

without Power Set, with Collection and Choice), of the statements that ρ is regular uncountable, $\sigma = \rho^+$, $\rho^{<\rho} = \rho$ and $2^\rho = \sigma$, and of the statements that R is the code of a Γ-iteration: the relation $\prec$ is a wellorder, the coded sequence is a $< \rho$-support iteration along this wellorder, the sets enumerated by the maps e_α are dense, and the names for the iterands are forced to be $< \rho$-compact preorders on σ for which the given names are names for linking functions and for lower bounds. All these are first-order statements about set forcing and the components of the code. If R is the code of a Γ-iteration, then every elementary submodel of a sufficiently large H_θ which contains ρ, σ and R satisfies $\mathcal{T}$.

Lemma 5.8. *Let M be a transitive model of $\mathcal{T}$ such that $H_\sigma \subseteq M$ and $H_\sigma, \sigma \in M$, and such that the constants ρ, σ are interpreted by the true ρ, σ. Let δ be the length of the iteration coded by R in M. For $\alpha \leq \delta$ let P_α^M be the α-th stage of this iteration as defined in M, and let P_α be the α-th stage of the $< \rho$-support iteration defined in V from the names in R. Then for every $\alpha \leq \delta$:*

$$P_\alpha^M \subseteq P_\alpha, \quad \text{and} \quad M \models p \leq_{P_\alpha^M} q \iff p \leq_{P_\alpha} q \quad (p, q \in P_\alpha^M),$$
$$D \in M,\ M \models D \text{ is dense in } P_\alpha^M \implies D \text{ is dense in } P_\alpha,$$
$$P_\alpha \text{ is } < \rho\text{-closed and satisfies the stationary } \sigma\text{-chain condition},$$
$$H_\sigma^{M[G]} = H_\sigma^{V[G]} \quad (G \subseteq P_\alpha \text{ generic over } V).$$

Proof. The relation $\prec$ on $J \subseteq \sigma$ is a wellorder in V: a descending sequence $s \in {}^\omega J$ would belong to $H_\sigma \subseteq M$, whereas M satisfies that $\prec$ is well-founded. We prove the four assertions by induction along this wellorder. At stage α, the induction hypothesis $H_\sigma^{M[G_\alpha]} = H_\sigma^{V[G_\alpha]}$ gives

$$[\sigma]^{<\rho} \cap M[G_\alpha] = [\sigma]^{<\rho} \cap V[G_\alpha].$$

The order of the iterand and its linking function on σ are the same in $M[G_\alpha]$ and $V[G_\alpha]$. Whether finitely many conditions have a common lower bound is absolute, so every centered family of size $< \rho$ in $V[G_\alpha]$ belongs to $M[G_\alpha]$ and has a lower bound there; and conditions with the same value of the linking function are compatible. Thus the iterands are $< \rho$-compact and ρ-linked in $V[G_\alpha]$, and Corollary 2.5 applies to the iterations P_α.

For the assertion on the order, let $\beta < \alpha$. By the induction hypothesis, P_β^M is a dense subset of P_β with the induced order, and every dense subset of P_β^M in M is dense in P_β. Hence a P_β^M-name in M is a P_β-name, which gives $P_\alpha^M \subseteq P_\alpha$, and every filter G which is P_β-generic over V induces a filter which is P_β^M-generic over M; the two filters evaluate the names in M in the same way. Let $p, q \in P_\alpha^M$, and let θ be the statement $p(\beta) \leq q(\beta)$ about the order of the iterand at β. Its truth value is the same in $M[G]$ and in $V[G]$. Since every condition of P_β has an extension in P_β^M, it follows that $p \restriction \beta$

forces θ over P_β^M in M if and only if it forces θ over P_β in V. As the order of the iteration is defined coordinatewise from these forcing statements, $p \leq q$ holds in M if and only if it holds in V.

For the density assertion, consider first a successor stage. A nice name for an ordinal below σ is given by a maximal antichain of size at most ρ in the dense set of the preceding stage, together with ordinal values. Replacing the conditions by the ordinals which represent them, we obtain a code in $H_\sigma \subseteq M$. If only part of a maximal antichain is given, extend it to a maximal antichain and assign the value 0, which represents the largest condition, to the new elements; the resulting name is forced by the trivial condition to be a condition of the iterand. At a limit stage, code the value at every coordinate of the support in this way. For the resulting code c_p of a condition p,

$$|\operatorname{supp}(p)| < \rho, \qquad |\operatorname{tc}(c_p)| \leq \rho, \qquad c_p \in H_\sigma \subseteq M.$$

The condition coded by c_p in M has the same values at all coordinates, since every ordinal below σ represents a condition of the iterand. Thus P_α^M is dense in P_α. Since the two orders agree on P_α^M, every set which is dense in P_α^M in the sense of M is dense in P_α. Note that only the codes of the conditions need to belong to M.

Finally, let $x \in H_\sigma^{V[G]}$. Choose a well-founded extensional relation $E \subseteq \gamma^2$ and a point $e < \gamma < \sigma$ with $x = \pi_E(e)$, where π_E is the transitive collapse. A nice name for E consists of at most ρ conditions; its code belongs to H_σ^V, hence to M, and it has the same value in $M[G]$. The relation E is well-founded in $M[G]$, and by Collection its transitive collapse exists there. By uniqueness of the collapse, $x = \pi_E(e) \in M[G]$, and x has hereditary size $< \sigma$ in $M[G]$. Conversely, an enumeration in $M[G]$ which witnesses that a set has hereditary size $< \sigma$ witnesses the same in $V[G]$. This proves the equality and completes the induction. □

Proposition 5.9. *Assume* $(*)$. *There is a first-order formula* $\mathrm{Wit}(R, \chi)$ *over* H_σ, *with class variables* R, χ, *such that for every class* R *over* H_σ,

$$R \text{ is the code of a } \Gamma\text{-iteration} \quad \Longleftrightarrow \quad \exists \chi \ \mathrm{Wit}(R, \chi).$$

The formula has no parameters; in particular it does not depend on $\vec{T}$ *or on the stationary sets* S_t^n.

Proof. The formula $\mathrm{Wit}(R, \chi)$ requires R to consist of ordinals and expresses that χ codes a set $A \subseteq \sigma$, a binary relation E on A, a relation Sat, a function $j : H_\sigma \to A$ and elements $h, s, r \in A$ with the following properties. The relation E is well-founded and extensional, j is an embedding of $(H_\sigma, \in)$ into (A, E), and

$$\operatorname{pred}_E(j(a)) = j[a] \quad (a \in H_\sigma),$$
$$\operatorname{pred}_E(h) = j[H_\sigma], \qquad \operatorname{pred}_E(s) = j[\sigma].$$

The element r represents the code R, with all its components, in (A, E); this is expressed by comparing the elements of R and of the set represented by r one by one. The relation Sat satisfies the Tarski conditions for the satisfaction relation of (A, E), and

$$\forall n \in \omega \, (\mathrm{Ax}_{\mathcal{T}}(n) \Rightarrow \mathrm{Sat}(n)),$$

where $\mathrm{Ax}_{\mathcal{T}}$ is the recursive set of Gödel numbers of the axioms of $\mathcal{T}$, with the constants interpreted by $j(\rho)$, s and r.

All these conditions are first order over H_σ. In particular, well-foundedness is expressed by

$$\neg\exists f \in H_\sigma \, (f : \omega \to A \ \wedge \ \forall n < \omega \ f(n+1) \ E \ f(n)),$$

since every ω-sequence of elements of $A \subseteq \sigma$ belongs to H_σ.

Suppose that $\mathrm{Wit}(R, \chi)$ holds, and let π be the transitive collapse of (A, E). By induction on the rank,

$$\pi(j(a)) = a \quad (a \in H_\sigma), \qquad \pi(h) = H_\sigma, \qquad \pi(s) = \sigma,$$

and $\pi(r) = R$. Thus the collapse is a transitive model of $\mathcal{T}$ which contains H_σ, σ and R, and Lemma 5.8 shows that R is the code of a Γ-iteration.

Conversely, let R be the code of a Γ-iteration. Take a sufficiently large regular θ and $N \prec H_\theta$ with $|N| = \sigma$, $H_\sigma \subseteq N$, and $H_\sigma, \sigma, R \in N$. The transitive collapse of N is the identity on H_σ and therefore maps R to itself. The collapsed model and its satisfaction relation yield a class χ with $\mathrm{Wit}(R, \chi)$. □

The code R and the witness χ can be combined into one class. Thus an existential quantifier over Γ-iterations can be expressed by an existential quantifier over pairs of classes (R, χ) with $\mathrm{Wit}(R, \chi)$, and a universal quantifier over Γ-iterations by a universal quantifier over such pairs. The model coded by χ is used only to verify that R codes an iteration and to compute the order of the conditions enumerated by the last map e_δ of the code: by Lemma 5.8, the order computed in this model is the true one. Thus the preorder on σ induced by e_δ, which is a dense subset of the last forcing, is first-order definable over H_σ from R and χ, and it does not depend on the choice of χ. All other class quantifiers in the formulas below range over all classes of the universe and are not relativized to this model. Proposition 5.9 concerns iterations in the usual sense. The fact that the quotients of D are forcing equivalent to such iterations was proved in Proposition 4.9.

5.5 Summary of the properties of the forcing

Theorem 5.10. *For each $B = B_{\vec{T}}$, the forcing $D = D_{\vec{T}}$ defined in Section 4 has the following properties. Its bookkeeping function handles the names for forcings in the classes Γ_i and for term forcings as stated in Proposition 5.6, and its definition is uniform in $\vec{T}$.*

(M1) *For every $j \geq 1$, $D_{\geq j}$ is a complete subforcing of D and is $< \omega_j$-closed. Moreover, for every $p \in D$ and every $q \leq p \restriction \bigcup_{k \geq j} I_k$ in $D_{\geq j}$ there is $p' \leq p$ whose restriction to $\bigcup_{k \geq j} I_k$ is q and which agrees with p on the other coordinates.*

(M2) *Let $i \geq 1$. In W_{i+1}, the quotient $D_{\geq i}/G_{D_{>i}}$ is forcing equivalent to a $< \omega_i$-support iteration with the stationary ω_{i+1}-chain condition. For every $C \in \mathcal{C}_i$ and every class X over H_{ω_i} in W_i there is $C' \in \mathcal{C}_i$ with $C \subseteq C'$ and $X \in N_{C'}$. For all $C \subseteq C'$ in $\mathcal{C}_i$, the quotient $P_{C'}/G_C$ is forcing equivalent to a Γ_i-iteration. For every $C \in \mathcal{C}_i$, every forcing $Q \in N_C$ which belongs to Γ_i in N_C and every $q \in Q$, there are $C \subseteq C_0 \subseteq C'$ in $\mathcal{C}_i$ and a filter H which contains q and is Q-generic over N_{C_0}, such that $N_{C'} = N_{C_0}[H]$. Here the sets in $\mathcal{C}_i$ need not be initial segments of I_i.*

(M3) *Every finite family of classes over* HC *in $B[G_D]$ belongs to $B[G_C]$ for some closed set $C \in B$ with $|C \cap I_i| \leq \omega_{i+1}$ for every i. The set C may contain coordinates of infinitely many levels.*

(M4) *D has cardinality at most $\aleph_{\omega+2}$ and is $\aleph_{\omega+2}$-cc. It preserves all cardinals and adds no reals.*

(M5) *For every $i \geq 1$, the Γ_i-iterations have codes which are classes over $H_{\omega_{i+1}}$, and there is a first-order formula $\mathrm{Wit}(R, \chi)$ over $H_{\omega_{i+1}}$ such that, in W_{i+1} and in N_C for every $C \in \mathcal{C}_i$, a class R is the code of a Γ_i-iteration if and only if $\exists \chi\ \mathrm{Wit}(R, \chi)$. This formula has no parameters; in particular it mentions neither the sets T_n nor the clubs added by $C_{\vec{T}}$.*

(M6) *Let $j \geq 1$ and let C, P and $\dot{E}$ be as in Lemma 5.5. Then the term forcing $\mathrm{Term}(P, \mathbb{A}_{\omega_j}(\dot{E}))$ is appended at a stage after all coordinates on which P and $\dot{E}$ depend. The generic filter induced on P is generic over W_j, which contains the generic filter for the term forcing.*

Proof. At each stage of the recursion, the bookkeeping only uses the forcing relation of a forcing D_M which has already been defined. The results of Section 4 hold for every iteration with partial memory, whether or not its bookkeeping handles all names; hence they hold at every stage of the recursion and for D itself, and so do the results of the present section. The uniformity in $\vec{T}$ was shown after Definition 4.5.

(M1) is Lemma 4.6. The first assertion of (M2) is Proposition 4.9, and the remaining ones are Lemma 5.7. (M4) is Lemmas 4.12 and 4.13. (M5) is Proposition 5.9, together with the fact that $(*)$ holds for $\rho = \omega_i$ in the models in question. (M6) is Proposition 5.6(2).

For (M3), let X be a class over HC in $B[G_D]$. In $W_2 = B[G_{D_{>1}}]$, X has a nice name over the quotient at level one which consists of ω_1 conditions.

Code this name by a sequence of ordinals of length ω_1, and apply Proposition 5.3 to this sequence. Together with the closure of the supports of the conditions, this gives a closed set $C \in B$ which satisfies the bounds at every level, such that $X \in B[G_C]$. For a finite family of classes, take the union of the corresponding sets. □

6 Forcing and the reduction of complexity

6.1 The class Γ_i

Fix $i \geq 1$, and write $\rho = \omega_i$ and $\sigma = \rho^+$. All models N in which the class Γ_i is considered in this section satisfy the hypothesis $(*)$ of Subsection 5.4, that is,

$$\rho^{<\rho} = \rho, \qquad 2^\rho = \sigma.$$

Each $P \in \Gamma_i$ is given with an iteration code as in Subsection 5.4. Neither the stationary sets S^n_t nor the sets T_n occur in the definition.

Lemma 6.1. *Every forcing in Γ_i is $< \rho$-closed, is σ-cc, has a dense subset of cardinality at most σ, and preserves $(*)$. Moreover:*

1. *If $P, Q \in \Gamma_i$ in N, then in $N[G_P]$ the forcing Q of N densely embeds into the iteration which is given there by the same names for the iterands. In particular $P \times Q$ is forcing equivalent to a member of Γ_i in N.*

2. *If $P \in \Gamma_i$ and $P \Vdash \dot{Q} \in \Gamma_i$, then $P * \dot{Q}$ is forcing equivalent to a member of Γ_i.*

3. *If $P \in \Gamma_i$ and $p \in P$, then $P \restriction p$ is forcing equivalent to a member of Γ_i.*

Proof. By Corollary 2.5, every Γ_i-iteration and each of its initial segments is $< \rho$-closed and σ-cc. By the chain condition, the value of a condition at a coordinate is given by a nice name with antichains of cardinality at most ρ. By induction, a condition which is given in this way is coded by a set of at most ρ ordinals below σ. Since $\sigma^\rho = \sigma$, the iteration has a dense subset of size at most σ.

The iteration adds no sequences of ordinals of length less than ρ. It preserves σ, and a nice name for a subset of ρ is coded by a set of at most ρ ordinals below σ; there are at most $\sigma^\rho = \sigma$ such names. Thus it preserves both equations in $(*)$.

For (1), apply Lemma 4.14 with $R = P$ to the iteration for Q. Concatenate P with a name for the lifted iteration. The same concatenation proves (2): strengthen a condition to decide the support of its second component, using the $< \rho$-distributivity of P. If the length of the second iteration is

variable, the σ-chain condition gives a bound below σ^+, and we extend the iteration to that length by trivial iterands. For (3) use the remark on cones following Definition 4.1. □

6.2 The forcing relation for second-order formulas

We now show that the forcing relation for second-order statements over H_τ is definable by a second-order formula of the same complexity. In all forcing statements below, class quantifiers range over all classes of the generic extension, and not only over the classes which belong to a model that is used to verify that a code is the code of a forcing.

Lemma 6.2. *Let $\tau > \omega$ be regular with $\tau^{<\tau} = \tau$ in N, and let $Q \in N$ be τ-cc with a dense subset which is given as a preorder on τ. For every second-order formula φ over $H_\tau^{N[G_Q]}$, the relation $q \Vdash_Q \varphi$ is expressed over H_τ^N by a second-order formula with the same prefix of class quantifiers, which is obtained recursively from φ. In particular, the forcing relation of a first-order formula is first order, the forcing relation of a Σ^2_m formula is Σ^2_m, and the forcing relation of a Π^2_m formula is Π^2_m. The parameters of the translated formula are classes coding the dense preorder and the names for the class parameters of φ.*

Proof. Put $H = H_\tau^N$, and let Q_0 be the given dense preorder on τ.

Names for elements. Every element x of $H_\tau^{N[G_Q]}$ is the image of a point under the transitive collapse of a well-founded extensional relation on some $\gamma < \tau$. We code a name for x by a triple a consisting of nice Q_0-names for the field and for the relation, and a maximal antichain deciding the point. Such a name involves fewer than τ statements, each of which is decided by an antichain of size $< \tau$; hence $a \in H$. Every $x \in H_\tau^{N[G_Q]}$ has a name of this form: by the chain condition, γ can be bounded below a given condition p, and mixing with a name for the empty set below the conditions incompatible with p gives a name which is forced by the trivial condition to be of the required form.

Let Obj(a) express that a is such a code, and write x_a for the element named by a. The predicate Obj is first-order definable over H. To see this, note that a function $f : \omega \to \gamma$ in the extension has a nice name for its graph which is an element of H, and that every descending ω-sequence witnessing ill-foundedness has such a name; these names are used to express well-foundedness, and they do not involve the codes of elements just defined. The Boolean values $\|x_a = x_b\|$ and $\|x_a \in x_b\|$ are expressed by means of names for isomorphisms between the corresponding relations below their distinguished points: by the maximum principle such an isomorphism has a name consisting of fewer than τ pairs, and conversely an isomorphism shows that the collapses agree. Hence the Boolean values of atomic formulas are first-order definable over H.

Boolean values. We represent Boolean values by antichains $A \in H$ of Q_0, modulo equality of the values $[A]$ which they represent. Their order is first order:

$$[A] \leq [B] \quad \iff \quad \forall a \in A\, \forall q \leq a\, \exists r \leq q\, \exists b \in B\ (r \leq b).$$

Consequently equality and complements are first order. The supremum of a family of Boolean values which is given by a class is its least upper bound in this order; here all quantifiers range over antichains in H. The supremum exists by the chain condition and the completeness of the Boolean completion. Note that no definable choice of representatives and no additional class quantifier is needed.

Names for classes. A Q-name for a class over $H_\tau^{N[G_Q]}$ is coded by a function χ which assigns a Boolean value to every a with $\mathrm{Obj}(a)$, such that

$$\|x_a = x_b\| \wedge \chi(a) = \|x_a = x_b\| \wedge \chi(b) \quad (\mathrm{Obj}(a), \mathrm{Obj}(b)).$$

Let $\mathrm{Name}(\chi)$ express that χ is such a function, where the values are determined up to equality of Boolean values; this is first order. Every name in N for a class over $H_\tau^{N[G_Q]}$ gives rise to such a class χ over H. Conversely, every χ with $\mathrm{Name}(\chi)$ determines a name for a class, and the value of this name does not depend on the choice of the codes a.

First-order formulas. For a first-order formula θ, the Boolean value $\|\theta\|$ is defined by recursion on θ, using the Boolean operations and, for the quantifiers, suprema and infima over all a with $\mathrm{Obj}(a)$. Since θ is a fixed formula, this gives a first-order definition of $\|\theta\|$, and $p \Vdash \theta$ means $[p] \leq \|\theta\|$. No class coding a satisfaction relation or an assignment of Boolean values to all formulas is used.

Class quantifiers. Finally, the maximum principle and mixing with a name for the empty class below the conditions incompatible with p give

$$\begin{aligned} p \Vdash \exists X\, \psi(X) &\iff \exists \chi\, [\mathrm{Name}(\chi) \wedge p \Vdash \psi(\chi)], \\ p \Vdash \forall X\, \psi(X) &\iff \forall \chi\, [\mathrm{Name}(\chi) \Rightarrow p \Vdash \psi(\chi)]. \end{aligned}$$

Here χ ranges over all classes over H in N. By recursion on the finite prefix of class quantifiers we obtain a formula with the same prefix. □

We apply this lemma with $\tau = \sigma$. The hypotheses hold because $2^\rho = \sigma$ implies $\sigma^{<\sigma} = \sigma$, and because every forcing in Γ_i is σ-cc and has a dense subset of size at most σ. Note that a name for a subset of H_ρ is coded by an *element* of H_σ, since it has hereditary size at most ρ. By contrast, a name for an arbitrary class over H_σ of the extension is coded by a *class* over H_σ. This difference is the reason for the drop in complexity in Lemma 6.6.

6.3 The translation T_i

We consider second-order formulas in prenex form. Using a fixed pairing function, we contract every block of class quantifiers of the same kind to a single class quantifier. The translation T_i is defined by recursion on the number of class quantifiers. The base case consists of the formulas with at most one class quantifier, that is, of the first-order formulas (with class parameters) and the $\Sigma^2_1(H_\rho)$ and $\Pi^2_1(H_\rho)$ formulas.

Definition 6.3. *Let N be a model of $(*)$, and let the class parameters belong to N.*

1. *If φ is a first-order, $\Sigma^2_1(H_\rho)$ or $\Pi^2_1(H_\rho)$ formula, let $T_i(\varphi)$ be its first-order translation over H_σ: relativize the first-order quantifiers to the definable set H_ρ, and replace the class quantifier over H_ρ, if there is one, by a first-order quantifier over the subsets of H_ρ, all of which belong to H_σ. The class parameters of φ are elements of H_σ and are used as such.*

2. *For formulas with more than one class quantifier put*

$$T_i(\neg\varphi) = \neg T_i(\varphi).$$

3. *If $\exists X\, \varphi$ has at least two class quantifiers, let $T_i(\exists X\, \varphi)$ assert that there are an iteration code with last forcing $Q \in \Gamma_i$, a condition $q \in Q$, and a Q-name $\dot{X}$ for a subset of H_ρ such that*

$$q \Vdash_Q T_i(\varphi)(\dot{X}).$$

A universal class quantifier is treated as $\neg\exists\neg$. In (3), the forcing relation is the usual forcing relation for the formula $T_i(\varphi)$, which has already been defined; its class quantifiers range over all classes over H_σ of the generic extension.

Lemma 6.4. *Let N be a model of $(*)$, let P be forcing equivalent in N to a member of Γ_i, and let G_P be P-generic over N. Then for every formula φ as above with parameters in N,*

$$N \models T_i(\varphi) \quad \Longleftrightarrow \quad N[G_P] \models T_i(\varphi).$$

Proof. Transfer the generic filter to a representative and assume $P \in \Gamma_i$. We argue by induction on the number of class quantifiers of φ. In the base case, $T_i(\varphi)$ is equivalent to a first-order, Σ^2_1 or Π^2_1 statement over H_ρ; since P is $< \rho$-closed, H_ρ is the same in N and $N[G_P]$, and Fact 2.8 gives the equivalence. The step for negation is immediate.

Consider $\exists X\, \varphi$ with at least two class quantifiers, and put $\psi = T_i(\varphi)$. Suppose first that $Q, q, \dot{X} \in N$ witness $T_i(\exists X\, \varphi)$ in N. In $N[G_P]$ let $e : Q \to$

$\widehat{Q}$ be the dense embedding into the lifted iteration given by Lemma 6.1. For any $\widehat{Q}$-generic filter over $N[G_P]$ containing $e(q)$, let G_Q be the induced Q-generic filter and put $X = \dot{X}^{G_Q}$. The filters G_P, G_Q are product-generic over N. Thus $N[G_Q] \models \psi(X)$, and P is forcing equivalent in $N[G_Q]$ to a member of Γ_i by the same lemma. By induction $N[G_Q][G_P] \models \psi(X)$. Hence $\widehat{Q}$, $e(q)$ and the $\widehat{Q}$-name for X which corresponds to $\dot{X}$ under e witness $T_i(\exists X\, \varphi)$ in $N[G_P]$.

Conversely, suppose that $T_i(\exists X\, \varphi)$ holds in $N[G_P]$. Let $\dot{Q}, \dot{q}, \ddot{X} \in N$ be names for witnesses, and choose $p \in G_P$ which forces that they are as required. By Lemma 6.1, choose $S \in \Gamma_i$ forcing equivalent to $(P \restriction p) * (\dot{Q} \restriction \dot{q})$. Under this equivalence, $\ddot{X}$ corresponds to an S-name $\dot{X}^*$ with $1_S \Vdash_S \psi(\dot{X}^*)$. Thus $S, 1_S, \dot{X}^*$ witness $T_i(\exists X\, \varphi)$ in N. □

Lemma 6.5. *Let $C \in \mathcal{C}_i$. Then for every second-order formula φ over H_ρ and all parameters $\vec{a} \in N_C$,*

$$W_i \models \varphi(\vec{a}) \quad \iff \quad N_C \models T_i(\varphi)(\vec{a}).$$

Proof. Recall that $W_i = B[G_{D_{\geq i}}]$ and $N_C = W_{i+1}[G_C] \subseteq W_i$. We use the following facts.

(i) W_i is an extension of N_C by a $< \rho$-closed forcing (Lemma 4.7).

(ii) For every $X \subseteq H_\rho$ in W_i there is $C' \in \mathcal{C}_i$ with $C \subseteq C'$ and $X \in N_{C'}$, and the quotient from N_C to $N_{C'}$ is forcing equivalent to a member of Γ_i (Lemma 5.7(a)).

(iii) For every forcing $Q \in N_C$ which belongs to Γ_i in N_C and every $q \in Q$ there are $C \subseteq C_0 \subseteq C'$ in $\mathcal{C}_i$ and a filter H with $q \in H$ which is Q-generic over N_{C_0}, such that $N_{C'} = N_{C_0}[H]$ (Lemma 5.7(b)).

We argue by induction on the number of class quantifiers of φ, simultaneously for all $C \in \mathcal{C}_i$ and all parameters. The base case follows from (i) and Fact 2.8, and the step for negation is immediate.

Suppose that $W_i \models \exists X \varphi(X, \vec{a})$. By (ii), choose C' such that a witness X belongs to $N_{C'}$. The induction hypothesis gives $N_{C'} \models T_i(\varphi)(X, \vec{a})$. Use (ii) to choose $R \in \Gamma_i$ representing the quotient, with $N_{C'} = N_C[G_R]$. By the forcing theorem there are $q \in G_R$ and an R-name $\dot{X}$ with $q \Vdash_R T_i(\varphi)(\dot{X}, \vec{a})$. Thus $(R, q, \dot{X})$ witnesses $N_C \models T_i(\exists X \varphi)(\vec{a})$.

Conversely, let $Q, q, \dot{X} \in N_C$ witness $T_i(\exists X \varphi)(\vec{a})$ in N_C, and choose C_0, C' and H as in (iii). By (ii), applied to $C \subseteq C_0$, there are $S \in \Gamma_i$ in N_C and an S-generic filter g over N_C with $N_{C_0} = N_C[g]$. Since H is Q-generic over $N_C[g]$, the pair (g, H) is generic for $S \times Q$ over N_C; in particular H is Q-generic over N_C, and g is S-generic over $N_C[H]$. Put $X = \dot{X}^H$. Since $q \in H$,

$$N_C[H] \models T_i(\varphi)(X, \vec{a}).$$

By Lemma 6.1(1), S is forcing equivalent in $N_C[H]$ to a member of Γ_i. Hence Lemma 6.4, applied to the model $N_C[H]$ and the formula φ, gives

$$N_{C'} = N_C[H][g] \models T_i(\varphi)(X, \vec{a}), \qquad \text{and hence} \qquad W_i \models \varphi(X, \vec{a})$$

by the induction hypothesis for C'. Hence $W_i \models \exists X \varphi(X, \vec{a})$. □

Lemma 6.6. *For $k \geq 2$ the translation can be defined uniformly so that*

$$T_i(\Sigma^2_k(H_{\omega_i})) \subseteq \Sigma^2_{k-1}(H_{\omega_{i+1}}),$$
$$T_i(\Pi^2_k(H_{\omega_i})) \subseteq \Pi^2_{k-1}(H_{\omega_{i+1}}).$$

The translated formulas have no parameters for $\vec{T}$, for the clubs added by $C_{\vec{T}}$, or for the bookkeeping function.

Proof. Since $\mathcal{P}(H_\rho) \subseteq H_\sigma$ and H_ρ, ρ are definable in H_σ, the first-order formulas and the $\Sigma^2_1(H_\rho)$ and $\Pi^2_1(H_\rho)$ formulas are translated into first-order formulas over H_σ, and their class parameters are elements of H_σ.

For $\exists X \varphi \in \Sigma^2_k(H_\rho)$, we express $T_i(\exists X \varphi)$ by means of a single existential class quantifier $\exists Z$, where Z codes four classes

$$Z_0 = R, \qquad Z_1 = \chi, \qquad Z_2 = \{q\}, \qquad Z_3 = \{a\},$$

by a fixed pairing function. Here $R, \chi \subseteq H_\sigma$ are a code of Q and a witness for it, $q, a \in H_\sigma$, and a codes the name $\dot{X}$. By Proposition 5.9, the condition $\mathrm{Wit}(R, \chi)$ is first order. The conditions on q and a are also first order. The elements q and a are uniquely determined by Z, independently of the classes quantified later, so the first-order quantifiers which define them from Z can be moved inside the remaining class quantifiers.

Lemma 6.2 requires a dense subset of Q which is given as a preorder on σ. We use the preorder induced by the last map e_δ of the code R. Its order is not a component of R, but it is computed correctly in the model coded by χ (Lemma 5.8); hence it is first-order definable over H_σ from R and χ, as noted after Proposition 5.9. By the induction hypothesis and Lemma 6.2, the statement $q \Vdash_Q T_i(\varphi)(\dot{X})$ is therefore first order over H_σ for $k = 2$, and $\Pi^2_{k-2}(H_\sigma)$ for $k > 2$. Together with the leading quantifier $\exists Z$ this gives a $\Sigma^2_{k-1}(H_\sigma)$ formula in both cases. Taking negations gives the assertion for Π^2_k.

All these formulas refer to the class Γ_i of the model in which they are evaluated. The model coded by χ is used only in $\mathrm{Wit}(R, \chi)$ and in the definition of the preorder on σ; all other class quantifiers range over all classes over H_σ. The translation is defined by recursion on the formula, using only the definable cardinals ρ, σ and the class Γ_i; hence it is recursive and introduces no parameters. □

6.4 Reduction to $B_{\vec{T}}$

Proposition 6.7. *For every Π^2_n formula A on the reals there is a $\Pi^2_1(H_{\omega_n})$ formula Φ_A, obtained recursively from A and independent of $\vec{T}$, such that*

$$B_{\vec{T}}[G_{D_{\vec{T}}}] \models A(a) \quad \iff \quad B_{\vec{T}} \models \Phi_A(a) \tag{1}$$

for all reals a. The corresponding statement holds for Σ^2_n formulas, with $\Sigma^2_1(H_{\omega_n})$ in place of $\Pi^2_1(H_{\omega_n})$. The only parameters of $\Phi_A(a)$ are the real parameters a.

Proof. Apply Lemmas 6.5 and 6.6 successively for $i = 1, \dots, n-1$, each time with $C = \varnothing$. At step i we pass from $W_i = B_{\vec{T}}[G_{D_{\geq i}}]$ to $N_\varnothing = W_{i+1} = B_{\vec{T}}[G_{D_{\geq i+1}}]$. This model satisfies $(*)$ for $\rho = \omega_i$, because $B_{\vec{T}}$ satisfies GCH and $D_{\geq i+1}$ is $< \omega_{i+1}$-closed. At each step the complexity drops by one and the domain of the first-order quantifiers increases from H_{ω_i} to $H_{\omega_{i+1}}$.

The resulting statement is $\Pi^2_1(H_{\omega_n})$, and it is evaluated in $W_n = B_{\vec{T}}[G_{D_{\geq n}}]$. The forcing $D_{\geq n}$ is $< \omega_n$-closed, so Fact 2.8 shows that the statement holds in W_n if and only if it holds in $B_{\vec{T}}$. For $n = 1$ only this last step is needed. The same argument applies to Σ^2_n formulas. The class Γ_i and the formula Wit do not depend on $\vec{T}$, and the translation is defined by recursion on formulas. Hence Φ_A is obtained recursively from A and has no parameters other than a. □

7 The formulas Code_n

7.1 The case $n = 1$

Put

$$\mathrm{Code}_1(t) \iff \exists C \subseteq \omega_1 \, [C \text{ is club and } C \cap S^1_t = \varnothing].$$

By Proposition 3.7(S2), Code_1 is Σ^2_1 over HC.

7.2 Iterated almost disjoint coding

Recall that $\langle B^r_\eta : \eta < \omega_{r+1} \rangle$ is the canonical constructible almost disjoint family of subsets of ω_r. For $X \subseteq \omega_1$ write $Z_1(X) = X$ and

$$Z_{r+1}(X) = \{\eta < \omega_{r+1} : Z_r(X) \cap B^r_\eta \text{ is bounded in } \omega_r\}.$$

The set $Z_r(X)$ belongs to every model considered below which contains X, and its definition is absolute between these models.

Lemma 7.1. *Let $V \subseteq V'$ be transitive models of* ZFC*, let $P \in V$ be a forcing, and put $T = \mathrm{Term}^V(P, \mathbb{A}_\rho(\dot{E}))$ for a P-name $\dot{E} \in V$, where ρ, P and $\dot{E}$ satisfy the hypotheses of Subsection 2.7 in V. Suppose $H \in V'$ is T-generic over V and G_P is P-generic over V'. Then (G_P, H) is $P \times T$-generic over V, and $V'[G_P]$ contains a subset of ρ which codes $\dot{E}^{G_P}$ by almost disjoint coding.*

Proof. Since $V[H] \subseteq V'$, the filter G_P is P-generic over $V[H]$. The two-step iteration $T * \check{P}$ is the product $T \times P$. Apply the projection $P \times T \longrightarrow P * \mathbb{A}_\rho(\dot{E})$ from Theorem 2.10. The generic filter induced by this projection gives the required set. Whether the intersection of two given subsets of an ordinal is bounded is absolute between transitive models. □

Lemma 7.2. *Fix $j \geq 1$ and work in $W_j = B[G_{D_{\geq j}}]$. Let C be a set of coordinates of levels below j, with at most ω_{k+1} coordinates of level k for each $k < j$, such that $C \cup \bigcup_{k \geq j} I_k$ is closed. Let P be the preorder on ω_j presenting the quotient $D_{C \cup \bigcup_{k \geq j} I_k} / G_{D_{\geq j}}$ as in Lemma 5.5. Then P is ω_j-cc. For $j = 1$ it is forcing equivalent to the trivial forcing.*

If $\dot{E}$ is a nice P-name for a subset of ω_{j+1}, then the term forcing $\mathrm{Term}(P, \mathbb{A}_{\omega_j}(\dot{E}))$ is appended in the construction of D at a coordinate of level j, after all coordinates on which P and $\dot{E}$ depend. Its generic filter and the P-generic filter induced by G_D give a generic filter for $P * \mathbb{A}_{\omega_j}(\dot{E})$.

Proof. The quotient is a finite iteration of the restrictions to C of the quotients at levels $j-1, \ldots, 1$. Each is ω_j-cc by Proposition 4.9, so P is ω_j-cc.

By Lemma 5.5, P and $\dot{E}$ belong to $B[G_H]$ for a suitable closed set H of coordinates of levels at least j. In this model $\omega_j^{<\omega_j} = \omega_j$ and $\omega_{j+1}^{<\omega_j} = \omega_{j+1}$ hold, since neither the forcing $D_{>j}$ nor an initial segment of the quotient at level j adds sequences of ordinals of length $< \omega_j$. Theorem 2.10 therefore shows that the term forcing, defined in $B[G_H]$, is $< \omega_j$-compact and ω_j-linked and has a dense subset of size at most ω_{j+1}.

By Proposition 5.6(2), this term forcing is appended at some stage. Let H' be its generic filter. The filter induced on P is P-generic over W_j, and W_j contains H'; here we use that the restriction to the closed set $C \cup \bigcup_{k \geq j} I_k$, which contains all coordinates of levels at least j, is a projection. By Lemma 7.1, the two filters are generic for the product over the model in which the term forcing was defined, and the projection of Theorem 2.10 gives a generic filter for $P * \mathbb{A}_{\omega_j}(\dot{E})$. Note that the restriction to a set of coordinates which does not contain all coordinates of levels at least j would not give this. □

Proposition 7.3. *Fix $j \geq 1$, put $W_j = B[G_{D_{\geq j}}]$, and let*

$$C \subseteq \bigcup_{1 \leq k < j} I_k, \qquad |C \cap I_k| \leq \omega_{k+1} \quad (1 \leq k < j),$$

be such that $C \cup \bigcup_{k \geq j} I_k$ is closed. Put

$$Q_j = D_{C \cup \bigcup_{k \geq j} I_k} / G_{D_{\geq j}}.$$

Then for every $E \subseteq \omega_{j+1}$ in $W_j[G_{Q_j}]$, the final model $B[G_D]$ contains a set $X \subseteq \omega_1$ with $Z_{j+1}(X) = E$.

Proof. For $1 \leq r \leq j$, put $C_r = C \cap \bigcup_{1 \leq k < r} I_k$, and define W_r and Q_r as above, with C_r and r in place of C and j. Since the memory of a coordinate contains only coordinates of the same or higher levels, $C_r \cup \bigcup_{k \geq r} I_k$ is closed. By Lemma 7.2, we can identify Q_r with a preorder P_r on ω_r, by means of an enumeration of its conditions, such that

$$|P_r| \leq \omega_r, \qquad P_r \text{ is } \omega_r\text{-cc};$$

the forcing Q_1 is trivial. Starting with $E_{j+1} = E$, we construct by downward recursion sets $E_r \in W_r[G_{Q_r}]$, $E_r \subseteq \omega_r$, for $r = j, \ldots, 1$, such that

$$E_{r+1} = \{\eta < \omega_{r+1} : E_r \cap B^r_\eta \text{ is bounded in } \omega_r\}.$$

At step r, a nice P_r-name $\dot{E}_{r+1}$ for E_{r+1} is coded by at most ω_{r+1} ordinals, since its antichains have size $< \omega_r$. By Lemma 5.5, P_r and $\dot{E}_{r+1}$ belong to $B[G_{H_r}] \subseteq W_r$ for a closed set $H_r \in B$ of coordinates of levels at least r, with the same order on P_r, and the term forcing $T_r = \mathrm{Term}^{B[G_{H_r}]}(P_r, \mathbb{A}_{\omega_r}(\dot{E}_{r+1}))$ is appended at a later stage, at level r. Since the restriction to $C_r \cup \bigcup_{k \geq r} I_k$ is a projection, the filter induced on Q_r is generic over W_r. It yields a P_r-generic filter over W_r, and W_r contains the generic filter for T_r. Lemma 7.1 gives the required set E_r. Note that the enumeration which identifies Q_r with P_r need not belong to $B[G_{H_r}]$.

For $r > 1$ we have

$$W_r[G_{Q_r}] \subseteq W_{r-1}[G_{Q_{r-1}}],$$

because $C_r \subseteq C_{r-1} \cup I_{r-1}$; in particular W_{r-1} contains the generic filter on the coordinates of level $r-1$ in C and the generic filters for the term forcings used so far. Thus the same procedure applies to E_r and C_{r-1}. At each step we choose a stage of the bookkeeping which is later than all stages used before. For $r = 1$ the forcing Q_1 is trivial, so T_1 is the almost disjoint coding forcing itself. With $X = E_1$, the recursion gives $Z_r(X) = E_r$ for $1 \leq r \leq j+1$, hence $Z_{j+1}(X) = E$. □

Lemma 7.4. *If $t \notin T_n$, the set S^n_t remains stationary in $B_{\vec{T}}[G_{D_{\vec{T}}}]$.*

Proof. The set S^n_t is stationary in $B_{\vec{T}}$ by Lemma 3.6. First force with $D_{\geq n}$, which is $< \omega_n$-closed. The statement that there is a club disjoint from S^n_t is $\Sigma^2_1(H_{\omega_n})$, with S^n_t as a class parameter. Fact 2.8 shows that no such club is added.

The remaining quotients at the finitely many levels below n form an ω_n-cc iteration. Such a forcing preserves stationary subsets of ω_n: given a name for a club, for each $\alpha < \omega_n$ take a maximal antichain deciding the first point of this club above α. There are fewer than ω_n possible values; let $f(\alpha) < \omega_n$ be a bound. The set of limit ordinals closed under f is a club in the ground model which is forced to be contained in the given club, and it meets S^n_t. For $n = 1$ there is no quotient below level n. □

7.3 Codes of models

In this subsection we use only the following properties of the final model $B[G_D]$.

(C1) All cardinals are preserved, and for every finite family of classes over HC there is a closed set $C \in B$ with $|C \cap I_k| \leq \omega_{k+1}$ for every k such that the family belongs to $B[G_C]$.

(C2) Let $r \geq 2$, and let $E \subseteq \omega_r$ belong to a model $W_{r-1}[G_{Q_{r-1}}]$, where Q_{r-1} is as in Proposition 7.3 for $j = r-1$. Then there is a class $X \subseteq \omega_1$ with $Z_r(X) = E$.

(C3) Each stationary set S^n_t with $t \notin T_n$ is stationary in the final model. For $t \in T_n$ the final model contains a club $C \subseteq \omega_n$ disjoint from S^n_t and the set $P^n_t(C)$ of Lemma 3.3; both belong to $B_{\vec{T}}$.

Here (C1) is Theorem 5.10(M3) and (M4), (C2) is Proposition 7.3, and (C3) is Lemma 7.4 together with the definition of $C_{\vec{T}}$. We refer to the models $W_{r-1}[G_{Q_{r-1}}]$ in (C2) as the *models of (C2) for* r. Note that if C is as in (C1) and $j \geq 1$, then $C \cap \bigcup_{k<j} I_k$ satisfies the hypotheses of Proposition 7.3, and $B[G_C]$ is contained in the corresponding model $W_j[G_{Q_j}]$. All these models have the same cardinals as L.

We now fix the coding of models by sets of ordinals. Fix once and for all constructible functions which code finite tuples of ordinals, finite sequences, and terms of the language introduced below by ordinals. For every infinite cardinal τ, these functions map objects built from ordinals below τ to ordinals below τ.

The language $\mathcal{L}$ is defined in ω steps. Let $\mathcal{L}_0$ consist of $\in$, of two function symbols F_j, H_j for each $j < r$, and of the constants

$$b_0, \ldots, b_{r+2}, \quad z_1, \ldots, z_k, \quad t_1, \ldots, t_l, \quad d_1, \ldots, d_m.$$

Given $\mathcal{L}_m$, let $\mathcal{L}_{m+1}$ be obtained by adding a function symbol f_φ for every formula of $\mathcal{L}_m$ of the form $\exists v\, \varphi(v, \vec{u})$, with one argument for each of the free variables $\vec{u}$. Put $\mathcal{L} = \bigcup_{m<\omega} \mathcal{L}_m$. This language is countable and recursive, and every formula $\exists v\, \varphi(v, \vec{u})$ of $\mathcal{L}$ has its function symbol f_φ in $\mathcal{L}$; we call the functions which interpret these symbols Skolem functions. In addition we use the constants

$$c_\alpha, \ g_\alpha \qquad (\alpha < \omega_r).$$

Here the constants c_α are intended to denote the ordinals $\alpha < \omega_r$, the constants g_α name all elements of the model, the b_j denote the cardinals ω_j, the z_i denote subsets of ω_r which are of the form $Z_r(X)$, and the t_i denote real parameters. The d_i are finitely many further constants, for objects such as the order types and isomorphisms occurring in a witness below.

A *code* of a model for this language, in which every element is the value of a closed term, is a set of ordinals below ω_r which consists of three disjoint parts, coding the equality relation between closed terms, the membership relation between closed terms, and the satisfaction relation of the model. An instance of the satisfaction relation is given by the Gödel number of a formula of the countable language and a finite sequence of closed terms which are assigned to its free variables; thus the uncountably many constants occur only in these assignments, and not in the recursive enumeration of formulas. We write $q_{\mathrm{eq}}(u, v)$, $q_{\mathrm{mem}}(u, v)$ and $q_{\mathrm{sat}}(e, \vec{u})$ for the ordinals which code the corresponding instances of equality, membership and satisfaction. These functions, and the function assigning to a function symbol and codes of terms the code of the resulting term, are fixed in advance; in particular they do not depend on the model which is coded. Similarly, a *code of an ordinal* consists of two disjoint parts, coding the field and the strict order of a wellorder; the field is coded separately so that the ordinal 0 is included.

For $0 < a < \omega_{j+1}$ let $f_{j,a}$ be the $<_L$-least surjection from ω_j onto a. The Σ_1 definition of the initial segments of $<_L$ in Section 3.1 gives a functional Σ_1 definition of the map $a \mapsto f_{j,a}$, uniformly in j: one asserts the existence of a surjection f and of the set of all its $<_L$-predecessors, such that no $<_L$-predecessor of f is a surjection with the same domain and range. The fixed coding functions have Σ_1 definitions of the same kind. The theories below require these Σ_1 definitions to define total functions over L_β on the relevant domains. By the upward absoluteness of Σ_1 formulas, the values computed over L_β are then the true values.

Definition 7.5. *For each $r \geq 1$ let T_r be the theory in the above language which consists of ZFC^- and the following axioms.*

(i) $b_j = \omega_j$ *for* $0 \leq j \leq r + 2$, *and* $b_{r+2} = \omega^L_{r+2}$. *We write* $\rho = b_r$ *and* $\beta = b_{r+1}$. *The set* L_β *exists.*

(ii) The definitions Φ_D and Φ_B of the canonical diamond sequence and of the canonical almost disjoint family at ρ are total and functional over L_β on exactly the domains specified in Definition 3.2. The same holds for the definitions of the fixed coding functions and their inverses, and their restrictions to β satisfy the corresponding identities for the even and odd parts of subsets of β. For each $j \leq r$, the definition of the map $a \mapsto f_{j,a}$, evaluated over the same L_β, is total and functional for $0 < a < b_{j+1}$.

(iii) Every z_i is a subset of ρ, and every t_i is a subset of ω.

(iv) For $j < r$ and $0 < a < b_{j+1}$, $F_j(a)$ is a function from b_j onto a, and, for $u \in a$,

$$H_j(a, u) \in b_j, \qquad F_j(a)(H_j(a, u)) = u.$$

The axiom schemes of ZFC^- *are taken for the formulas of the language of set theory only. The finitely many axioms listed here and the axiom schemes are given by a recursive set of Gödel numbers. The Skolem functions are additional functions on the model; they need not be elements of the model or definable in it, and they are subject only to the conditions on the satisfaction relation stated in the proof of Lemma 7.7.*

Note that a transitive model N of T_r with $b_r^N = \omega_r$ is suitable for $r+1$ in the sense of Definition 3.2.

We shall define formulas $V_r(Y; \vec{X}, \vec{t})$ expressing that the set decoded from Y is a code of a well-founded model of T_r with certain properties, and we shall prove two things: every Y satisfying V_r yields such a model (soundness), and every such model which belongs to a model of (C2), and has cardinality ω_r there, is obtained in this way (completeness). The models used for completeness are transitive models of T_r of cardinality ω_r in which $b_j = \omega_j$ holds for all $j \leq r$; they are obtained as transitive collapses of the elementary submodels considered below. In the soundness direction, the fact that the constants b_j, $j \leq r$, are interpreted by the true cardinals ω_j will be a consequence of the formula V_r, and not an additional assumption.

Lemma 7.6. *There is a first-order formula* $V_1(Y; X_1, \dots, X_k, t_1, \dots, t_l)$ *over* HC *with the following properties. If* $V_1(Y; \vec{X}, \vec{t})$ *holds, then* $Y \subseteq \omega_1$ *is a code of a well-founded extensional model, together with its satisfaction relation and its Skolem functions, whose transitive collapse* N *satisfies* T_1*,* $b_1^N = \omega_1$*,* $z_i^N = X_i$ *and* $t_i^N = t_i$*. Conversely, every transitive model of this kind which has cardinality* ω_1 *has a code* Y *satisfying* V_1*.*

Proof. The equality relation, the membership relation and the satisfaction relation are given by the three parts of the class $Y \subseteq \omega_1$. The finitely many conditions which express that Y is the code of a model, and the Tarski conditions for the satisfaction relation, quantify over countable ordinals only, and hence are first order over HC. Every descending ω-sequence of closed terms is coded by an element of HC, so well-foundedness is first order as well. The model N is the transitive collapse of the quotient of the set of closed terms by the equality relation.

We require $c_0 = \varnothing$ and $c_{n+1} = c_n \cup \{c_n\}$ for $n < \omega$. Thus the constants c_n, $n < \omega$, are interpreted by the natural numbers. For each $0 < \alpha < \omega_1$, we require that c_α is an ordinal, that $F_0(c_\alpha)$ has domain ω and range c_α, and that H_0 gives the corresponding preimages. We require, for $n < \omega$,

$$F_0(c_\alpha)(c_n) = c_{f_{0,\alpha}(n)}.$$

The surjections $f_{0,\alpha}$ are hereditarily countable, so these conditions are first order over HC. By induction on α, we see that c_α is interpreted by α. We require all these constants to belong to b_1. Since N satisfies $b_1 = \omega_1$, it

follows that b_1^N is the true ω_1: it cannot be smaller because it contains every countable ordinal; if it were larger, N would contain a surjection from ω onto the true ω_1.

The conditions $z_i \subseteq b_1$ and $c_\alpha \in z_i \leftrightarrow \alpha \in X_i$ ensure $z_i^N = X_i$. The analogous conditions for the constants c_n ensure $t_i^N = t_i$. Conversely, given a transitive model as in the statement, interpret the constants g_α by an enumeration of its elements and the constants c_α by the ordinals α, put $F_0(\alpha) = f_{0,\alpha}$ for $0 < \alpha < \omega_1$, and choose interpretations of the function symbols f_φ by recursion on the languages $\mathcal{L}_m$. This gives a code Y satisfying V_1. □

We now define the corresponding formulas for $r \geq 2$ by simultaneous recursion. Finitely many classes which are quantified together are always combined into one class by the fixed pairing function. A variable E is intended to range over codes of ordinals: we write $|E|_r$ for the order type of the wellorder coded by $Z_{r-1}(E)$. The notation $|E|_r$ is used only in our arguments, not in the formulas themselves. We write Π^2_0 for the class of first-order formulas.

Lemma 7.7. *For $r \geq 2$ there are formulas*

$$O_r, C_r^+, C_r^- \in \Pi^2_{r-2}, \qquad V_r \in \Pi^2_{r-1},$$

defined uniformly in r, with the following properties in the final model.

(a) $O_r(E)$ holds if and only if $Z_{r-1}(E)$ is a code of a wellorder of a subset of ω_{r-1}, and $\{|E|_r : O_r(E)\} = \omega_r$.

(b) If $O_r(E)$ and $a = |E|_r$, then

$$a \in Z_r(X) \iff \exists H\, C_r^+(X, E; H),$$
$$a \notin Z_r(X) \iff \exists H\, C_r^-(X, E; H).$$

In particular, exactly one of the two formulas on the right has a witness H.

(c) If $V_r(Y; \vec{X}, \vec{t})$ holds, then $Z_r(Y)$ is a code of a well-founded extensional model whose transitive collapse N satisfies T_r and

$$b_j^N = \omega_j \ (j \leq r), \qquad z_\ell^N = Z_r(X_\ell), \qquad t_\ell^N = t_\ell.$$

The same holds if finitely many further constants are required to denote sets of the form $Z_r(X)$.

(d) Conversely, let N be a transitive model of ZFC^-, with interpretations of the constants of $\mathcal{L}_0$, which satisfies the axioms (i)–(iii) of T_r and the conditions displayed in (c). Suppose that N belongs to a model of

(C2) for r and has cardinality ω_r in that model. Then the constants c_α, g_α and the function symbols can be interpreted in such a way that N becomes a model of T_r in which every element is the value of a closed term, and there is Y with $V_r(Y; \vec{X}, \vec{t})$ such that $Z_r(Y)$ is the code of this model.

Proof. We argue by induction on r. For each r we define O_r, then C_r^+ and C_r^- together with auxiliary formulas W_r^θ, and finally V_r. For $r = 2$ the formulas of the previous step are replaced by V_1 and by first-order statements about subsets of ω_1.

The formula O_r. $O_r(E)$ expresses that the relation coded by $Z_{r-1}(E)$ is a wellorder. For $r = 2$ we have $Z_1(E) = E$, and this is first order over HC. For $r > 2$, membership of an ordinal below ω_{r-1} in $Z_{r-1}(E)$ is expressed by means of a code E' of this ordinal and a witness H for $C^+_{r-1}(E, E'; H)$ or $C^-_{r-1}(E, E'; H)$. The axioms of linear orders, which concern finitely many elements, are required to hold for all codes of ordinals and all such witnesses. Well-foundedness is expressed by requiring that there is no class which codes a descending ω-sequence of codes of ordinals together with witnesses, in the sense of C^+_{r-1}, showing that each of them precedes the previous one in the order coded by $Z_{r-1}(E)$. Since the formulas of the previous step are Π^2_{r-3}, we obtain $O_r \in \Pi^2_{r-2}$. Every $a < \omega_r$ is the order type of a constructible wellorder of a subset of ω_{r-1}, whose code is a constructible subset of ω_{r-1}. If $r > 2$, (C2) for $r - 1$ gives E such that $Z_{r-1}(E)$ is this code; for $r = 2$ the code itself is a subset of ω_1. This proves (a).

The formulas $C_r^\pm$ and W_r^θ. In $C_r^\varepsilon(X, E; H)$, for $\varepsilon \in \{+, -\}$, the class H is required to code the following objects: a class Y' which satisfies V_{r-1} and thus yields a transitive model N of T_{r-1}; constants of this model which denote the sets decoded from the given classes, an ordinal a and an isomorphism π; and, for each of the finitely many instances of the satisfaction relation of N which are needed below, a witness in the sense of C^+_{r-1} that this instance belongs to the code $Z_{r-1}(Y')$. In terms of the model N, this means

$$\begin{gathered} \rho = \omega_{r-1}, \qquad \beta = (\rho^+)^N, \\ z = Z_{r-1}(X) \in N, \qquad e = Z_{r-1}(E) \in N, \\ a < \beta, \qquad \pi : e \cong a, \end{gathered}$$

where π is an isomorphism between the wellorder coded by e and the ordinal a; the same applies if finitely many classes X_i, E_i are given. The formula V_{r-1} ensures that the constants denote the sets z and e. Since the witnesses for the instances of the satisfaction relation are part of H, no existential class quantifier is needed to express that these instances hold in N. Hence the conditions on the model, on the constants and on the instances of satisfaction are Π^2_{r-2}. For $r = 2$, the model is coded directly by a subset of ω_1 and these conditions are first order.

In addition, $C_r^\varepsilon(X, E; H)$ requires that N satisfies

$$\sup(z \cap B_a^{r-1}) < \rho \quad (\varepsilon = +), \qquad \sup(z \cap B_a^{r-1}) = \rho \quad (\varepsilon = -).$$

Since the definition of the almost disjoint family is total and functional over L_β^N and $a < \beta$, the upward absoluteness of Σ_1 formulas shows that N computes B_a^{r-1} correctly. Whether $z \cap B_a^{r-1}$ is bounded in ρ is absolute between N and the universe, and $a = |E|_r$ because $\pi \in N$. Hence a witness for $C_r^+(X, E; H)$ shows $a \in Z_r(X)$, and a witness for $C_r^-(X, E; H)$ shows $a \notin Z_r(X)$. In the same way we obtain formulas $W_r^\theta \in \Pi^2_{r-2}$ such that $\exists K\, W_r^\theta(\vec{E}; K)$ expresses that the relation θ holds between the ordinals $|E_1|_r, |E_2|_r, \ldots$, for each of the following relations θ: equality and order of ordinals, being smaller than the true ω_j (for $j < r$), the graphs of the fixed coding functions and of the functions $f_{j,a}$, and the recursive relations on ω which describe the syntax of $\mathcal{L}$ and the axioms of T_r. These relations are absolute between N and the universe, in contrast to arbitrary statements which hold in N.

For the converse implications in (b), let X and E be given, with $O_r(E)$ and $a = |E|_r$. By (C1), X and E belong to $B[G_C]$ for a closed set C as in (C1). If $r > 2$, we work in the corresponding model of (C2) for $r - 1$, which contains $B[G_C]$; if $r = 2$, we work in the final model. Let $\mathcal{G}$ be the set of the graphs of the fixed coding functions, including the pairing function and the coding of terms. With $\rho = \omega_{r-1}$, $z = Z_{r-1}(X)$, $e = Z_{r-1}(E)$ and the isomorphism $\pi : e \cong a$, choose

$$M \prec H_\vartheta, \quad |M| = \rho, \quad \rho \cup (a+1) \subseteq M,$$
$$\rho, z, e, a, \pi, L_{\omega_r}, \mathrm{Sat}(L_{\omega_r}), \mathcal{G} \in M,$$

for a sufficiently large ϑ. Let $\bar\pi : M \to N$ be the transitive collapse. Condensation gives

$$\bar\pi(\rho) = \rho, \qquad \beta = \bar\pi(\omega_r) = (\rho^+)^N > a, \qquad \bar\pi(L_{\omega_r}) = L_\beta,$$

and by elementarity the definitions mentioned in T_{r-1} are total over L_β. The model N belongs to the model in which we work and has cardinality ρ there. For $r > 2$, part (d) of the induction hypothesis gives Y' satisfying V_{r-1} such that $Z_{r-1}(Y')$ is a code of N, and part (b) of the induction hypothesis gives the witnesses for the required instances of the satisfaction relation. For $r = 2$, a code of N is itself a subset of ω_1, and we use Lemma 7.6. This yields a witness H for C_r^+ or C_r^-, according to whether $a \in Z_r(X)$ or not. The same argument applies to finitely many classes and to the formulas W_r^θ. This proves (b).

The formula V_r. Membership of ordinals in $Z_r(Y)$ is again expressed by means of codes of ordinals and witnesses. The formula V_r is a conjunction of finitely many conditions, each of the following kind: for all ordinals $\alpha_1, \ldots, \alpha_m < \omega_r$ which stand in a given relation θ of the kind considered above, a given propositional combination A of statements of the form

$\alpha_i \in Z_r(Y)$ or $\alpha_i \in Z_r(X_\ell)$ is true. We express such a condition by saying that there is no witness for a violation of it. Here a witness for a violation consists of

- codes $E_1, \dots, E_m$ of ordinals, that is, classes with $O_r(E_i)$;
- a class K with $W_r^\theta(\vec{E}; K)$, which shows that the ordinals $|E_i|_r$ stand in the relation θ;
- signs $\varepsilon_i \in \{+, -\}$, one for each of the statements which occur in A, such that A is false when these statements are given the truth values ε_i, and classes H_i with $C_r^{\varepsilon_i}(Y, E_i; H_i)$ or $C_r^{\varepsilon_i}(X_\ell, E_i; H_i)$, respectively.

Since there are only finitely many choices of signs, the statement that $(\vec{E}, K, \vec{H})$ is a witness for a violation is a finite disjunction of conjunctions of Π^2_{r-2} formulas; we denote it by $\mathrm{Viol}_r(Y; \vec{E}, K, \vec{H})$. The condition is expressed by

$$\forall \vec{E}\, \forall K\, \forall \vec{H}\, \neg\, \mathrm{Viol}_r(Y; \vec{E}, K, \vec{H}), \qquad \mathrm{Viol}_r \in \Pi^2_{r-2},$$

which is Π^2_{r-1}. By (a), every ordinal below ω_r has a code. As shown in the proof of (b), every true instance of θ has a witness in the sense of W_r^θ, and for every code E exactly one of the two signs has a witness. Hence the displayed formula expresses the intended condition on $Z_r(Y)$. The ordinals $q_{\mathrm{eq}}(u, v)$, $q_{\mathrm{mem}}(u, v)$ and $q_{\mathrm{sat}}(e, \vec{u})$ and the codes of terms enter these conditions through the relation θ, which contains the graphs of the fixed coding functions.

In this form, V_r requires the axioms of equality, including the congruence conditions for membership, for the function symbols and for satisfaction; the Tarski conditions for the satisfaction relation, with existential quantifiers treated by means of the function symbols f_φ; and the axioms of T_r. Writing S for the part of the code which codes satisfaction, the conditions for the existential quantifier are

$$\begin{aligned} &S(\exists v\, \varphi(v, \vec{u})) \Rightarrow S(\varphi(f_\varphi(\vec{u}), \vec{u})), \\ &\quad S(\varphi(w, \vec{u})) \Rightarrow S(\exists v\, \varphi(v, \vec{u})) \quad \text{(for all closed terms } w\text{)}. \end{aligned}$$

Thus an element which satisfies an existential formula is always given by the term $f_\varphi(\vec{u})$, and no existential quantifier over closed terms is needed.

For well-foundedness, V_r requires that there is no class which codes, for every $n < \omega$, the following classes: codes E_n and E'_n of ordinals; a witness, in the sense of W_r^θ, that $|E_n|_r$ and $|E_{n+1}|_r$ are codes of closed terms u_n and u_{n+1} and that $|E'_n|_r = q_{\mathrm{mem}}(u_{n+1}, u_n)$; and a witness, in the sense of C_r^+, that $|E'_n|_r \in Z_r(Y)$. These countably many classes are combined into one class. The condition on this class is Π^2_{r-2}, since a quantifier over $n < \omega$ can be moved inside the class quantifiers. Hence the existence of such a class is a Σ^2_{r-1} statement, and its negation is Π^2_{r-1}.

Next, V_r requires, in the same form,

$$c_0 = \varnothing, \qquad c_{n+1} = c_n \cup \{c_n\}\ (n < \omega), \qquad c_\alpha \text{ is an ordinal.}$$

Thus the constants c_n, $n < \omega$, are interpreted by the natural numbers. For $j < r$, $0 < \alpha < \omega_{j+1}$ and $i < \omega_j$, it requires

$$\begin{gathered} c_\alpha \in b_{j+1}, \qquad F_j(c_\alpha) : b_j \twoheadrightarrow c_\alpha, \\ u \in c_\alpha \Rightarrow \big[H_j(c_\alpha, u) \in b_j \ \wedge\ F_j(c_\alpha)(H_j(c_\alpha, u)) = u\big], \\ F_j(c_\alpha)(c_i) = c_\gamma \quad (\gamma = f_{j,\alpha}(i)). \end{gathered}$$

The last equation is required for all γ and all witnesses, in the sense of W_r^θ, for $\gamma = f_{j,\alpha}(i)$. The condition $\alpha < \omega_{j+1}$ is expressed by W_r^θ if $j+1 < r$, and it holds for all codes of ordinals if $j+1 = r$; the condition $i < \omega_j$ is treated in the same way. The witnesses and the closed terms u are universally quantified. Note that the term $H_j(c_\alpha, u)$ provides the preimage of an element of c_α; therefore no existential quantifier over codes of ordinals is needed.

We now verify that these requirements have the intended effect. Let N be the transitive collapse of the model coded by $Z_r(Y)$. Then $b_0^N = \omega$. If $b_j^N = \omega_j$, where $j < r$, then induction on $0 < \alpha < \omega_{j+1}$ gives

$$c_\alpha^N = \{c^N_{f_{j,\alpha}(i)} : i < \omega_j\} = \{f_{j,\alpha}(i) : i < \omega_j\} = \alpha.$$

Thus $b_{j+1}^N \geq \omega_{j+1}$. If the inequality were strict, then, since $N \models b_{j+1} = (b_j)^+$, the model N would contain a surjection from ω_j onto ω_{j+1}. Hence $b_j^N = \omega_j$ for all $j \leq r$.

Finally, V_r requires $z_i \subseteq b_r$ and, for all codes E of ordinals and all witnesses for the two sides, that

$$|E|_r \in Z_r(X_i) \quad \Longleftrightarrow \quad q_{\mathrm{mem}}(c_{|E|_r}, z_i) \in Z_r(Y);$$

here the ordinal $q_{\mathrm{mem}}(c_{|E|_r}, z_i)$ is identified by W_r^θ. By (a) and (b), every ordinal below $\omega_r = b_r^N$ has a code and the corresponding witnesses exist, so we obtain $z_i^N = Z_r(X_i)$. The analogous conditions for the constants c_n, $n < \omega$, ensure $t_i^N = t_i$. All these conditions are Π^2_{r-1}. This proves (c).

For (d), let N be as in the statement, and work in the given model of (C2) for r. Interpret the constants g_α by an enumeration of N of length ω_r and the constants c_α by the ordinals $\alpha < \omega_r = b_r^N$. For $j < r$ and $0 < a < b_{j+1}^N$ put $F_j(a) = f_{j,a}$, which belongs to N by axiom (ii), and let $H_j(a, u)$ be the least preimage of u under $f_{j,a}$. Finally choose interpretations of the function symbols f_φ, by recursion on m for the symbols of $\mathcal{L}_{m+1}$, such that $f_\varphi(\vec{u})$ satisfies $\varphi(v, \vec{u})$ whenever some element does. The resulting structure is a model of T_r in which every element is the value of a closed term. Its code, consisting of the equality, membership and satisfaction relations, is a subset of ω_r which belongs to the model of (C2), and the structure satisfies all

conditions required by V_r. By (C2) there is Y in the final model such that $Z_r(Y)$ is this code. The witnesses are sound by (b): a witness for a violation of one of the conditions of V_r would show that the structure violates this condition, and a class as in the condition on well-foundedness would yield a descending $\in$-sequence in N. Hence $V_r(Y; \vec{X}, \vec{t})$ holds.

This completes the definition of O_r, of $C_r^{\pm}$ and W_r^{θ}, and of V_r, and the proof of their properties. The formulas are obtained recursively from r. The sets $Z_r(X)$ and the true cardinals ω_j were used above only to describe the meaning of the formulas; the formulas themselves refer to them only through the witnesses and the constants, and they have no parameters. Whenever countably many classes are needed, for sequences or for the axiom schemes, they are combined into one class, using $\omega_1 \times \omega = \omega_1$. □

7.4 The definition of Code_n for $n \geq 2$

For each $r \geq 1$ fix the following first-order formula $\chi_r(\rho, \beta, z, t)$ of the language of set theory. It asserts that there are sets P, C such that

$$P = \{\eta < \beta : z \cap B^r_\eta \text{ is bounded in } \rho\}, \qquad C = \mathrm{Even}(P),$$

where the almost disjoint family is computed over L_β by means of Φ_B, and such that $\psi_{\mathrm{club}}(\rho, \beta, C, t)$ holds. Let q_r be the ordinal which codes, in the coding fixed above, the instance of the satisfaction relation for the formula χ_r and the terms b_r, b_{r+1}, z_1, t_1. Thus q_r is determined by the fixed coding of formulas and terms; it does not depend on the model which is coded.

For $r = 1$, let $D_1(Y, t)$ be the first-order statement that q_1 belongs to the part of Y which codes satisfaction. For $r \geq 2$ put

$$D_r(Y, t) \quad \Longleftrightarrow \quad \exists E, H, K \, [W_r^{=q_r}(E; K) \wedge C_r^{+}(Y, E; H)],$$

where $W_r^{=q_r}(E; K)$ is the formula W_r^{θ} which expresses $|E|_r = q_r$. The three class variables can be combined into one. Hence D_1 is first order and $D_r \in \Sigma^2_{r-1}$ for $r \geq 2$. If $V_r(Y; X, t)$ holds and N is the transitive collapse of the model coded by $Z_r(Y)$, then

$$D_r(Y, t) \quad \Longleftrightarrow \quad N \models \chi_r(\rho, \beta, z, t),$$

where $\rho = b_r^N = \omega_r$, $\beta = b_{r+1}^N$ and $z = z_1^N = Z_r(X)$. Moreover, since the definition of the almost disjoint family is total over L_β, the set P defined in N is $Z_{r+1}(X) \cap \beta$.

For $r \geq 1$, put

$$\mathrm{Code}_{r+1}(t) \quad \Longleftrightarrow \quad \exists X \, \forall Y \, [V_r(Y; X, t) \longrightarrow D_r(Y, t)].$$

For $r = 1$, $V_1(Y; X, t) \to D_1(Y, t)$ is first order; for $r \geq 2$, V_r is Π^2_{r-1} and D_r is Σ^2_{r-1}. Thus $\mathrm{Code}_{r+1} \in \Sigma^2_{r+1}$, uniformly in r.

Proposition 7.8. *In the final model, for every $n \geq 1$ and $t \in I$,*

$$\mathrm{Code}_n(t) \quad \Longleftrightarrow \quad t \in T_n.$$

Proof. Let $n = 1$. If $t \in T_1$, then $C_{\vec{T}}$ adds a club disjoint from S^1_t, so $\mathrm{Code}_1(t)$ holds. If $t \notin T_1$, then S^1_t is stationary by (C3), so $\mathrm{Code}_1(t)$ fails.

Let $n = r + 1 \geq 2$ and $t \in T_n$. By (C3) there is a club $C \subseteq \omega_n$ disjoint from S^n_t such that $P = P^n_t(C) \subseteq \omega_n$ belongs to $B_{\vec{T}}$. By (C2) there is X with $Z_n(X) = P$. Suppose that $V_r(Y; X, t)$ holds, and let N be the corresponding transitive model. Then $\rho = b^N_r = \omega_r$, $\beta = b^N_{r+1} < \omega_n$, and $z = Z_r(X) \in N$. As noted above, the set $\{\eta < \beta : z \cap B^r_\eta \text{ is bounded in } \rho\}$, computed in N, is $P \cap \beta$; in particular $P \cap \beta \in N$. Since N is suitable for n, Lemma 3.3 gives $N \models \psi_{\mathrm{club}}(\rho, \beta, C \cap \beta, t)$. Thus $N \models \chi_r(\rho, \beta, z, t)$, and $D_r(Y, t)$ follows. Since this holds for every Y with $V_r(Y; X, t)$, the set X witnesses $\mathrm{Code}_n(t)$.

Conversely let $n = r + 1 \geq 2$ and $t \notin T_n$, and suppose that X witnesses $\mathrm{Code}_n(t)$. By (C1), X belongs to $B[G_C]$ for a closed set C as in (C1), and hence to the corresponding model $W_r[G_{Q_r}]$. We work in this model. For every $\delta < \omega_n$, choose an elementary submodel of a sufficiently large H_θ which has size ω_r and contains ω_r and $\delta + 1$ as subsets and the set $z = Z_r(X)$, the real t, L_{ω_n}, its satisfaction relation, and the graphs of the fixed coding functions, including the pairing function and the coding of terms, as elements. Its transitive collapse N_δ satisfies ZFC^- and the axioms (i)–(iii) of T_r, with $b^{N_\delta}_r = \omega_r$ and $\beta_\delta = b^{N_\delta}_{r+1} > \delta$.

The model N_δ belongs to $W_r[G_{Q_r}]$ and has cardinality ω_r there. If $r \geq 2$, recall from the proof of Proposition 7.3 that $W_r[G_{Q_r}]$ is contained in the model $W_{r-1}[G_{Q_{r-1}}]$ which is defined from the coordinates of C of levels below $r-1$; this is a model of (C2) for r. Hence Lemma 7.7(d) gives Y_δ with $V_r(Y_\delta; X, t)$ such that $Z_r(Y_\delta)$ codes N_δ. If $r = 1$, a code of N_δ itself is such a set Y_δ, by Lemma 7.6. Since X witnesses $\mathrm{Code}_n(t)$, we obtain $D_r(Y_\delta, t)$.

Put $P = Z_n(X)$ in the final model. As noted above, $D_r(Y_\delta, t)$ means that $\mathrm{Even}(P) \cap \beta_\delta$ is a club in β_δ which is disjoint from S^n_t. The ordinals β_δ, $\delta < \omega_n$, are unbounded in ω_n. By Lemma 3.4, $\mathrm{Even}(P)$ is a club in ω_n disjoint from S^n_t. This contradicts (C3). □

8 Comparing the forcings $F_{\vec{T}}$

Put $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$ and write $M_{\vec{T}} = B_{\vec{T}}[G_{D_{\vec{T}}}]$. Throughout this section, the stationary sets and the forcings $C_n(T)$ are those defined in L, and all products are formed in L. By Proposition 6.7, for each Π^2_n formula $A(a)$ there is a $\Pi^2_1(H_{\omega_n})$ formula $\Phi_A(a)$, independent of $\vec{T}$, such that

$$M_{\vec{T}} \models A(a) \quad \Longleftrightarrow \quad B_{\vec{T}} \models \Phi_A(a).$$

The corresponding assertion holds for Σ^2_n formulas. Note that this equivalence holds for every $D_{\vec{T}}$-generic filter over $B_{\vec{T}}$. Therefore we never have to compare the forcings $D_{\vec{R}}$ and $D_{\vec{T}}$ for different sequences directly.

Whenever $\vec{R} \leq \vec{T}$ and a $C_{\vec{T}}$-generic filter G over L is given, we let $B_{\vec{R}}$ be the extension of L by the restriction of G to the complete subforcing $C_{\vec{R}}$. Thus $B_{\vec{R}} \subseteq B_{\vec{T}}$. The statements about the forcing relations of $F_{\vec{R}}$ and $F_{\vec{T}}$ proved below do not depend on this choice of the generic filters.

Lemma 8.1. *Suppose that $\vec{R} \leq \vec{T}$ and $R_m = T_m$ for all $m < k$. Then, for generic filters chosen as above,*

$$H_{\omega_k}^{B_{\vec{R}}} = H_{\omega_k}^{B_{\vec{T}}}.$$

Proof. Write

$$P = \prod_{m<k} C_m(T_m), \qquad U = \prod_{m\geq k} C_m(R_m), \qquad R' = \prod_{m\geq k} C_m(T_m \setminus R_m).$$

Thus $C_{\vec{R}} = U \times P$ and $C_{\vec{T}} = U \times R' \times P$. By Lemma 3.6, $U \times R'$ is $< \omega_k$-distributive over L, so R' is $< \omega_k$-distributive over $L[G_U]$. Since $|P| < \omega_k$, Lemma 2.7 gives the same distributivity over $L[G_U][G_P]$. The two B-models therefore have the same sequences of ordinals of length $< \omega_k$, and hence the same H_{ω_k}. □

Proposition 8.2. *Suppose that $\vec{R} \leq \vec{T}$ and $R_m = T_m$ for all $m \neq n$. If $A(a)$ is a Π^2_n formula with real parameters and $F_{\vec{T}} \Vdash A(a)$, then $F_{\vec{R}} \Vdash A(a)$.*

Proof. Let $B_{\vec{R}} \subseteq B_{\vec{T}}$ be as above. By Lemma 8.1 with $k = n$, the two models have the same H_{ω_n}. Every class over H_{ω_n} in $B_{\vec{R}}$ is also a class in $B_{\vec{T}}$, and the first-order part of Φ_A has the same meaning in both models for such a class. Since Φ_A is $\Pi^2_1(H_{\omega_n})$,

$$B_{\vec{T}} \models \Phi_A(a) \quad \Longrightarrow \quad B_{\vec{R}} \models \Phi_A(a).$$

Now assume $F_{\vec{T}} \Vdash A(a)$, and let G' be an arbitrary $C_{\vec{R}}$-generic filter over L. Extend G' to a $C_{\vec{T}}$-generic filter by forcing with $C_n(T_n \setminus R_n)$. By the hypothesis and Proposition 6.7, $\Phi_A(a)$ holds in the larger model $B_{\vec{T}}$, and by the implication just proved it holds in $L[G']$. Since G' was arbitrary, Proposition 6.7 gives $F_{\vec{R}} \Vdash A(a)$. □

Proposition 8.3. *Suppose that $R_m = T_m$ for all $m \leq n$, and let $A(a)$ be a Σ^2_n or Π^2_n formula with real parameters. If $\vec{R} \leq \vec{T}$ and $B_{\vec{R}} \subseteq B_{\vec{T}}$ are as above, then $\Phi_A(a)$ has the same truth value in $B_{\vec{R}}$ and in $B_{\vec{T}}$. Consequently, whether or not $\vec{R} \leq \vec{T}$, $F_{\vec{R}}$ and $F_{\vec{T}}$ force the same Σ^2_n and Π^2_n statements with real parameters.*

Proof. For $\vec{R} \leq \vec{T}$, Lemma 8.1 with $k = n + 1$ gives equality of $H_{\omega_{n+1}}$ in the two B-models. In particular they have the same H_{ω_n} and the same subsets of it, since $|H_{\omega_n}| = \omega_n$. Thus $\Phi_A(a)$ has the same truth value in both models.

Every $C_{\vec{R}}$-generic filter extends to a $C_{\vec{T}}$-generic filter, and every $C_{\vec{T}}$-generic filter restricts to a $C_{\vec{R}}$-generic filter. Proposition 6.7 now gives equality of the forcing truth values. For incomparable sequences, compare each with their coordinatewise intersection. □

Proposition 8.4. *For every $\vec{T} \in L$, the forcing $F_{\vec{T}}$ preserves cardinals and adds no reals, and for every second-order formula φ and all reals $a \in L$, the trivial condition of $F_{\vec{T}}$ decides $\varphi(a)$. Thus $F_{\vec{T}}$ satisfies* (A1).

Proof. Every second-order formula is equivalent to a Σ^2_n or Π^2_n formula for some finite n. By Proposition 6.7, the truth value of such a formula $A(a)$ in the extension by $D_{\vec{T}}$ is the truth value in $B_{\vec{T}}$ of the formula $\Phi_A(a)$, whose only parameters are the constructible reals a. Since $C_{\vec{T}}$ is cone homogeneous by Lemma 3.6, Fact 2.3 shows that the trivial condition of $C_{\vec{T}}$ decides $\Phi_A(a)$. By Proposition 6.7, which holds for every $D_{\vec{T}}$-generic filter, the trivial condition of $F_{\vec{T}}$ decides $A(a)$.

The forcing $C_{\vec{T}}$ preserves cardinals and adds no reals by Lemma 3.6, and $D_{\vec{T}}$ preserves cardinals and adds no reals by Theorem 5.10(M4). Hence the two-step iteration $F_{\vec{T}}$ preserves cardinals and adds no reals. □

Remark 8.5. The map $A \mapsto \Phi_A$ of Proposition 6.7 is recursive. The statement $\Phi_A(a)$ is a statement about the structure $(H_{\omega_n}, \mathcal{P}(H_{\omega_n}), \in)$ of $B_{\vec{T}}$, which is a set, and the forcing relation of $C_{\vec{T}}$ for statements about such structures is definable in L, uniformly in $\vec{T}$ and in A. Hence the relation "$1_{F_{\vec{T}}} \Vdash A(a)$" is definable in L; no truth predicate for the universe is needed for this.

8.1 Verification of the four hypotheses

Theorem 8.6. *For $\vec{T} \in L$ put $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$. The family $\langle F_{\vec{T}} \rangle$ is definable in L and satisfies* (A1)–(A4) *of Theorem 2.2. The relation "$1_{F_{\vec{T}}} \Vdash \varphi(a)$" is definable in L, as a relation between Gödel numbers of second-order formulas, sequences $\vec{T}$ and real parameters.*

Proof. The definability of the family was shown after Definition 4.5. Proposition 8.4 gives (A1). The parameter-free formulas Code_n of Proposition 7.8, which do not depend on $\vec{T}$, give (A2). Propositions 8.2 and 8.3 give (A3) and (A4), respectively; in (A4), R_m and T_m may differ for infinitely many $m > n$. Remark 8.5 gives the definability of the forcing relation. None of these results depends on the recursive construction of the next section. □

9 The choice of $\vec{T}_*$ and the uniformizing functions

In this section we prove Theorem 2.2. The proof uses only the hypotheses (A1)–(A4) and does not depend on the particular definition of the forcings $F_{\vec{T}}$. Theorem 8.6 shows that these hypotheses hold for $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$.

9.1 The recursion in L

Fix a recursive enumeration $\langle A_{n,e} : e < \omega\rangle$ of the Π^2_n formulas $A_{n,e}(x,y)$. We work in L. For $n \geq 1$, $e < \omega$ and $x \in \mathbb{R}^L$, we say that the section of $A_{n,e}$ at x is *nonempty at* $\vec{T}$ if

$$1_{F_{\vec{T}}} \Vdash A_{n,e}(x,y) \qquad \text{for some } y \in \mathbb{R}^L.$$

By (A1), every statement $A_{n,e}(x,y)$ with $x, y \in \mathbb{R}^L$ is decided by the trivial condition of $F_{\vec{T}}$. Since $F_{\vec{T}}$ adds no reals, it follows that the section of $A_{n,e}$ at x is nonempty at $\vec{T}$ if and only if the set $\{y : A_{n,e}(x,y)\}$ is nonempty in the extension by $F_{\vec{T}}$.

Put $\vec{T}^0 = \langle I : n \geq 1\rangle$. Let $n \geq 1$ and suppose that $\vec{T}^{n-1}$ has been defined, with $T^{n-1}_n = I$. We define sequences $\vec{T}^{n,\alpha}$, which agree with $\vec{T}^{n-1}$ at all coordinates other than n, and a partial function a_n on the pairs (e,x), by recursion on α. Set $\vec{T}^{n,0} = \vec{T}^{n-1}$ and begin with the empty function. At stage α:

(i) if there is a pair (e,x) to which no value has been assigned yet and such that the section of $A_{n,e}$ at x is nonempty at $\vec{T}^{n,\alpha}$, take the $<_L$-least such (e,x) and the $<_L$-least real y with $1_{F_{\vec{T}^{n,\alpha}}} \Vdash A_{n,e}(x,y)$. Put $a_n(e,x) = y$ and remove only $\ell(e,x,y)$ from T_n, that is, $T^{n,\alpha+1}_n = T^{n,\alpha}_n \setminus \{\ell(e,x,y)\}$;

(ii) at a limit stage take the intersection of the sets $T^{n,\alpha}_n$ defined so far and the union of the partial functions;

(iii) the recursion stops if there is no pair as in (i).

Write $\vec{T}^n$ for the sequence at which the recursion stops and a_n for the resulting partial function. Finally put

$$T^*_n = T^n_n, \qquad \vec{T}_* = \langle T^*_n : n \geq 1\rangle.$$

Lemma 9.1. *For every n, the recursion stops before ω_2. It is definable in L, and $\vec{T}_* \in L$. Moreover, if $a_n(e,x) = y$, then*

$$1_{F_{\vec{T}^n}} \Vdash A_{n,e}(x,y),$$

and $a_n(e,x)$ is defined for every pair (e,x) such that the section of $A_{n,e}$ at x is nonempty at $\vec{T}^n$.

Proof. The least pair and the least value in (i) exist by the definition of nonemptiness, since $<_L$ is a wellorder. Every step of the form (i) assigns a value to a new pair (e,x). There are only ω_1 such pairs in L, so this cannot happen at every stage below ω_2.

Suppose that the value y has been assigned to (e,x) at stage α. For every later stage α' of this recursion, including limit stages and the last stage, we have $\vec{T}^{n,\alpha'} \leq \vec{T}^{n,\alpha}$, and these two sequences differ only at the coordinate n. Hence (A3) shows that $A_{n,e}(x,y)$ is forced by $F_{\vec{T}^{n,\alpha'}}$. When the recursion stops, there is no pair as in (i), which is the last assertion.

Since the relation "$1_{F_{\vec{T}}} \Vdash \varphi(a)$" is definable in L, the recursion is a definable recursion in L. Replacement gives the sequence $\langle \vec{T}^n : n \geq 1\rangle$ and the sequence $\vec{T}_*$. □

9.2 Verification

Proof of Theorem 2.2. Fix n. The sequences $\vec{T}^n$ and $\vec{T}_*$ have the same entries at all coordinates $m \leq n$. By (A4),

$$1_{F_{\vec{T}^n}} \Vdash A_{n,e}(x,y) \quad \Longleftrightarrow \quad 1_{F_{\vec{T}_*}} \Vdash A_{n,e}(x,y)$$

for all $e < \omega$ and all reals $x, y \in L$. In particular the same sections are nonempty at $\vec{T}^n$ and at $\vec{T}_*$. Note that (A4) is applied only once here, to two sequences which may differ at all coordinates $m > n$.

In an extension by $F_{\vec{T}_*}$ define

$$U_{n,e}(x,y) \quad \Longleftrightarrow \quad A_{n,e}(x,y) \ \wedge \ \neg\,\mathrm{Code}_n(\ell(e,x,y)).$$

This is Π^2_n. By (A2) and the injectivity of ℓ,

$$\neg\,\mathrm{Code}_n(\ell(e,x,y)) \quad \Longleftrightarrow \quad \ell(e,x,y) \notin T^*_n \quad \Longleftrightarrow \quad a_n(e,x) = y.$$

Indeed, at the beginning of the recursion for n the set T_n contains every code, and at each step of the form (i) exactly the code $\ell(e,x,a_n(e,x))$ is removed; the recursions for the other coordinates do not change T_n. By Lemma 9.1 and (A4), $A_{n,e}(x,a_n(e,x))$ holds in the extension whenever $a_n(e,x)$ is defined. Conversely, if the section $\{y : A_{n,e}(x,y)\}$ is nonempty in the extension, then the section of $A_{n,e}$ at x is nonempty at $\vec{T}_*$ and hence at $\vec{T}^n$, so $a_n(e,x)$ is defined. Consequently $U_{n,e}$ is a subset of $A_{n,e}$ and has exactly one value at every x with nonempty section.

Thus Π^2_n-UP holds for every $n \geq 1$. □

Proof of Theorem 1.1. Apply Theorem 2.2 to the family $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$, whose hypotheses are verified in Theorem 8.6. The sequence $\vec{T}_*$ belongs to L, and $F_{\vec{T}_*}$ is the required cardinal-preserving set forcing which adds no reals. □

10 Projective determinacy

We first carry out the construction over a ground model of the form $L[A]$, where $A \subseteq \omega_1$ is used as an additional predicate. We then show that for

$L(\mathbb{R})^{M_\omega}$ the predicate A can be chosen with parameter-free Δ^2_1 definitions, which allows us to eliminate it.

10.1 The construction over $L[A]$

In this subsection fix the ground model

$$\mathcal{W} = L[A] \models \mathsf{ZFC}, \qquad A \subseteq \omega_1^{\mathcal{W}}.$$

Throughout this section, all ground-model objects are computed in $\mathcal{W}$, and ω_n, $\aleph_\alpha$ denote the cardinals of $\mathcal{W}$. The forcing extensions used in the construction preserve these cardinals. Write $\Sigma^2_n[A]$ and $\Pi^2_n[A]$ for the point-classes over $(\mathrm{HC}, \in, A)$, with class quantifiers ranging over all subsets of HC in the extension under consideration. The predicate A always denotes the same set of the ground model.

Lemma 10.1. *The stationary sets, the almost disjoint families and the sets $P^n_t(C)$ of Section 3 have analogues in $\mathcal{W}$, defined with the canonical wellorder $<_{L[A]}$ in place of $<_L$. For $n = 1$, the diamond sequence is first-order definable over $(\mathrm{HC}, \in, A)$, by a definition which is absolute for extensions preserving* HC. *For $n \geq 2$, the definitions corresponding to Φ_D and Φ_B are $\Sigma_1(A)$ over $L_{\omega_n}[A]$, and Lemma 3.3 holds with $L_\beta[A]$ in place of L_β, for suitable models N containing A.*

Proof. Condensation for the hierarchy $L_\alpha[A]$ gives GCH in $\mathcal{W}$ and

$$H^{\mathcal{W}}_{\omega_n} = L_{\omega_n}[A] \qquad (n \geq 1).$$

Indeed, for $n \geq 2$ consider elementary submodels of size ω_{n-1} which contain ω_{n-1} as a subset and A as an element. Their transitive collapses map A to itself, since $A \subseteq \omega_1$, and this shows that every member of $H^{\mathcal{W}}_{\omega_n}$ belongs to $L_{\omega_n}[A]$. For $x \in \mathrm{HC}^{\mathcal{W}}$, take a countable elementary submodel which contains A and ω_1 as elements and the transitive closure of x as a subset. Its transitive collapse shows that $x \in L_\beta[A \cap \delta]$ for countable ordinals β, δ, and $L_\beta[A \cap \delta]$ belongs to a countable level of the hierarchy $L_\alpha[A]$; this proves the equality for $n = 1$. Counting the levels of the hierarchy gives CH, and for $n \geq 2$ the submodels considered above give $2^{\omega_{n-1}} = \omega_n$.

We use the good Σ_1 wellorder $<_{L[A]}$ throughout. For $n = 1$, the diamond sequence is defined as follows. For a countable limit ordinal δ, let (u, c) be the least pair in $L[A \cap \delta]$ such that

$$u \subseteq \delta, \quad c \subseteq \delta \text{ is club}, \quad \forall \xi \in c \; (u \cap \xi \neq d_\xi),$$

and put $d_\delta = u$; if there is no such pair, or if δ is not a limit ordinal, put $d_\delta = \varnothing$. Every pair in $L[A \cap \delta]$ which satisfies this condition occurs in a countable level of that hierarchy. Indeed, with $\lambda = \max(\omega, |\delta|^{L[A\cap\delta]})$, condensation

with δ fixed puts the pair below $(\lambda^+)^{L[A\cap\delta]} \leq \omega_1^{\mathcal{W}}$. The recursion through δ is hereditarily countable. That a pair is the least one satisfying the condition, or that there is no such pair, is verified by quantifying over the countable levels of $L[A\cap\xi]$ at each $\xi \leq \delta$. These levels and the sets $A\cap\xi$ are first-order definable over $(\mathrm{HC}, \in, A)$, so the recursion is first order over this structure.

This is a diamond sequence. Otherwise, let (X, C) be the $<_{L[A]}$-least counterexample, and take a countable elementary submodel of a sufficiently large limit level $L_\theta[A]$ which contains A, ω_1, $\langle d_\xi : \xi < \omega_1\rangle$, X and C. Let δ be its intersection with ω_1. The transitive collapse maps A to $A \cap \delta$, the diamond sequence to $\langle d_\xi : \xi < \delta\rangle$, and (X, C) to $(X \cap \delta, C \cap \delta)$. The image of (X, C) is the least pair in $L[A \cap \delta]$ which satisfies the condition displayed above: all its predecessors in the canonical wellorder belong to the collapsed level, and the condition is absolute for this level. Hence $d_\delta = X\cap\delta$, contradicting $\delta \in C$. The stationary sets S^1_t and S^1_* are defined from this sequence as in Section 3.

For $n \geq 2$ use the recursion from Lemma 3.1 with $<_{L[A]}$. In the condensation argument the elementary submodels contain A as an element and ω_{n-1} as a subset, so their collapses fix A. The witnesses for Φ_D and for the analogous definition Φ_B belong to $H^{\mathcal{W}}_{\omega_n} = L_{\omega_n}[A]$.

For the sets $P^n_t(C)$, choose the club $E \subseteq \mathrm{Lim}(C)$ in $L[A, C]$ and the codes w_d of wellorders using $<_{L[A]}$. As before, the odd part of $P \cap \beta$ gives $\beta \in E$, and the proof of Lemma 3.3 applies with $L_\beta[A]$ in N. The collapses used in the converse fix A and compute Φ_D and Φ_B correctly, so Lemma 3.4 applies. □

Lemma 10.2. *The formulas of Section 7 can be modified so that the models which they describe contain A as an element, with the same bounds on their complexity: there are formulas*

$$V_1^A \text{ (first order)},$$
$$O^A_r,\ C^{A,+}_r,\ C^{A,-}_r \in \Pi^2_{r-2}[A], \qquad V^A_r \in \Pi^2_{r-1}[A] \qquad (r \geq 2)$$

with the properties stated in Lemmas 7.6 and 7.7, and such that in the models described by V^A_r an additional constant a is interpreted by A.

Proof. Add a constant a to the language of the theories T_r, replace L_β by $L_\beta[a]$, and replace the statements about cardinals of L by the corresponding statements about $L[a]$. The surjections $f_{j,\alpha}$ and the other definable functions are defined using $<_{L[a]}$.

Let $r = 1$, and let Y be a code with equality relation $=_Y$ and membership relation E_Y. We can require the constants c_ξ to denote the ordinals ξ by the first-order conditions

$$E_Y(u, c_\xi) \iff \exists\eta < \xi\ (u =_Y c_\eta) \qquad (\xi < \omega_1).$$

Indeed, for the transitive collapse N, induction gives $c_\xi^N = \xi$. If all these constants belong to b_1 and $N \models b_1 = \omega_1$, then b_1^N is the true ω_1: a larger value would give a surjection from ω onto the true ω_1 in N. Now require

$$N \models a \subseteq b_1, \qquad E_Y(c_\xi, a) \iff A(\xi) \quad (\xi < \omega_1). \tag{P1}$$

These are first-order conditions over $(\mathrm{HC}, \in, A)$, and they imply $a^N = A$. This gives the formula V_1^A and the analogue of Lemma 7.6.

For $r \geq 2$ we follow the proof of Lemma 7.7. By induction, the models described by the formulas of the previous step contain A, and hence they compute the functions defined from $<_{L[A]}$ correctly. In the definition of V_r^A, the condition (P1) is expressed in the same way as the conditions which ensure $z_i^N = Z_r(X_i)$, by means of codes of ordinals and witnesses. Here ξ ranges over countable ordinals and $A(\xi)$ is an atomic formula, so no additional class quantifier is needed, and the complexity of the formulas is computed as before. For the completeness assertions, the elementary submodels are chosen so as to contain A as an element and ω_r as a subset; their transitive collapses map A to itself and satisfy (P1). □

Theorem 10.3. *There is a cardinal-preserving forcing $\mathbb{P}_A \in \mathcal{W}$ adding no reals and forcing $\Pi^2_n[A]$-UP for every finite $n \geq 1$.*

Proof. Let I be the set of codes $\ell(e, x, y)$ with $x, y \in \mathbb{R}^{\mathcal{W}}$, and carry out the constructions of Sections 3–5 in $\mathcal{W}$, using the stationary sets and almost disjoint families of Lemma 10.1 and the wellorder $<_{L[A]}$ wherever $<_L$ was used. The definitions of the conditions, the supports, the memories, the term forcings and the subforcings are unchanged. The proofs use only GCH, the closure and linkedness of the iterands, and the stationary sets S^n_*, all of which are available in $\mathcal{W}$. Thus Theorem 5.10 and the results on which it rests hold over this ground model.

Write

$$F^A_{\vec{T}} = C^A_{\vec{T}} * \dot{D}^A_{\vec{T}}, \qquad B^A_{\vec{T}} = \mathcal{W}[G_{C^A_{\vec{T}}}].$$

The proof of Proposition 6.7 goes through with A as an additional predicate. Fact 2.8 allows ground-model predicates, and Lemma 6.2 uses the canonical name for A. Hence each $\Pi^2_n[A]$ formula φ has a $\Pi^2_1(H_{\omega_n}, A)$ translation Φ^A_φ, independent of $\vec{T}$, such that

$$B^A_{\vec{T}}[G_{D^A_{\vec{T}}}] \models \varphi(x, y) \iff B^A_{\vec{T}} \models \Phi^A_\varphi(x, y).$$

Lemmas 10.1 and 10.2, together with the proof of Proposition 7.8, give formulas

$$\mathrm{Code}^A_n \in \Sigma^2_n[A], \qquad F^A_{\vec{T}} \Vdash (\mathrm{Code}^A_n(t) \iff t \in T_n). \tag{P2}$$

The forcing $C^A_{\vec{T}}$ is cone homogeneous, and Fact 2.3 applies to statements whose parameters are A and reals of $\mathcal{W}$. The proof of Proposition 8.4

therefore gives (A1), and (P2) gives (A2). For (A3), let $\vec{R} \leq \vec{T}$ differ only at the coordinate n. By Lemma 8.1 over $L[A]$ with $k = n$, the models $B^A_{\vec{R}}$ and $B^A_{\vec{T}}$ have the same H_{ω_n} and the same predicate A. Hence a $\Pi^2_1(H_{\omega_n}, A)$ statement which holds in the larger model holds in the smaller one, and we can argue as in Proposition 8.2. For (A4), use Lemma 8.1 over $L[A]$ with $k = n + 1$, as in Proposition 8.3. The forcing relation is definable in $\mathcal{W}$ as in Remark 8.5.

The proof of Theorem 2.2 can now be carried out in $\mathcal{W}$, with the set I defined above and with $<_{L[A]}$-least choices in place of $<_L$-least choices. There are ω_1 pairs (e, x), the recursion for each n stops before ω_2, and Replacement gives the sequence $\vec{T}_* \in \mathcal{W}$. For $\mathbb{P}_A = F^A_{\vec{T}_*}$, the uniformizing functions are given by

$$U_{n,e}(x, y) \iff \varphi_{n,e}(x, y) \wedge \neg\, \mathrm{Code}^A_n(\ell(e, x, y)),$$

where $\langle \varphi_{n,e} : e < \omega \rangle$ enumerates the $\Pi^2_n[A]$ formulas. The verification in Section 9 uses only (A1)–(A4), so it gives the conclusion, including the preservation of cardinals and the fact that no reals are added. □

10.2 Eliminating the predicate

Lemma 10.4. *Suppose that membership and nonmembership in $A \subseteq \omega_1$ have parameter-free Σ^2_1 definitions. Then every $\Sigma^2_n[A]$ or $\Pi^2_n[A]$ formula, $n \geq 1$, is equivalent to a Σ^2_n or Π^2_n formula, respectively. This holds in every model in which the two definitions are correct, and also for formulas with additional free class variables.*

Proof. Write the definitions as $\exists X\, \theta_+(\xi, X)$ and $\exists X\, \theta_-(\xi, X)$, where θ_+ and θ_- are first order. By the axiom of choice, witnesses for all $\xi < \omega_1$ can be collected into one class. Thus there is a first-order formula $D(B, Z)$ asserting $B \subseteq \mathrm{Ord} \cap \mathrm{HC}$ and

$$\forall \xi < \omega_1 \big[(\xi \in B \to \theta_+(\xi, Z_\xi)) \wedge (\xi \notin B \to \theta_-(\xi, Z_\xi))\big],$$

such that $\exists Z\, D(B, Z)$ holds exactly for $B = A$. For a formula $\varphi(A)$ which begins with an existential class quantifier use

$$\exists B \exists Z\, [D(B, Z) \wedge \varphi(B)];$$

for a formula which begins with a universal class quantifier use

$$\forall B \forall Z\, [D(B, Z) \to \varphi(B)].$$

The additional class quantifiers can be merged with the first block of class quantifiers of φ. □

10.3 The ground model $L(\mathbb{R})^{M_\omega}$

Assume the large-cardinal hypothesis of Theorem 1.2. Let $\mathcal{M} = M_\omega$, $\mathcal{R} = \mathbb{R}^{\mathcal{M}}$, and $\mathcal{W} = L(\mathcal{R})^{\mathcal{M}}$. At first we use only $\mathcal{W} \models \mathsf{ZF}$. In the following definition of A we identify reals with subsets of ω, using the fixed recursive coding when a relation is given on ω^ω. In first-order definitions over $\mathcal{W}$, the set $\mathcal{R}$ is allowed as a parameter, as in [20, Section 7, printed p. 66].

Lemma 10.5. *There is $A \subseteq \kappa = \omega_1^{\mathcal{W}}$ such that*

$$\mathcal{W} = L[A] \models \mathsf{ZFC} + \mathsf{CH} + \mathsf{PD}.$$

Both membership and nonmembership in A, for countable ordinals, are Σ_1 over $\mathcal{W}$ in the parameter $\mathcal{R}$, without further parameters.

Proof. Let $\prec$ be the wellorder of $\mathcal{R}$ given by the order of construction in $\mathcal{M}$. Steel states that this order is Δ_1 over $\mathcal{W}$ in the paragraph after Corollary 7.21 [20, printed p. 78]. Its graph is a set in $\mathcal{W}$ by Separation. Acceptability gives

$$\mathcal{P}(\omega)^{\mathcal{M}} \subseteq J^{\mathcal{M}}_{\omega_1^{\mathcal{M}}};$$

see [20, Definition 2.3 and the following paragraph]. The hierarchy is continuous, and each $J^{\mathcal{M}}_\beta$ for $\beta < \omega_1^{\mathcal{M}}$ is countable in $\mathcal{M}$, since it is obtained by closing under countably many finitary rudimentary functions. Thus each proper initial segment of $\prec$ is countable in $\mathcal{M}$. A bijection witnessing this is coded by a real and hence belongs to $\mathcal{W}$. Since $\mathcal{M}$ and $\mathcal{W}$ have the same reals, they have the same HC and the same ω_1.

Every proper initial segment of $\prec$ is therefore countable in $\mathcal{W}$. In ZF it follows that the order type of $\prec$ is at most $\omega_1^{\mathcal{W}}$, and by Cantor's theorem it is uncountable. Thus the unique increasing enumeration

$$e : \kappa \longrightarrow \mathcal{R}$$

belongs to $\mathcal{W}$. Neither the axiom of choice nor CH in $\mathcal{W}$ has been used so far. Put

$$A = \{\omega \cdot \xi + n : \xi < \kappa,\ n \in e(\xi)\}. \tag{P3}$$

Then $A \in \mathcal{W}$ and $L[A] \subseteq \mathcal{W}$. Conversely every real $e(\xi)$ can be computed from A, so e and its range $\mathcal{R}$ belong to $L[A]$. By the minimality of $L(\mathcal{R})$ we obtain $\mathcal{W} = L[A]$. Hence the axiom of choice holds in $\mathcal{W}$, and e witnesses CH. Every level of projective determinacy holds in $\mathcal{M}$ by the projective determinacy theorem [16], using finitely many Woodin cardinals and a measurable cardinal below a later Woodin cardinal. Since $\mathcal{W}$ has the same reals as $\mathcal{M}$, it has the same projective sets and the same strategies for games on ω, so $\mathcal{W} \models \mathsf{PD}$.

For the computation of the complexity, let $E(\xi, r)$ assert

$$r \in \mathcal{R} \ \wedge\ \exists f\, [f : (\xi, \in) \cong (\{s \in \mathcal{R} : s \prec r\}, \prec)].$$

The statement that f is such an isomorphism uses only quantifiers bounded by $\mathcal{R}, \xi, f$ and the Δ_1 definition of $\prec$. By Collection, the class of Σ_1 formulas is closed under bounded quantification; hence E is Σ_1 over $\mathcal{W}$ in the parameter $\mathcal{R}$. For $\xi < \kappa$ it holds exactly if $r = e(\xi)$. For $\alpha = \omega \cdot \xi + n < \kappa$,

$$\begin{aligned} \alpha \in A &\iff \exists r \in \mathcal{R}\, [E(\xi, r) \wedge n \in r], \\ \alpha \notin A &\iff \exists r \in \mathcal{R}\, [E(\xi, r) \wedge n \notin r]. \end{aligned} \tag{P4}$$

Division of ordinals by ω with remainder is Δ_1-definable in ZF. Both formulas in (P4) are therefore Σ_1, and they are complementary, because for every $\xi < \kappa$ there is exactly one r with $E(\xi, r)$. If we substitute the definition of $\prec$, then neither e, κ nor the graph of $\prec$ occurs as a parameter. □

10.4 Definability of A in the forcing extensions

Lemma 10.6. *Suppose $\mathcal{V} \models \mathsf{ZFC} + \mathsf{CH}$, put $\mathcal{R}_\mathcal{V} = \mathbb{R}^\mathcal{V}$, and assume*

$$\mathcal{L} = L(\mathbb{R})^\mathcal{V} \models \mathsf{ZFC} + \mathsf{CH}, \qquad \mathrm{HC}^\mathcal{L} = \mathrm{HC}^\mathcal{V}.$$

For every Σ_1 formula $\varphi(\mathcal{R}_\mathcal{V}, a)$ there is a Σ^2_1 formula $\widehat{\varphi}(a)$, obtained recursively from φ, such that

$$\mathcal{L} \models \varphi(\mathbb{R}^\mathcal{V}, a) \iff \mathcal{V} \models \widehat{\varphi}(a) \qquad (a \in \mathrm{HC}^\mathcal{V}).$$

The formula $\widehat{\varphi}$ has no parameter for $\mathbb{R}^\mathcal{V}$.

Proof. Write φ as $\exists z\, \psi(z, \mathcal{R}_\mathcal{V}, a)$ with bounded ψ, and put $H = \mathrm{HC}^\mathcal{V}$ and $\tau = \omega_1^\mathcal{V} = \omega_1^\mathcal{L}$. Suppose that φ holds in $\mathcal{L}$. Choose a sufficiently large level $L_\theta(\mathcal{R}_\mathcal{V})$ which contains a witness, H and all levels of the hierarchy up to a level containing these sets. In $\mathcal{L}$, take an elementary submodel of size τ which contains $\mathcal{R}_\mathcal{V}, H, a$ as elements and every member of H. Its transitive collapse is the identity on $\mathrm{TC}(\{\mathcal{R}_\mathcal{V}\})$. Condensation for the hierarchy $L_\alpha(\mathcal{R}_\mathcal{V})$ gives a level $L_\beta(\mathcal{R}_\mathcal{V})$ of size at most τ, with $\beta < \tau^+$, which contains $\mathcal{R}_\mathcal{V}, H, a$ and a witness for ψ.

This instance of condensation is proved by collapsing the hierarchy together with the satisfaction relations of its levels. At a successor step, elementarity shows that every element of the next level has a definition over the preceding level, with a formula and parameters in the submodel, and that every set defined in this way belongs to the submodel. Hence the collapse of a successor level is exactly the set of all subsets of the collapsed preceding level which are definable over it. At limit steps the collapsed levels are unions. By induction, the collapsed levels are the levels of the hierarchy $L_\alpha(\mathcal{R}_\mathcal{V})$.

The level $L_\beta(\mathcal{R}_\mathcal{V})$, the sequence of the preceding levels and their satisfaction relations have hereditary size at most τ. Hence they can be coded, together with a witness for ψ, by a single class $X \subseteq H$. We let $\widehat{\varphi}(a)$ assert

that there is a class X with the following properties. X codes a well-founded extensional relation E_X on a set $D \subseteq H$, with a distinguished copy of H and a distinguished element representing $\mathcal{R}_\mathcal{V}$, whose members are exactly the copies of the reals. The copy of H consists of the elements $\langle 0, u\rangle$ for $u \in H$, and the E_X-predecessors of $\langle 0, u\rangle$ are exactly the elements $\langle 0, v\rangle$ with $v \in u$. The coded hierarchy starts with $\mathrm{TC}(\{\mathcal{R}_\mathcal{V}\})$, its successor levels consist exactly of the definable subsets of the preceding levels, its limit levels are unions, and the coded satisfaction relations satisfy the Tarski conditions, where formulas are represented by their Gödel numbers. Finally, the last level contains a and a witness for $\psi(z, \mathcal{R}_\mathcal{V}, a)$.

All these conditions are first order over H, because elements, indices of levels, parameters and Gödel numbers are represented by elements of H. Well-foundedness is first order as well, since it is expressed by

$$\neg\exists f \in H\,[f : \omega \to D \ \wedge\ \forall n < \omega\ f(n+1)\ E_X\ f(n)],$$

and every countable sequence of members of H belongs to H. All components are coded by the one class X, so $\widehat{\varphi}$ is Σ^2_1. We have shown that $\widehat{\varphi}(a)$ holds in $\mathcal{V}$ if φ holds in $\mathcal{L}$.

Conversely, suppose that X is as in $\widehat{\varphi}(a)$. The transitive collapse of the coded structure contains the true $\mathrm{TC}(\{\mathcal{R}_\mathcal{V}\})$ as its first level. By the conditions on successor and limit levels, induction shows that its levels are the levels $L_\gamma(\mathcal{R}_\mathcal{V})$. The witness therefore belongs to $\mathcal{L}$, and since bounded formulas are absolute, $\mathcal{L} \models \psi(z, \mathcal{R}_\mathcal{V}, a)$. □

Lemma 10.7. *For the set A in (P3), there are parameter-free Σ^2_1 formulas $\sigma_+(\alpha)$ and $\sigma_-(\alpha)$ such that, in every set-forcing extension $\mathcal{W}'$ of $\mathcal{W}$ preserving* HC*,*

$$\begin{aligned} \mathcal{W}' \models \sigma_+(\alpha) &\iff \alpha \in A, \\ \mathcal{W}' \models \sigma_-(\alpha) &\iff \alpha \notin A \end{aligned} \qquad (\alpha < \omega_1^\mathcal{W}).$$

Proof. Let σ_+ and σ_- be the translations of the two formulas in (P4) given by Lemma 10.6. Every such $\mathcal{W}'$ has the same reals and ordinals as $\mathcal{W}$, so

$$L(\mathbb{R})^{\mathcal{W}'} = L(\mathbb{R})^\mathcal{W} = \mathcal{W}.$$

The enumeration $e \in \mathcal{W}$ still witnesses CH in $\mathcal{W}'$. Hence Lemma 10.6 applies in $\mathcal{W}'$, with $\mathcal{L} = \mathcal{W}$. The two formulas in (P4) refer to the same order and the same enumeration, and hence define A and its complement. Note that by Lemma 10.6 a new class in $\mathcal{W}'$ cannot be a witness for $\sigma_+(\alpha)$ or $\sigma_-(\alpha)$ unless the corresponding formula in (P4) holds in $\mathcal{W}$. □

Proof of Theorem 1.2. Take $\mathcal{W} = L[A]$ from Lemma 10.5 and the forcing $\mathbb{P}_A$ of Theorem 10.3. In the extension by $\mathbb{P}_A$, every $\Pi^2_n[A]$ relation has a $\Pi^2_n[A]$ uniformization. By Lemma 10.7, the hypothesis of Lemma 10.4 holds

in this extension, so A can be eliminated without increasing the complexity. Thus the extension satisfies $\mathbf{\Pi}^2_n$-UP for every finite $n \geq 1$.

The forcing preserves all cardinals of $\mathcal{W}$ and adds no reals. It preserves CH, and it changes neither the projective sets nor the set of strategies for games on ω; hence it preserves PD. This proves the theorem. □

10.5 The two uniformization patterns

Proposition 10.8. *Both L and $\mathcal{W} = L(\mathbb{R})^{M_\omega}$ satisfy $\mathbf{\Sigma}^2_n$-UP for every finite $n \geq 1$.*

Proof. First work in $\mathcal{W} = L[A]$ from Lemma 10.5, with A as an additional predicate, and put $H = \mathrm{HC}^{\mathcal{W}}$ and $\kappa = \omega_1^{\mathcal{W}}$. Every $X \subseteq H$ belongs to

$$H^{\mathcal{W}}_{\kappa^+} = L_{\kappa^+}[A].$$

The initial segments of the canonical wellorder $\prec = <_{L[A]}$ on $\mathcal{P}(H)^{\mathcal{W}}$ have size at most κ. This wellorder is $\mathbf{\Sigma}^2_1[A]$-good in the following sense. There is a first-order formula $G_A(T, J, C)$ over $(H, \in, A)$ such that

$$\exists C\, G_A(T, J, C) \quad \Longleftrightarrow \quad \{J_i : i \in D_J\} = \{S \subseteq H : S \prec T\}, \tag{P5}$$

where the class J codes the family of classes J_i, for i in an index set $D_J \subseteq H$. Moreover, for every $T \subseteq H$ there are such J and C. The enumeration $\langle J_i : i \in D_J \rangle$ of the predecessors of T is arbitrary.

To see this, choose a limit ordinal $\beta < \kappa^+$ such that $L_\beta[A]$ contains T, H, A, J and the set of all $<_{L[A]}$-predecessors of T; such an ordinal exists because $<_{L[A]}$ is a good $\mathbf{\Sigma}_1(A)$ wellorder. Let C code, by classes over H, the level $L_\beta[A]$, the sequence of the preceding levels with their satisfaction relations, and an enumeration of the predecessors of T which are subsets of H. The formula G_A requires that the coded structure contains copies of the true H and A, and that the elements representing T and the classes J_i have exactly the copies of the elements of T and of J_i as members. It requires that every successor level consists exactly of the subsets of the preceding level which are definable over it with the predicate A. Finally, it requires a witness for the $\mathbf{\Sigma}_1$ definition of $\mathrm{Init}_A(t, I)$, the analogue for $L[A]$ of Init in Section 3.1, and that the enumerated classes are exactly the elements of $I \cap \mathcal{P}(H)$. These are first-order conditions, as in Lemma 10.6. The transitive collapse of a well-founded structure satisfying them is a level of the hierarchy $L_\alpha[A]$, and by the upward absoluteness of $\mathbf{\Sigma}_1(A)$ formulas the set I is the set of all $<_{L[A]}$-predecessors of T. This proves (P5), and the existence of J and C for every T and for any given enumeration J of the predecessors of T. Lemma 10.4 applies to formulas with the free class variables T, J, so the relation in (P5) is also parameter-free $\mathbf{\Sigma}^2_1$ in $\mathcal{W}$. Equivalently, using the canonical bijection between H and κ, we obtain a good wellorder of $\mathcal{P}(\omega_1)^{\mathcal{W}}$.

Let $\mathcal{E}(x,y)$ be a $\Sigma^2_n[A]$ relation. Combine the class quantified by the leading existential quantifier with y into one class T, using a fixed recursive coding. Write

$$\mathcal{E}(x,y) \iff \exists T\,[\mathrm{out}(T) = y \wedge \psi_*(x,T)],$$

where ψ_* includes the statement that T codes a pair of this form, is first order if $n = 1$, and belongs to $\Pi^2_{n-1}[A]$ if $n \geq 2$. We uniformize $\mathcal{E}$ by choosing the $\prec$-least T satisfying $\psi_*(x,T)$:

$$\begin{aligned} U(x,y) \iff \exists T, J, C\,[&G_A(T,J,C) \wedge \mathrm{out}(T) = y \wedge \psi_*(x,T) \\ &\wedge \forall i \in D_J\ \neg\psi_*(x,J_i)]. \end{aligned} \tag{P6}$$

For $n = 1$, the part of this formula after the existential class quantifiers is first order. For $n \geq 2$, the statement $\forall i \in D_J\ \neg\psi_*(x,J_i)$ is $\Sigma^2_{n-1}[A]$: the quantifier over i is first order, and by the axiom of choice the witnesses for all $i \in D_J$ can be collected into one class. Its conjunction with $\psi_* \in \Pi^2_{n-1}[A]$, preceded by the existential class quantifiers, is $\Sigma^2_n[A]$. Thus (P6) defines a $\Sigma^2_n[A]$ uniformization of $\mathcal{E}$.

By Lemma 10.7 and Lemma 10.4, A can be eliminated without increasing n. This proves the assertion for $\mathcal{W}$. For L use the same argument without the predicate A; in this case nothing has to be eliminated. □

Lemma 10.9. *For each finite $n \geq 1$, boldface Σ^2_n uniformization and boldface Π^2_n uniformization cannot both hold in a model of* ZFC.

Proof. There is a universal Π^2_n relation $\mathcal{U}(e,x,y)$, where the real e codes a formula and its real parameter. To handle arbitrary first-order formulas, a class coding their satisfaction relation is added to the last block of class quantifiers: it is characterized by the first-order Tarski conditions, which are added as a conjunct if the last block is existential and as the hypothesis of an implication if it is universal. Since the satisfaction relation exists and is unique, this does not increase the number of alternations.

Assume $\mathbf{\Pi}^2_n$ uniformization and let $\mathcal{F}(e,x,y)$ be a $\mathbf{\Pi}^2_n$ uniformization of $\mathcal{U}$. If $\mathcal{U}_e$ is the graph of a function, then $\mathcal{F}_e = \mathcal{U}_e$. Now

$$\mathcal{D}(e,y) \iff \neg\mathcal{F}(e,e,y)$$

is $\mathbf{\Sigma}^2_n$ and every section is nonempty, since $\mathcal{F}(e,e,\cdot)$ has at most one value. If $\mathbf{\Sigma}^2_n$ uniformization also held, $\mathcal{D}$ would have a uniformization which is a total function with graph $\mathcal{K} \in \mathbf{\Sigma}^2_n$. Since this function is total,

$$\neg\mathcal{K}(e,y) \iff \exists z\,[\mathcal{K}(e,z) \wedge z \neq y],$$

so $\mathcal{K}$ is also $\mathbf{\Pi}^2_n$. Choose e_0 with $\mathcal{U}_{e_0} = \mathcal{K}$. Then $\mathcal{F}_{e_0} = \mathcal{K}$, whereas the value y of this function at e_0 satisfies both $\mathcal{K}(e_0,y)$ and $\neg\mathcal{F}(e_0,e_0,y)$, a contradiction. □

Proof of Corollary 1.3. Proposition 10.8 gives Σ^2_n-UP in the two ground models, and Theorems 1.1 and 1.2 give Π^2_n-UP in their extensions. Lightface uniformization gives boldface uniformization by pairing the real parameter with the input and restricting the resulting uniformizing function to that parameter. Lemma 10.9 gives the failures stated in the corollary. The forcings add no reals and preserve ω_1, hence preserve HC and all projective truth. □

In the extensions, the wellorder of the ground model still wellorders the subsets of HC which belong to the ground model. The new subsets of HC are not in $L[A]$, so this wellorder is not a wellorder of all classes over HC. The argument for Σ^2_n uniformization in the ground model therefore does not apply in the extension.

Use of generative AI. I used ChatGPT Astra to help develop the forcing construction and to draft parts of the proofs. The main idea of this work, the strategy of removing codes, is an old idea of mine, which I originally developed in another context (where it failed to work). The second-order context leaves enough room between what we code and how the reals behave for it to work here. I also used ChatGPT to improve the language, to revise the exposition, to avoid typos, and to locate references and relevant literature, in particular the paper [2], which made a much neater proof possible. I take responsibility for the contents of this paper.

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